



LUANGWA COMMUNITY FORESTS PROJECT



Document Prepared by BioCarbon Partners

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1 PROJECT DETAILS

1.1 Summary Description of the Implementation Status of the Project

The Luangwa Community Forests Project is a large-scale REDD+ grouped project implemented on communal land and private ranches in Eastern and Lusaka Provinces, Zambia with an initial project activity instance and second instance area of 943,646 ha and 92,990 ha respectively (Figure 1). The third project activity instance (subject to verification in this monitoring period) is 199,622 ha in project area extent. The total project area, at the end of this monitoring period, is estimated at 1,236,258 ha.

Implementation is in partnership with the traditional authorities and the government of the Republic of Zambia. The project generates emissions reductions through avoided deforestation with mitigation activities that include a combination of direct conservation support, engagement and capacity building with key Government and community stakeholders. The baseline scenario is unplanned deforestation driven primarily by subsistence agriculture. Communities are impoverished and biodiversity declines due to deforestation and increased poaching.

The project's community objective is instituting poverty alleviation activities benefiting at least 20,000 households, specifically targeting poor and vulnerable households. The biodiversity objectives are maintaining a wildlife corridor between several national parks in the catchment of Zambia's 4th largest river system, conserving and maintaining vulnerable and endangered species through habitat protection and reduction in poaching. The climate objectives are to avoid deforestation in the project area and assist communities and biodiversity with climate change adaptation strategies through income diversification, improved farming techniques, crop diversification, and maintenance of habitat corridors. Project area boundaries were marked at validation and since then, aerial and ground patrols are conducted regularly to monitor compliance. During each monitoring period, 20% of accounting and proxy area biomass plots and 100% of leakage area monitoring plots are inventoried. Remote sensing is used to monitor disturbance and fires while wildlife and other biodiversity indicators are monitored in bi-annual fixed transects surveys and random ground patrols.

The project implementation and crediting period start date is 1 May 2015 and will be operational until 30 April 2045. The second and third activity instances started on the date of 11 August 2021 and 1 January 2022, respectively. All three instances will also be operational until 30 April 2045. In the current monitoring period, The project delineated a new reference area representative of all the 3 PAI and updated the CDM. The estimated gross emissions reduction for the 30-year project span with all three project activity instances combined is 102,251,032 tCO₂e, at an average of 3,408,368 tCO₂e/yr. The combined estimated gross emissions reduction of the 3 project activity instances for the current monitoring period (1 January 2022 to 31 December 2022) is 2,163,019 tCO₂e, however due to insufficient data available to close finding #9 raised by Verra during its accuracy review (see the PRR), a 30% discount is applied to the verified volume.

Table 1. Summary of the audit history of the LCFP

Audit Type	Period	Program	VVB Name	Number of years
Validation	14 October 2019	VCS	SCS Global Services (SCS)	-
Verification	1-May-2015 to 31-Dec-2018	VCS	SCS Global Services (SCS)	3.8
Verification	1-January-2019 to 31-December-2019	VCS	Aenor International S.A.U.	1
Verification	1-January-2020 to 31-December-2020	VCS	Aenor International S.A.U.	1
Verification	1-January-2021 to 31-December-2021	VCS	Aenor International S.A.U.	1
Total				6.8

1.2 Sectoral Scope and Project Type

The project falls under scope 14 of the VCS, namely Agriculture, Forestry, and Other Land Use (AFOLU). The AFOLU project category is Reducing Emissions from Deforestation and Forest Degradation, using Methodology for Avoided Ecosystem Conversion, VM0009, v3.0 and qualifies as baseline type F-U3, unplanned mosaic or frontier deforestation. This is a grouped REDD+ project implemented in Eastern and Lusaka Provinces, Zambia.

1.3 Project Proponent

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Organization name	Msoro Community Forest Management Group
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Organization name	Mwape Community Forest Management Group
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Organization name	Mphuka Community Forest Management Group
Contact person	Lillian Njekwa
Title	Board Chairperson

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1.4 Other Entities Involved in the Project

Organization name	Munyamadzi Game Ranch
Role in the Project	Landowner of Munyamadzi project area
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Organization name	Nyakolwe Game Ranch
Role in the project	Landowner of Nyakolwe project area
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Title	Managing Director
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Email	sable@sabletsp.com

Organization name	Nyamvu Game Ranch
Role in the project	Landowner of Nyamvu project area
Contact person	Zaid Patel

Title	Director
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Email	Zaid137@hotmail.com

Organization name	Kazumba Game Ranch
Role in the project	Landowner of Kazumba project area
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Organization name	United States Agency for International Development (USAID)
Role in the project	Initial Funder
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Organization name	Forestry Department (FD)
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Organization name	Pia Manzi Wildlife Limited and Luansemfwa Safaris Limited
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Organization name	Bee Sweet
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Organization name	Ministry of Agriculture
Role in the project	Agricultural Extension Support
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Organization name	Ministry of Agriculture
Role in the project	Agricultural Extension Support
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Organization name	Ministry of Agriculture
Role in the project	Agricultural Extension Support
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1.5 Project Start Date

MR.2

The project start date is May 1st, 2015.

The first activity instance start date coincides with the project start date while the second and third activity instances start dates are August 11th 2021 and January 1st 2022 respectively.

1.6 Project Crediting Period

MR.3

The project crediting period is from May 1st, 2015 to April 30th, 2045 for a total of 30 years. The second and third project activity instances start on August 11th 2021 and January 1st 2022, both lasting up to April 30th 2045 for a period of 23.7 and 23.3 years respectively.

1.7 Project Location

All the areas identified under the first, second and third activity instances (Table 2 and Table 3), lie in the Eastern, Lusaka and Central Provinces of Zambia and fall into the Luangwa Zambezi River ecological zone and the Central Southern and Eastern Plateau Zone (Figure 1). In addition, all activity instances are situated in the Game Management Areas (GMA) within the respective Chiefdom areas. The project sites in Eastern Province are bordered by the South Luangwa National Park and Luangwa river on the north-western border. The project area consists of five sites in Lusaka Province in Rufunsa, Luangwa and Kafue Districts, one on each side of the great east road in Shikabeta and Mpanshya chiefdoms, two sharing the Lower Zambezi National Park's (LZNP) eastern boundary in Mburuma and Mphuka chiefdoms and one sharing a portion of the LZNP's western boundary in Chiawa Chiefdom. The second instance also includes the Mpanshya Wildlife Corridor, an area identified to function as an ecological corridor between two disconnected project areas of Mpanshya and with Shikabeta.

The other nine first instance project sites lie in the eastern province of Zambia in Mambwe, Lundazi, and Nyimba Districts. Project sites in Mambwe Districts are Mnkanya, Jumbe, Nsefu, Malama, and Msoro. In Lundazi District there is one project site in Mwanya Chiefdom. The project area in Mwanya is bordered by Lukusuzi National Park and Luambe National Park area. There are two sites in Nyimba District one each in Nyalugwe and Luembe chiefdom. The geographic boundaries of the project area and a full description of the boundaries of each site can be found in the shapefiles and narratives in Appendix I. Below is the location map of the project area.

MR.4

Table 2. Summary of Project Area: REDD+ Project Zones under communal tenure, for all three activity instances.

Project Activity Instance	Community Resource Board	Game Management Area	Chiefdom	Project Area (ha)	Forest Area (ha)
1	Mpanshya	Rufunsa	Mpanshya	84,530	82,183
1	Shikabeta	Luano	Shikabeta	118,102	111,968
1	Mwanya	Lower Lumimba	Mwanya	81,668	59,030
1	Luembe	West Petauke	Luembe	288,726	254,582
1	Nyalugwe	West Petauke	Nyalugwe	60,893	58,809
1	Msoro	Lower Lupande	Msoro	24,688	23,523
1	Malama	Lower Lupande	Malama	21,577	17,220
1	Mnkanya	Upper Lupande	Mnkanya	33,623	29,081

1	Jumbe	Upper Lupande	Jumbe	15,443	11,066
1	Nsefu	Upper Lupande	Nsefu	39,801	25,341
1	Chitungulu	Lumimba	Chitungulu	38,764	28,681
1	Sandwe	Sandwe	Sandwe	92,362	85,767
2	Mpanshya Corridor	Rufunsa & Luano	Mpanshya & Shikabeta	15,271	13,430
3	Kazembe	Upper Lumimba	Kazembe	18,515	13,241
3	Mwape	Chisomo	Mwape	20,500	18,845
3	Chiawa	Rufunsa	Chiawa	68,206	64,343
3	Mphuka	Rufunsa	Mphuka	25,881	24,541
3	Mburuma	Rufunsa	Mburuma	66,520	62,537
	Total			1.115,070	984,188

Table 3. Summary of Project Area: REDD+ Project Zones on Private Land for both project activity instances.

Project Activity Instance	Game Ranch	Chiefdom	Project Area (ha)	Forest Area (ha)
1	Munyamadzi	Luembe	18,671	16,449
1	Pia Manzi	Nyalugwe	24,798	22,393
2	Nyakolwe	Luembe	57,706	32,013
2	Nyamvu	Luembe	15,238	7,176
2	Kazumba	Luembe	4,775	2,732
	Total		121,188	80,763



As described in detail in section 1.1.2 of the PD, a GIS analysis of deforestation map from 2007 to 2017 was used to identify key constraints to conversion. The two factors that stood out most prominently are distance from roads and slope. These factors make sense since roads provide access to forested areas for farmers and charcoal makers. Farmers who are responsible for most deforestation prefer to cultivate on relatively flat ground which will better hold soil moisture and is easier to work with oxen (which some farmers have). The analysis found constraints to conversion at greater than 45% slope and greater than 6.5 km distance from a road. Therefore, to calculate the project accounting area, a polygon buffer covering 6.5 km from roads was clipped from the project area. Then, areas of high slope and areas that were not forest at each project instance start date, were also removed from the accounting area polygon.

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The project is bound between Lat: 12° 1'3.50"S, Long: 28°30'25.86"E (NW Corner) , lat: 12° 1'3.50"S, long: 33° 0'2.08"E (NE Corner), lat: 16° 0'2.92"S, long: 28°30'25.86"E (SE Corner), and lat= 16° 0'2.92"S, long: 33° 0'2.08"E (SW corner).

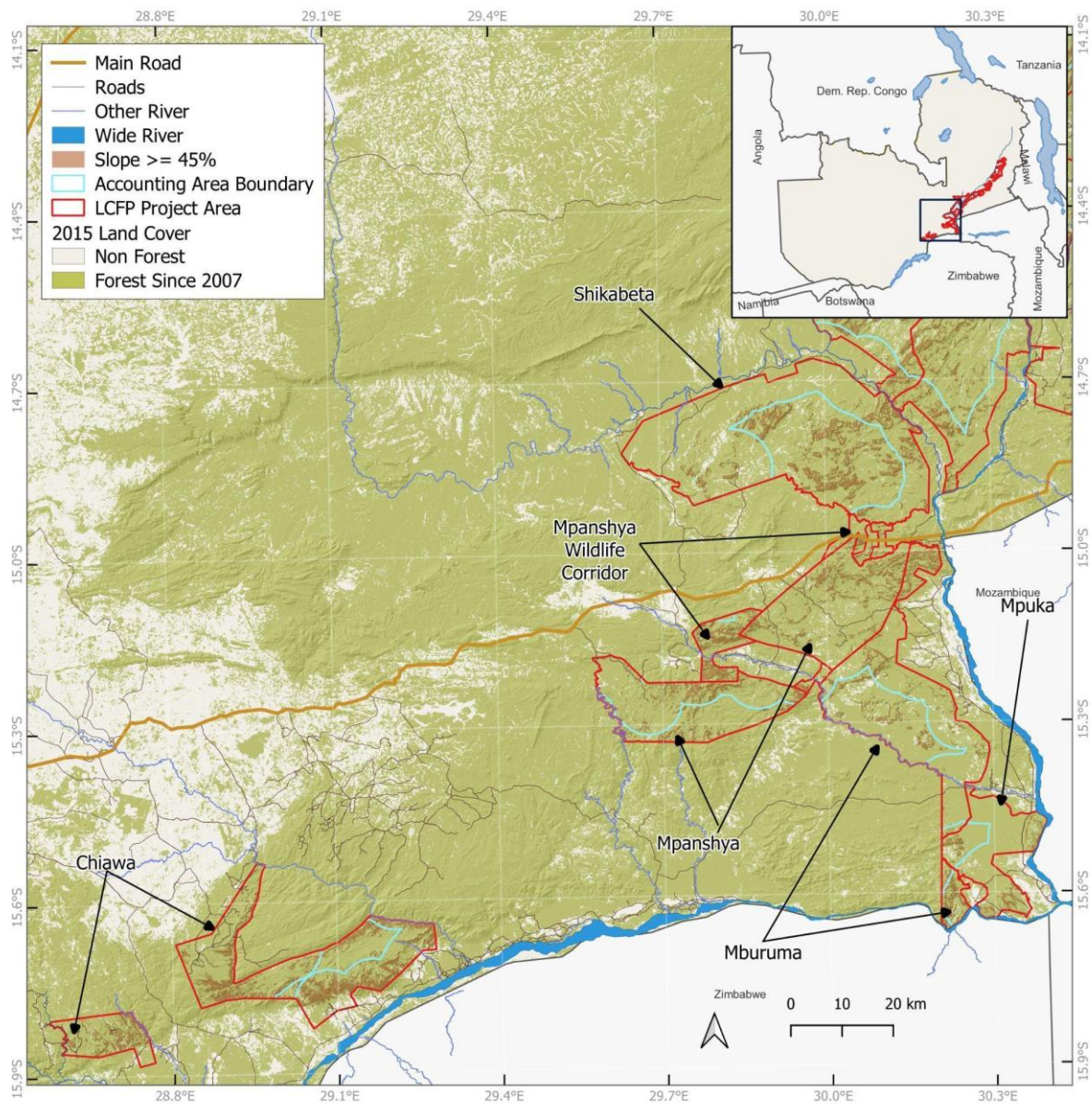


Figure 2. Shikabeta, Mpanshya, Mpanshya Wildlife Corridor, Mburuma, Mphuka and Chiawa Accounting Areas

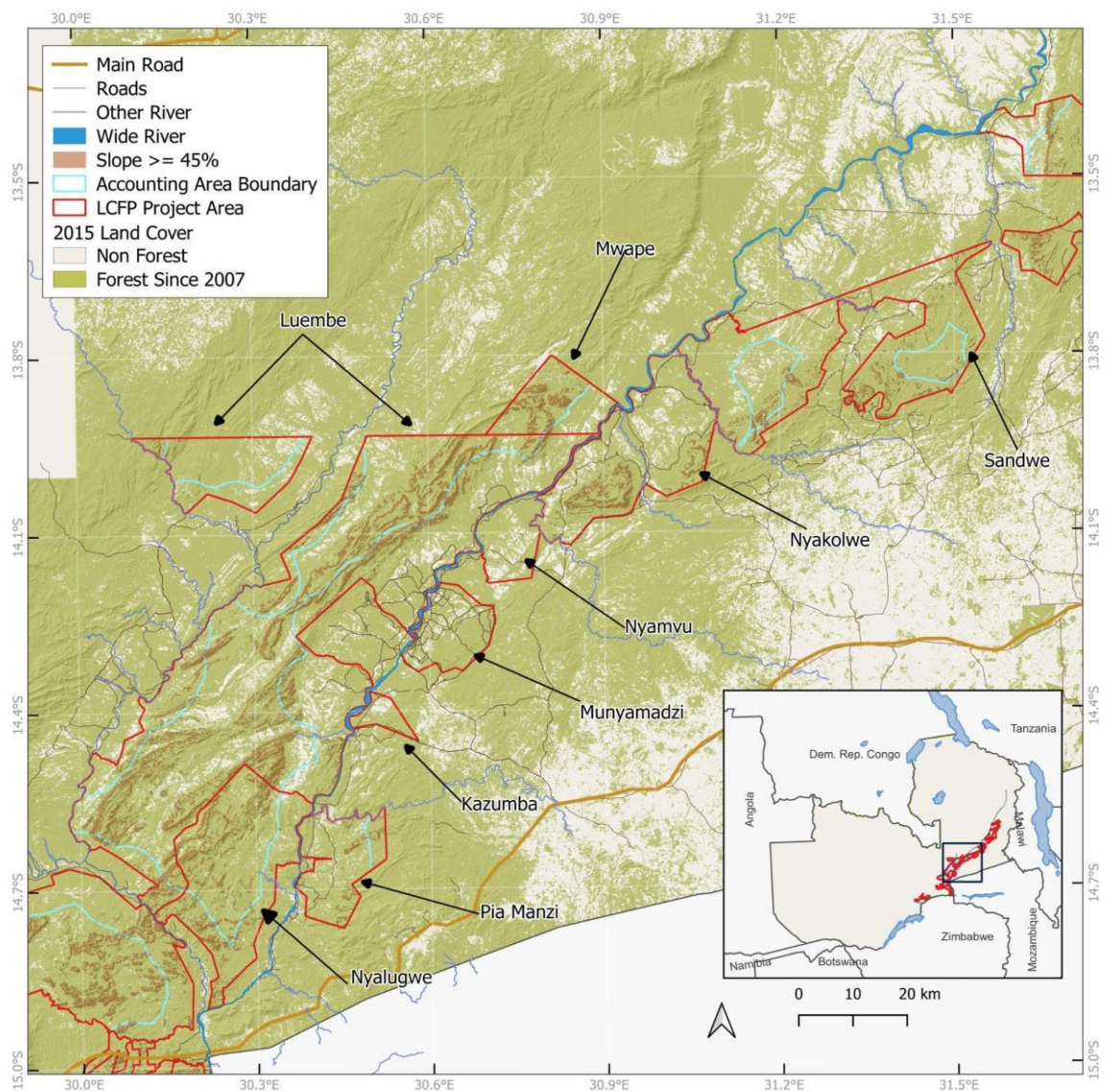


Figure 3. Luembe, Kazumba, Nyamvu, Nyakolwe, Munyamadzi and Sandwe Accounting Areas

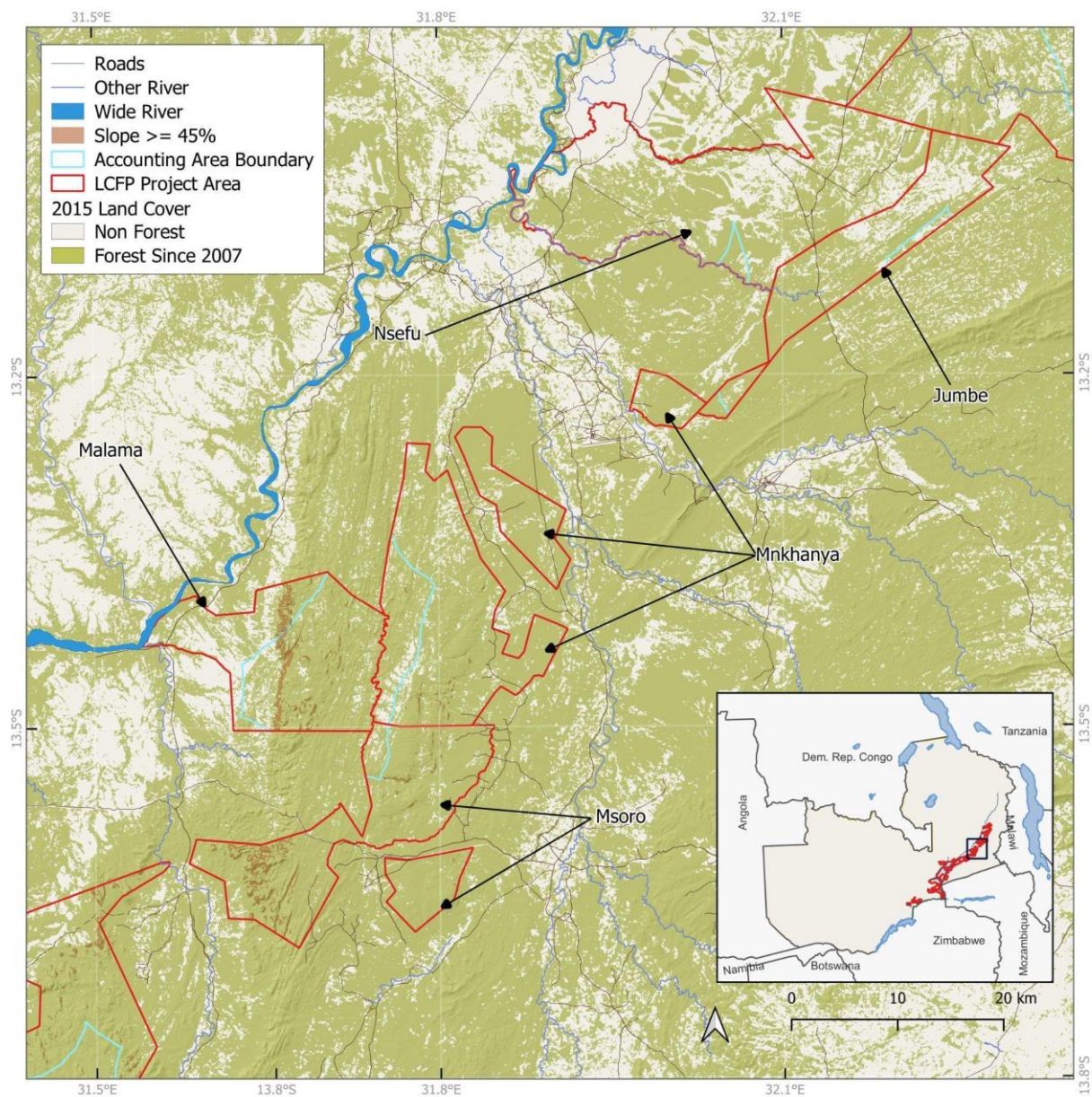


Figure 4. Malama, Msoro, Mnkhangya, Jumbe and Nsefu Accounting Areas

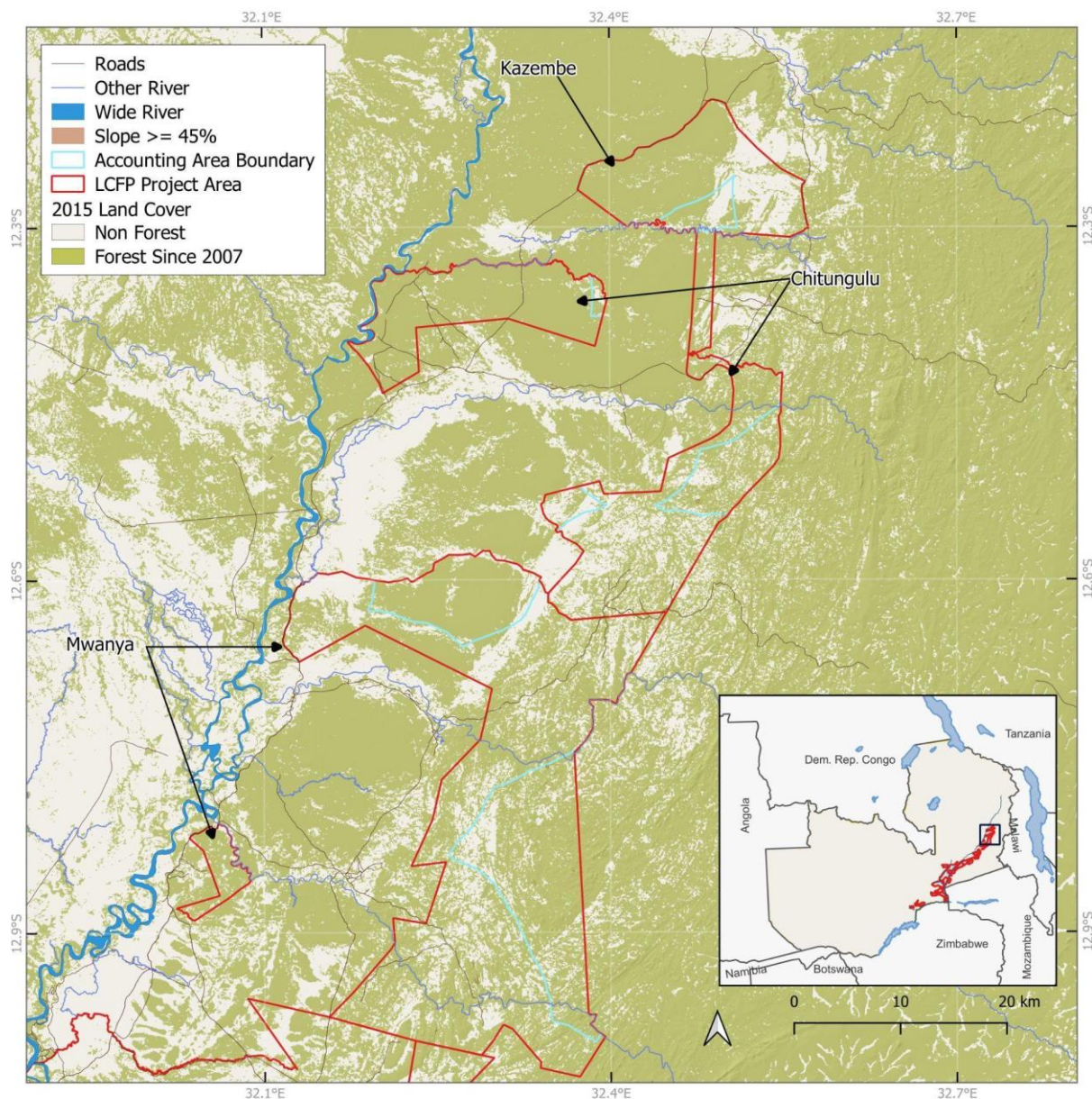


Figure 5. Mwanya, Chitungulu and Kazembe Accounting Area

1.8 Title and Reference of Methodology

The reference methodologies and tools applied to the project are the following:

- VCS VM0009 V3.0 Methodology for Avoided Ecosystem Conversion.
- VCS VT0001 Tool for the Demonstration and Assessment of Additionality in VCS Agriculture, Forestry, and Other Land Use (AFOLU) Project Activities, version 3.0.
- VCS AFOLU Non-Permanence Risk Tool v4.0

1.9 Participation under other GHG Programs

The project has not applied to any other GHG program.

1.10 Other Forms of Credit and Supply Chain (Scope 3) Emissions

The LCFP is not subject to binding emissions limits or emissions trading programs.

The project is validated under the CCB Standard Version 3.

The LCFP is not involved in any Supply Chain, and therefore does not impact the emissions of goods and services in a supply chain.

1.11 Sustainable Development Contributions

The LCFP contributes to 16 of the 17 UN Sustainable Development Goals (Table 4). There is no requirement for any formal monitoring and reporting of activities that pertain to these goals.

SDG 1. No Poverty: Disbursement of conservation fees, livelihood investments, local employment

SDG 2. Zero Hunger - Improvement of honey, climate smart agriculture and horticulture production

SDG 3. Good Health and Well-being - Support to Health Care facilities and services

SDG 4. Quality Education - Support to Education facilities and services

SDG 5. Gender Equality - Empowering of local women

SDG 6. Clean Water and Sanitation - Borehole drilling

SDG 7. Affordable and Clean Energy - Provision of solar equipment

SDG 8. Decent Work and Economic Growth - Small business development and income diversification

SDG 9. Industry, Innovation and Infrastructure - Forest carbon science innovation supports technology and industry development within developing countries such as Zambia and supports rural communities gaining access to global markets through various livelihood initiatives such as bee keeping.

SDG 10. Reduced Inequalities - Carbon Fees disbursement to local communities

SDG 11. Sustainable Cities and Communities - Watershed Protection and Climate SMART Agriculture development

SDG 12. Responsible Consumption and Production - Sustainable farming development

SDG 13. Climate Action – Reduced Deforestation and Degradation

SDG 15. Life on Land - Reduced deforestation and degradation

SDG 16. Peace, Justice and Strong Institutions - Support good governance of local institutions

SDG 17. Partnerships for the Goals - Partnership with Forest Department and CFMGs

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Table 4. Contributions of the LCFP to the Sustainable Development Goals

Row number	SDG	SDG Indicator	Net Impact on SDG Indicator	Current Project Contributions	Contributions Over Project Lifetime
1)	1.1	1.1.1 Proportion of population below the international poverty line	Implemented activities to decrease	At least 47,317 households benefited directly from Carbon and livelihood funds. A total of USD 926,417 were paid in livelihood investments in LCFP during 2022	The LCFP indicated a 220% change in household income since inception in ZMW. \$10,168,000+ paid to LCFP communities where 47,317 Households benefit directly from carbon and livelihood fees for at least 222,894 community beneficiaries Since project inception, more than 200 community impact projects have been implemented or initiated.

Row number	SDG	SDG Indicator	Net Impact on SDG Indicator	Current Project Contributions	Contributions Over Project Lifetime
2)	2.3 2.4	2.3.1 Volume of production per labour unit by classes of farming/pastoral/forestry enterprise size 2.3.2 Average income of small-scale food producers, by sex and indigenous status 2.4.1 Proportion of agricultural area under productive and sustainable agriculture	Implemented activities to increase	5,041 beehives have been delivered. 3,627.17 kg of honey was harvested. Average maize grain yield recorded from 1.8ton/ha to 4.0 ton/ha. Number of lead farmers increased by 101, increasing the size of land area under conservation farming by 113 ha 484 Lead farmers were trained, adding to the previously trained. Total # of farmers enrolled under conservation farming increased to 2,627. 210 Honey harvesters were trained. 130 Horticulture farmers trained	1386 Lead Farmers have been trained in CSA technologies, and 9,000+ follower farmers; and provided with farming inputs. 4 Agroforestry Chiefdom nursery has been established in 4 Chiefdoms. 8,045 beehives were deployed at end of 2022 for honey production Total # of farmers enrolled under conservation farming increased 9,888 from project start

Row number	SDG	SDG Indicator	Net Impact on SDG Indicator	Current Project Contributions	Contributions Over Project Lifetime
3)	3.8	3.8.1 Coverage of essential health services	Implemented activities to increase	Completion of Maternity Annex in Sandwe, Construction of a health post in Luembe and Construction of incinerator Nthumbe Health post in Chitungulu.	15 Clinics or health facilities have been improved/constructed or initiated since inception of LCFP
4)	4.6	Access to education services (BCP defined indicator) Or 4.6.1 Percentage of population in a given age group achieving at least a fixed level of proficiency in functional (a) literacy and (b) numeracy skills, by sex	Implemented activities to increase	Over 22 social projects were implemented in 2022 across the chiefdoms of Sandwe, Nyalugwe, Luembe, Mwanya, Chitungulu. Mwanya CFMG introduced a School Scholarship and Teachers' Support program, which have helped 25 students and 12 teachers, respectively. Chitungulu CFMG initiated a project that provides tertiary education support in bricklaying and tailoring to people in the community. This initiative has already	Constructed, improved or initiated 37 Education facilities including Teachers' houses and classroom blocks. Scholarship and Teacher's support programs in Mwanya Tertiary education support program in Chitungulu

Row number	SDG	SDG Indicator	Net Impact on SDG Indicator	Current Project Contributions	Contributions Over Project Lifetime
				benefited 13 community members.	
5)	5.5	5.5.2 Proportion of women in managerial positions	Implemented activities to increase	More than 30% of the BCP Senior management team have female representation	24 % female representation on BCP payroll 134 of 225 beekeeping farmers trained between 2019 and 2022 are female (59%)
6)	6.6	6.1.1 Proportion of population using safely managed drinking water services	Implemented activities to increase	In LCFP 10 boreholes were drilled and completed in Sandwe	136 WASH and boreholes projects completed or initiated since inception of LCFP
7)	7.1 7.2	7.1.2 Proportion of population with primary reliance on clean fuels and technology 7.2.1 Renewable energy share in the total final energy consumption	Implemented activities to increase	Installation of solar lighting in Chisanga, Luembe Chiefdom	Collaborated with the Forestry Department to ensure the community rights to sustainable production empowering communities, avoiding deforestation. Provided solar lighting at Kamphasa school (LCFP)

Row number	SDG	SDG Indicator	Net Impact on SDG Indicator	Current Project Contributions	Contributions Over Project Lifetime
8	8.1 8.3 8.5	8.1.1 Annual growth rate of real GDP per capita 8.3.1 Proportion of informal employment in non-agriculture employment, by sex 8.5.1 Average hourly earnings of female and male employees, by occupation, age and persons with disabilities	Implemented activities to increase	Support at least 47,317 households who benefited directly from carbon and livelihood fees A total of USD 926,417 were paid in livelihood investments in LCFP 5,401 beehives have been provided Mean maize grain yields increased from 1.8 tons/ha under conventional agriculture during the baseline year to 4.9 tons/ha and 4.0 tons/ha under CSA in 2020/2021 and 2021/2022 farming seasons respectively	The 2022 data showed that average household income increased 220% over the five-year period with agricultural production, alternative sources and forest-based products being the key sources of income for the households
9	8.8	8.8.1 Frequency rates of fatal and non-fatal occupational injuries, by sex and migrant status	Implemented activities to decrease	Project functions under the BCP HR manual which ensures correct policies at the workplace and protects	Project functions under the BCP HR manual which ensures correct policies at the workplace and protects

Row number	SDG	SDG Indicator	Net Impact on SDG Indicator	Current Project Contributions	Contributions Over Project Lifetime
				the health and safety of all employees.	the health and safety of all employees.
10	9.1 9.4	9.a.1 Total official international support (official development assistance plus other official flows) to infrastructure	Implemented activities to increase	Supported CFMG with Annual Work Plan development and Needs Assessments .	Completed 16 Stakeholder Facility improvement projects. In LCFP the project constructed for partnership stakeholders CFMG office blocks x 4 (Luembe, Nsefu, Sandwe, Shikabeta); police post x 2 (Mwanya, Shikabeta); scout houses x 4 (Luembe), guest house completion (Mnkhanya).
11	10.1	10.1.1 Growth rates of household expenditure or income per capita among the bottom 40 percent of the population and the total population	Implemented activities to increase	47,317 households benefited directly from carbon and conservation fees A total of USD 926,417 were paid in livelihood investments in LCFP in 2022	The 2022 data showed that average household income increased 220% over the five-year period with agricultural production, alternative sources and forest-based products being the key sources of income for the households

Row number	SDG	SDG Indicator	Net Impact on SDG Indicator	Current Project Contributions	Contributions Over Project Lifetime
					More than USD 1,971,122 has been invested in the communities of LCFP since inception.
12	11.1 11.5	11.1.1 Proportion of urban population living in slums, informal settlements or inadequate housing 11.5 (own indicator) Number of hectares of watershed forests protected to protect ecological system, and as such reduce impact of natural disaster 11.5 (own indicator) Number of community members partaking in Climate Smart Agriculture techniques to limit drought disaster impact	Implemented activities to increase	A total of 60,302 females and 6,752 vulnerable groups have self-reported to have been receiving benefits from different projects 51.48% of community members are female. LCFP protects 1,236,258 ha of REDD+ forests Total # of farmers enrolled under conservation farming increased to 2,627.	USD 12,129,831, has been invested into communities via carbon and livelihood funds, to develop infrastructure and improve sustainability of community member's livelihoods

Row number	SDG	SDG Indicator	Net Impact on SDG Indicator	Current Project Contributions	Contributions Over Project Lifetime
13	12.2	Proportion of land with sustainable management of natural resources (BCP defined indicator)	Implemented activities to increase	Sustainable forest management and use of ecological services are primary topics during community engagement meetings. New waste management SOP implemented on company level	8 years of agreeable sustainable management and efficient use of natural resources with partnership communities Collaborated with the Forestry Department to ensure the community rights to sustainable production empowering communities, avoiding deforestation.
14	13.0	Tonnes of greenhouse gas emissions avoided or removed	Implemented activities to increase	By conserving hectares of forest, LCFP has reduced carbon dioxide emission by 2.1 million tons in 2022 REDD+ topics and explanations to community members at VAG level meetings	Obtained 10,137,575 VCUs under LCFP.

Row number	SDG	SDG Indicator	Net Impact on SDG Indicator	Current Project Contributions	Contributions Over Project Lifetime
15	15.1 15.2	15.1.1 Forest area as a proportion of total land area 15.1.2 Proportion of important sites for terrestrial and freshwater biodiversity that are covered by protected areas, by ecosystem type 15.2.1 Progress towards sustainable forest management 15.5.1 Red List Index of species survival 15.9.1 Progress towards national targets established in accordance with Aichi Biodiversity Target 2 of the Strategic Plan for Biodiversity 2011-2020	Implemented activities to increase	In 2022, community scouts achieved 16,595 km distance patrol, which resulted in 1 person apprehended, 31 snares and 4 firearms confiscated. Three line transect biodiversity surveys were completed in conjunction with camera trapping surveys in two of them. Three critically endangered white-backed vultures have been equipped with GPS transmitters in order to monitor their movement and be notified through immobility alerts which can indicate a poisoning incident	The systematic data collection system using Spatial Monitoring and Reporting Tool (SMART) was implemented in June 2020, as well as the camera trap surveys, whereas the transect survey started in 2016 in Munyamadzi and 2022 in Shikabeta and Mwanya. BCP partners with Oxford affiliate, Lion Landscapes, in order to support biodiversity monitoring efforts within our project zones through the development of a monitoring plan that is scientifically sound and provides statistically robust results. The trends in wildlife abundance are encouraging, although not yielding significant levels yet.

Row number	SDG	SDG Indicator	Net Impact on SDG Indicator	Current Project Contributions	Contributions Over Project Lifetime
16	16.6 16.7	16.6.1 Primary expenditures as a proportion of original approved budget 16.6.2 Proportion of population satisfied with their last experience of public services 16.7.1 Proportions of positions in local institutions by sex, age, persons with disabilities and population groups 16.7.2 Proportion of population who believe decision-making is inclusive and responsive, by sex, age, disability and population group	Implemented activities to increase	Capacity building of CFMGs in regards to Needs Assessment, Annual Work Planning and performance review, funds management, Monitoring, Evaluation and Reporting, Accountability, Gender Equality. Community needs assessments conducted to ensure community inclusion / participation in decision making at community level inclusive of marginalized groups	Contribution to the good governance of the CFMG and inclusivity in decision-making at community level in all partnering chiefdoms.

Row number	SDG	SDG Indicator	Net Impact on SDG Indicator	Current Project Contributions	Contributions Over Project Lifetime
17	17.3 17.9 17.11 17.15	17.3.1 Foreign direct investments (FDI), official development assistance and South-South Cooperation as a proportion of total domestic budget 17.9.1 Dollar value of financial and technical assistance (including through North-South, South-South and triangular cooperation) committed to developing countries 17.11.1 Developing countries' and least developed countries' share of global exports 17.15.1 Extent of use of country-owned results frameworks and planning tools by providers of development cooperation	Implemented activities to increase	Sale of carbon credits contributes to the exports of the country, ensuring inbound flow of financial resources Supports through various meetings the jurisdictional process towards national REDD+ Supported DNPW with a Vehicle and Firearms. Supported DNPW with camera traps and GPSs Wildlife Camp support – Kabalika, Malijane & Chinzete	Total funds paid to communities during the years 2020 and 2022 amounts to USD 10,168,000 Donated a full GIS system to the Forestry Department, and provided related training. Supported DNPW with donation of a Vehicle and Firearms. Supported DNPW with donation of camera traps and GPSs

2 SAFEGUARDS

2.1 No Net Harm

The project was validated for positive climate, community and biodiversity impacts under the CCB Standard version 3 in 2019 (<https://registry.verra.org/app/projectDetail/CCB/1775>) and any potential negative impacts such as conflicts related to the equitable distribution of benefits are summarized in section 2.1.11 of the joint VCS/CCB Project Description. It is important to note that the negative impacts listed are potential impacts commonly associated with similar projects and therefore, should not necessarily be considered as significant. Furthermore, the project aims at reducing emissions from unplanned deforestation and forest degradation and specifically also aims to conserve the astounding biodiversity of the project area and benefit smallholder farmer's dependent on the project area with a variety of project activities aimed at improving livelihoods through poverty reduction such as conservation agriculture, horticulture and beekeeping. The project has potential to bring about positive environmental and socio-economic impacts. However, to mitigate any potential negative impacts including;

- Negative environmental impacts – Leakage and land tenure conflicts
- Socio-economic impacts- Livelihood disruption, unequitable distribution of benefits and governance issues.

The project is Implementing the following;

- Robust Community Engagement strategies with local communities and indigenous groups: i.e. FPIC, Community Consultations (Project planning and benefit sharing), Distribution of Project Information, Conflicts and Grievances resolution.
- Capacity building among stakeholders for effective project implementation. – Capacity building in good governance for community based organizations(CRBs/CFMGs) in the project zone.
- Fostering collaboration among community based organizations(CRBs/CFMGs), governments (FD and DNPW), NGOs, communities, and the private sector to achieve project goals.
- Ensuring that no invasive Species use in our climate smart agricultural activities
- Implementation of alternative Livelihood /Leakage mitigation activities: i.e. Conservation agriculture, Beekeeping, Horticulture, village banking and intensive sensitizations of community members.

By following these mitigation strategies and principles, the project effectively reduces carbon emissions, protect forests, and improve the well-being of local communities while minimizing potential social and environmental negative impacts.

2.2 Local Stakeholder Consultation

Stakeholder identification took place as part of the initial REDD+ viability assessment process that BCP undertook to identify new/potential REDD+ sites for the LCFP, beginning in 2014. The process followed can be broadly outlined as follows:

Workshops / Meetings with key Government and Community Authorities

Beginning at a national level, and then moving down to provincial, district, and Chiefdom levels, BCP held a series of workshops and meetings, in which we outlined REDD+ and explained the REDD+ viability assessment process, including two key criteria for REDD+:

- 1) Selection of a viable forest area for protection;
- 2) Documented consent from all relevant stakeholders / rights-holders.

Permissions were then received from Government and Chiefdom authorities to proceed with on-the-ground viability assessment activities, including participatory forest identification and stakeholder engagement in line with our standard Free, Prior, and Informed Consent (FPIC) protocol (Appendix E).

Initial “high level” stakeholder identification and mapping

During the initial District and Chiefdom-level workshops and meetings that were held, BCP solicited inputs from relevant Government and community authorities (namely: traditional authorities, including Chiefs / Chiefs’ representatives, and Community Resource Boards (CRBs), as legally recognized, Chiefdom-level community institutions), about who could be counted as potential stakeholders to any proposed REDD+ project, and where they were/are located, relative to the proposed forest areas under consideration for REDD+ protection. It was explained that identifying and consulting with any potential stakeholders is a verification requirement for REDD+, and more importantly, adequate stakeholder sensitization and consultation prior to project implementation is in everyone’s interest, to help ensure the sustainability of the project, by ensuring that all relevant stakeholders were consulted about, and consented to, participation in the project. It was explained that stakeholders could broadly be defined as anyone who could potentially be impacted by the proposed project (either positively or negatively), and/or anyone with potential decision-making power about whether or not to cut trees within the identified/proposed forest areas for protection.

At the District/Chiefdom level, Government and community authorities (traditional leaders) provided us with consistent feedback that, so long as a REDD+ project would be taking place on “community” land (including within a GMA), then all community members of the Chiefdom

could be considered to be stakeholders since traditional authorities recognize that community forests belong to the entire community within the Chiefdom, and therefore, all recognized residents within the Chiefdom would have recognized rights, within the traditional leadership structure, to access and use forest resources within community forest areas. Initial community maps were drawn by these key stakeholders, and also confirmed and supplemented by satellite imagery, which confirmed the location of villages and settlements, near to proposed forest areas within each Chiefdom that had expressed willingness/interest to participate in the program.

Detailed Chiefdom/Village level stakeholder mapping - PRA/PLA FPIC activities

More detailed stakeholder identification and mapping took place during significant PRA/PLA techniques that were undertaken in each of the 12 partner Chiefdoms, as part of the FPIC processes that were undertaken by BCP during 2015-2017, at a community level. Our FPIC records show that these meetings were held at a Chiefdom level, in each of the 12 Chiefdoms, and that representatives from the then all 62 registered VAGs within the 12 Chiefdoms were involved in the FPIC activities. During the PRA/PLA activities that took place, community stakeholder identifications were confirmed, and maps were further refined, to ensure that significant community inputs were received to identify and include any potential stakeholder communities. Local (District) level Government officers were involved in this process, as well as representatives from Chiefdom level institutions (Chiefdom royal establishments; CRBs), and village-level representatives from VAGs and traditional leaders (Headmen). Meetings were also open to members of the wider community, who also participated and provided inputs. As part of the stakeholder identification process, it was described, again, that “stakeholders” included anyone who could potentially be affected by the project, and/or anyone with a recognized “right,” or even ability, to influence decision-making about whether or not to cut trees within the identified forest area for protection. Again, stakeholders at all levels confirmed that community forests were/are viewed to be collectively “owned” community resources, and as such, “stakeholders” should include all Chiefdom inhabitants, in all villages within participating Chiefdoms, as their rights to access and use forest resources were/are recognized by members of the local community, as well as the Chiefdom authorities (traditional leaders) and Government – who recognize the authority of community institutions and traditional leaders, on community or “customary” land, even within GMAs. As such, it was agreed that a registry of Chiefdom inhabitants and villages could essentially function as a stakeholder registry. Initial lists of villages were received from Headmen and were shared with BCP through VAG and CRB structures. These were used to inform data-collection plans, to complete an on-the-ground confirmation activity, to confirm, register, and map all stakeholders for REDD+ projects within each of the 12 partner Chiefdoms of the LCFP.

Once FPIC processes were completed, community consent was received and documented in community consent letters for REDD+ implementation, signed by Chiefdom authorities

with legally recognized authority to manage community resources (Chiefs, CRBs). Once these letters were received, BCP undertook a “Stakeholder Household Registry” process, with the aim of confirming, documenting, registering, mapping, and counting community stakeholders to each REDD+ project, in every participating Chiefdom, in every VAG and village within that Chiefdom. This process was undertaken through CRBs and Village Action Groups (VAGs) as local-level community institutions and involved significant inputs, guidance, and participation of local traditional leaders (such as Headmen) to ensure that all members of their villages were registered as participants, stakeholders and beneficiaries to the project.

The data collection process was led by locally hired staff, working hand-in-hand with local traditional leaders such as village headmen, as well as representatives from local VAGs, CRBs, and Government.

The LCFP “Household Registry” now provides a comprehensive database, and map, of all stakeholders to REDD+ projects supported under the LCFP. As the processes above described, it has been informed by significant inputs from community authorities and government partners, solicited through a multitude of workshops, meetings, and PRA/PLA activities that have been undertaken as part of FPIC processes during the past three years, during which input was solicited, and received, concerning who should be counted as “stakeholders,” based on their recognized rights and abilities to inform decisions about forest protection within identified areas. The actual data collection process has taken approximately one year and has involved dozens of data collectors across all 12 Chiefdoms.

Other stakeholders are hunting concession holders that operate within the REDD+ project areas.

Project documentation has been made available to stakeholders throughout the project development process, including FPIC stakeholder sensitization activities that were initiated in 2014, and which have continued through 2021. Informational handouts clearly outlining the proposed project model, and accurately describing REDD+ activities, were developed and have been widely distributed to community (and Government) implementing partners, throughout this time. These handouts and other informational materials (such as informational posters about REDD+ and BCP) have been distributed to each of the 12 partner CRBs that BCP is working with, to implement REDD+ activities under the LCFP. Throughout the years, from 2015-2021, BCP has provided capacity building training to CRB members, as well as local Government implementing partners, and locally hired staff, to ensure their ability to accurately explain and disseminate information about REDD+, in line with FPIC requirements. BCP staff also developed and launched a standard “sensitization program,” that provided critical information about REDD+ and the proposed activities of the LCFP, and which made information available in a multitude of local languages, presented by locally hired and trained staff, who ensured that this information was disseminated in locally appropriate ways. In combination with these sensitization programs, BCP has ensured that

informational materials have been made available to local community stakeholders in each Chiefdom and that there are trained staff and implementing partners available at all times, in each Chiefdom, to undertake ongoing sensitization activities, to accurately disseminate information about REDD+, and to be available to answer any questions or concerns that stakeholders may have, at any given time.

BCP kept detailed records of all community and stakeholder meetings and further details of stakeholder consultation can be found in the LCFP Project Description.

After project validation, BCP has continued engagement with both higher-level and village level stakeholders. The details regarding the desired frequency of meetings and the records of actual meetings held are kept in an engagement tracker. See Appendix E for examples.

Monitoring results that point to serious concerns, such as encroachment within accounting and leakage areas are immediately reported once discovered to the relevant local stakeholders via our engagement teams. These critical results are then thoroughly investigated and discussed by local stakeholders and BCP. The results of the completed annual monitoring process are communicated to local stakeholders at the end of the monitoring period, soon after verification is completed. Since the results are tied to community benefits, a thorough explanation of monitoring results are communicated via our engagement teams to local stakeholders to substantiate their extent of the performance based benefits accrued. These communications occur during scheduled meetings with CFMG's and relevant community members and authorities.

None of these regular meetings and consultations with stakeholders, and points addressed therein, led to the modification of the project design during this monitoring period. In addition, there have been no relevant changes to the risks and costs to the communities associated with the project implementation in this monitoring period to communicate.

Local stakeholders have been made fully aware (via the information meetings and materials referred to above) of the requirements for an auditing team to verify the findings of the annual monitoring report, as is standard procedure, and what this auditing process will entail. This communication is part and parcel of the REDD+ project information dissemination which occurs since the start of community engagement and continues to be communicated during refresher information sharing meetings. Appendix A contains evidence of agreements between BCP and CRBs outlining BCPs responsibility to communicate monitoring and verification findings (among others).

Similar to the first instance project areas, BCP undertook a thorough process of obtaining Free Prior Informed Consent and conducted consultative mapping processes across the 3rd project activity instance chiefdoms in 2020 (evidence of the consultative meetings can be found in Appendix B). In this framework the respective communities voluntarily designated areas as REDD+ Zones and began the process of complying with the community forest regulations, Statutory Instrument(SI) number 11 of 2018. At the start of the third project

activity instance, all local natural resource governance institutions i.e Community Resource Boards (CRBs) in the new areas had been recognized as Community Forest Management Groups (CFMG), and already entered into a Community Forest Management Agreements with the Government of Zambia in accordance with SI 11 of 2018. However, it should be noted that one of the community forest areas, Chiawa, was recently encroached by outsiders from the plateau, i.e Lusaka, who set up temporary shelters, fields and consequently engaged in illegal charcoal production in a small portion of the area. The Chiawa CFMG and Traditional leadership have since been made aware of the situation and have been asked to develop and implement an action plan to address these issues within their REDD+ Zone. These two activities are scheduled for 2023 (M6).

2.3 AFOLU-Specific Safeguards

Negative Local Stakeholder Impact Mitigation

The measures to mitigate negative impacts to poachers are driven by a combination of job creation and livelihood alternatives. The project has created more than 60 community jobs, mainly for Community Scouts (CS). These CS are hired through BCP support via CRB contracts and their role is to protect community resources, including wildlife habitat. The income from these jobs goes directly into the Chiefdom, and it is fair to say that poachers have an equal, if not advantage in applying for CS jobs. The CS role requires physical endurance, knowledge of bush-craft, and a willingness to spend one's work time outdoors. People with poaching experience are naturally inclined to be high performing CS with a change in mindset towards legally earned income and with the passing of training.

BCP's regular investment in service training of CS serves as a deterrent to would-be poachers. The information regarding recurrent training is shared with communities via BCP's community engagement team. These deterrents help would be poachers with decision-making tools to not invest effort into illegal activities with a higher risk of CS intervention.

A key part of the strategy employed is a heavy investment in community projects which improve well-being. For example, investments in water, education, and health services, as well as community transport (such as boats) provide households with indirect improvements in quality of life. Direct improvements in quality of life are provided through BCP's investment in scalable food security safety net and income diversification schemes such as honey and conservation farming.

A further measure to reduce negative impacts is BCP's use of 'Community Covenants' for community projects BCP invests in. These provide a mechanism by which the community can engage with and help would-be poachers with social support as an incentive to stop poaching and risk undermining the Covenant or the CFMA which includes a definition of non-performance as poaching.

These measures above are also applicable to illegal charcoal producers. In addition, BCP has initiated a pilot Eco-Charcoal project which supplies legal, sustainable charcoal to the market. The model is being implemented in Rufunsa District under the Lower Zambezi REDD+ Project (LZRP) but the project model is scalable to Open Areas and operates under the principle of “Poachers to Gamekeepers”. This principle is to provide illegal charcoal producers with a pathway to becoming legal charcoal producers. Currently, there are no restrictions on people joining the Eco-Charcoal project, and the vision in time is to offer it for expansion to community areas where there is demand and a desire for alternative livelihoods. The scaling of Eco Charcoal production to LCFP has not commenced yet but still on the plan.

Impact on Local Stakeholder Land Use Rights

The land tenure has not changed as a result of the project. As described in section 2.2, the project undertook an extensive consultation process with all of the relevant local stakeholders. The land was previously in GMAs managed by community resources boards (CRBs). The project has helped the CRBs gain additional status as community forest management groups (CFMGs) and helped them create community forest management agreements (CFMAs) with the government that gives them rights to new resources in the area (carbon).

To provide an incentive for better protection of the forests within the CFMAs, BCP has signed agreements with the CRBs stipulating to pay 40 % of net revenue to communities.

Ongoing communication with local stakeholders and grievance redress procedure

BCP maintains a constant presence in the participating chiefdoms and continues to be engaged in awareness-raising activities and various livelihood activities described in section 3.

Additionally, BCP has a well-established and publicly known grievance mechanism that has been designed to address and resolve any problems. All the CFMAs that have been signed by communities have a grievance procedure stipulated covering the contractual agreements regarding the project activity (Appendix A). In addition, a Stakeholder Grievance Response SOP is in place, signed by lead members of the engagement team and signed for as received by each CRB and kept on file (Appendix E). The grievance mechanism is clearly outlined and discussed in community sensitization meetings, and contact information for BCP representatives is included in all public communications. Meeting minutes (written records) from all of the community sensitization meetings document that the grievance mechanism has been explained to the local community representatives in attendance—including their right to receive a response within 30 days, and their right to contact third party arbitrators directly should they have any ongoing concerns. In 2018, BCP updated the Grievance Mechanism from the 2013 version.

Grievances that are submitted to BCP are documented, as are the responses to any such grievances. These documentations are copied to the relevant GRZ office to ensure a third party involvement, to prevent any conflict of interest. Project management will attempt to resolve all reasonable grievances raised, and we commit to providing a written response to grievances within 30 days. In order to track grievances, BCP has established a Grievance Tracker by Chiefdom (Appendix E) which keeps track of dates received and actions taken to close-out of a grievance.

Our SOPs Concerning 1) Community Engagement, and 2) Social Monitoring reference the 2018 Grievance Mechanism as a tool that accompanies these community engagement processes. The formal process for handling grievances is outlined in these documents which are made available to the VVB. The grievance mechanism is also written into all contracts (such as Community Covenants) that are signed between BCP and community representatives, such that BCP and local community members are able to hold each other to account with regard to the delivery of services as per the terms of agreements. The Community Engagement Teams are expected to serve as liaisons between stakeholder communities and the project team. In addition, they are expected to communicate any needs, questions or concerns that either side has.

In the event that community stakeholders have a serious concern or grievance about BCP's Community Engagement Strategy, or if there is an on-going or long-lasting grievance among community stakeholders that BCP has been unable to address, BCP's grievance mechanism allows for the involvement of a neutral third-party arbitrator (including legal counsel) that will be able to clarify or resolve the grievance for either or both sides in the dispute. The preferred third-party arbitrator, in the event of disputes, would be a representative from an appropriate Non-Governmental Organization with a local presence in Zambia—for example, in the event of grievances concerning land, an organization that deals with land-access and/or land-rights will be involved. If there is a grievance concerning food security, an organization such as the Conservation Farming Unit would be involved. In the event that the community prefers to involve local government, BCP has advised the communities that they are welcome to bring concerns to the District government which is headed by the District Commissioner. BCP has a working relationship with local governments in the Districts of the project zone, and local District government officials have been identified as a mutually-trusted third party for both the stakeholder communities and BCP.

All stakeholders are made aware of grievance procedures at stakeholder meetings, the CFMAs (Appendix A) have a grievance procedure incorporated and feedback has to be given to all grievances raised.

Grievances are documented through the receipt of grievance letters, or meeting minutes. We encourage any verbal grievances to be channeled through Government-recognized local governance institutions such as Community Resource Boards, Traditional Authorities, Cooperatives, or Village Action Groups. The Grievances received are filed digitally in a

BCP's 'Grievance Register' Tool (Appendix E). This tool is designed to monitor the grievance close-out process against BCP's '2018 Grievance Mechanism and Stakeholder Grievance Response SOP' (Appendix E). Grievance letters are signed for as received and responded to in writing within 30 days. The government and any other key stakeholders are copied in to ensure that a response is transparent and shared with community institutions. Where a Grievance is raised at a smaller scale, such as a VAG or Zone, the grievance response letter is followed up by a meeting to discuss the response and any follow-up actions needed for closing out the grievance.

Copies of the Stakeholder Grievance Mechanism are shared with Cooperatives, CRBs, VAGs, and signed for as received by each CRB and kept on file.

In addition, all community meetings now have an agenda item titled 'Grievances' to ensure that at each meeting there is a public opportunity to air grievances in a formal, more regular, and minute forum. In order to continue to reinforce the importance of the Grievance Mechanism, training is specified in the BCP SOP on Stakeholder Grievance Responses for community institutions on an annual basis. BCP's goal is to build community relationships to a very strong level through this process of providing pathways for Grievances to flow more easily, as well as binding BCP to 30-day response periods.

The project received a total of 7 grievances in 2022 (M5) in the Chiefdoms of Mpanshya, Luembe, Mwanya, Malama and Mnkhanya, respectively (Table 5). All grievances received are considered closed. Our engagement in the process of resolving these grievances. Details of the grievances are in Appendix E.

Table 5. Details of the grievances received during the current monitoring period (2022; M5)

Chiefdom	Description	Opening Date	Closing Date
Mpanshya	Complaint over BCP's Gender streaming approach advocated during community engagement meetings in a period of local CRB elections	18/07/22	27/07/22
Malama	Allegation made by community members of inadequacy of forest carbon fees disbursed	11/08/22	14/09/22
Mwanya	Allegation made by community members of misuse of funds by the CFMG	20/08/22	18/01/23
Project level	Complaint over a lack of District hire	26/10/22	19/12/22
Mnkhanya	Complaint over Community Scouts apprehending community members	13/11/22	14/11/22
Malama	Complaint over a lack of Chiefdom hire	14/11/22	17/11/22

Luembe	Allegation made by a community member of misuse of funds by the CFMG	05/12/22	22/12/22
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Provision of Sustainable Livelihoods: In order to reduce potential risks faced by local stakeholders due to project implementation and further create a sense of project ownership, the project ensures active participation of local communities, small-scale farmers, and Indigenous peoples in decision-making processes through ongoing community sensitization/community engagement. In the current monitoring period, over 600 community meetings were held across the project zone addressing a wide range of issues e.g FPIC, REDD+, grievance resolution, VCS/CCB verification awareness and annual work planning etc.

Additionally, the project supports initiatives aimed at enhancing knowledge and skills in areas such as effective governance, Agroforestry, climate-smart agriculture (Minimum soil disturbance, residue retention, crop rotation and crop diversification), horticulture, and beekeeping. This is accomplished by facilitating access, as well as providing training and distribution of Climate smart agricultural, Agroforestry, beekeeping, and horticultural resources. The project further supports extension services through collaborations with the Ministry of Agriculture and Natures Nectar. This provides an alternative to traditional charcoal production and traditional agricultural methods that are detrimental to the environment. The ultimate goal is to promote diverse livelihood opportunities, enhance resilience, and enable local communities within the project area to better adapt to the impacts of climate change.

3 IMPLEMENTATION STATUS

3.1 Implementation Status of the Project Activity

The LCFP has developed a wide range of deforestation mitigation and benefit-sharing activities with local stakeholders. These activities fit broadly into three different categories namely infrastructure support and development, improved livelihoods, and income diversification through value addition to non-timber forest products, and thirdly forest encroachment and wildlife monitoring and poaching reduction. Project activities happen in the project area, project zone, and leakage areas shown in Figure 6.

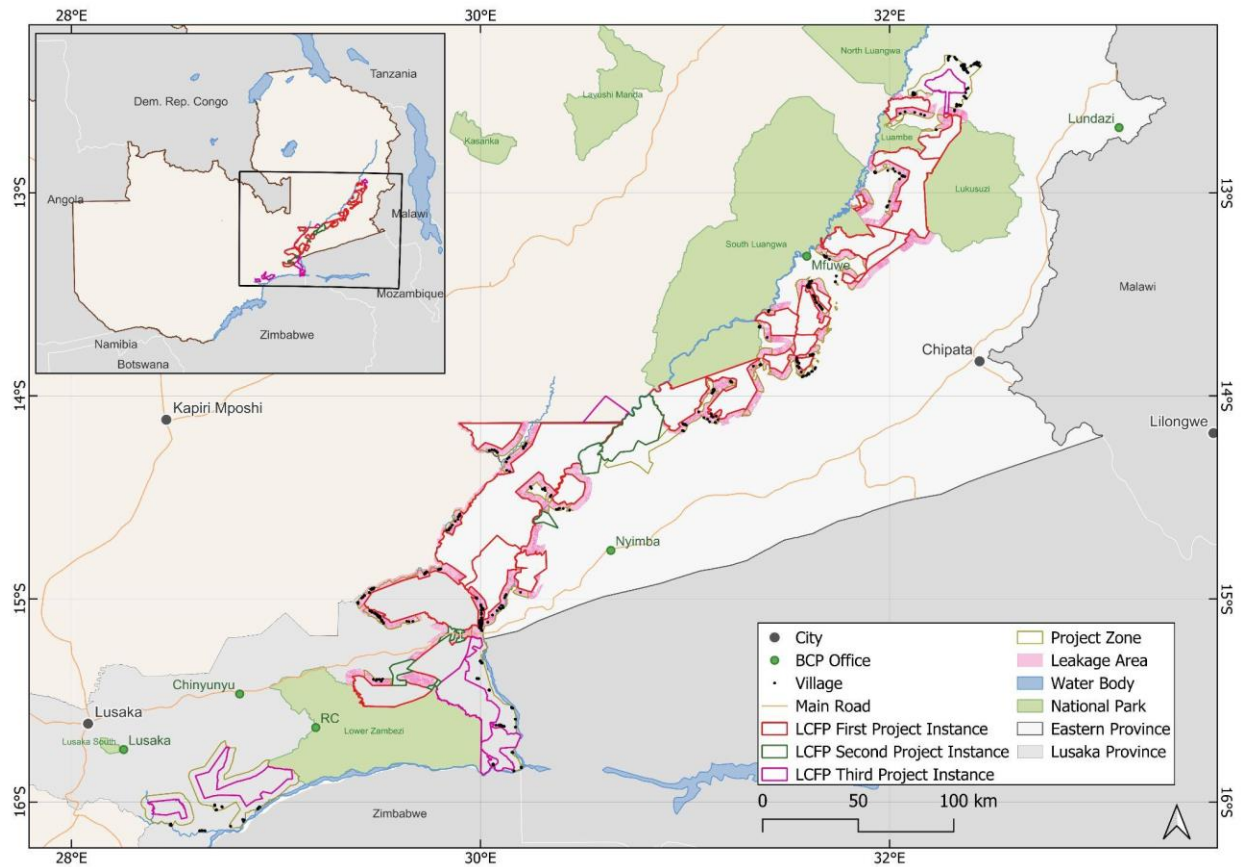


Figure 6. Project area, project zone, and leakage areas where project activities take place

Activity Area 1: Infrastructure Development, Social Service Delivery

Defined Activities:

- Construction and rehabilitation of school infrastructure; classroom blocks and latrines,
- Provision of safe drinking water; through the drilling of new boreholes, rehabilitation of existing boreholes, and improvement of existing wells.
- Provision of boats to facilitate safe river crossing
- Support the delivery of health care services through the provision of clinic furniture and fittings
- Provision of hammer mills, maize and rice shellers

Additional Activities Undergoing Investigation and Further Consultation:

- Roof-based water collection (Self Supply)
- Micro-credit project(s)
- Construction and rehabilitation of health care centers,
- Construction of culverts, drifts

Support to Infrastructure Development and Social Service Delivery

- Support to delivery of Community Social Services:

During the current monitoring period, 2022 (M5), Shikabeta CFMG utilized their conservation fees to procure and distribute 1,632 IBR iron sheets in Shimunguwo VAG, Shikabeta Chiefdom. This initiative benefited 136 households, with each household receiving 12 iron sheets. Similarly, Nyalugwe CFMG carried out a comparable scheme by procuring and distributing 1,143 iron sheets to 249 households. Among these, 115 households in Fundo, Mchimadzi, Kasansamula and Mvuwa VAGs receiving 5 iron sheets each while 34 households in Mvuwa VAG received 2 iron sheets each. In the same Monitoring period (M5), the project through the Nyalugwe and Chitungulu CFMG, procured banana boats (4 in Nyalugwe and 1 in Chitungulu). The banana boats were distributed to 4 VAGs in Nyalugwe Chiefdom, namely Mkoma, Nyalugwe, Fundo and Kasansamula) and 1 VAG (Kamila) of the Chitungulu Chiefdom.

In 2022 (M5), Mwanya CFMG introduced two programs, namely a School Scholarship and Teachers' Support program, which have helped 25 students and 12 teachers, respectively. Additionally, Chitungulu CFMG has also initiated a similar project that provides tertiary education support in bricklaying and tailoring to people in the community. This initiative has already benefited 13 community members.

In 2022 (M5), with an aim of curbing/reducing incidences of animal wildlife conflict in Mwanya and Chitungulu Chiefdoms, the project procured and elected through the CFMGs Solar fences in 4 VAGs in Mwanya (Mkasanga, Yakhobe, Lukusuzi and Chasera), 1 and 1 VAG in Mnkhanganya (Katemo) (See Appendix B).

In 2022 (M5), the Shikabeta CFMG project procured and distributed over 140 bicycles to the Lusemfwa VAG in Shikabeta chiefdom, acknowledging the significant distances that local community members travel on foot. This effort had a direct positive impact on 140 households.

- Improved water supply:

In 2022 (M5), the project through Sandwe CFMG, drilled and installed 10 boreholes in the different VAGs, namely Lumbuka (2), Mileka (1), Chinkhombe (1), Chikowa (2), Sengevwa (1), Chiwale (1) and Sandwe (1). The 10th borehole was drilled and installed at Manina Wildlife camp.

- Construction/Rehabilitation of Wildlife Camps

In the current monitoring period (M5), BCP Rehabilitated the DNPW Chinzete wildlife camp in Mpanshya, which fell into disrepair.

Infrastructure Projects that were initiated in the previous monitoring periods and are still ongoing or were completed in the current monitoring are shown in Table 6a below, whereas Table 6b presents the new projects that were initiated in the current monitoring period.

Table 6a. Infrastructure projects in the project zone (In progress /Completed in 2022).

Location (District)	Description of Intervention	Year started	Status/Date Completed
Mambwe (Malama)	Construction of 1X 2 classroom block	2021	In Progress
Mambwe (Mnkanya)	Improvement CRB office and 1x2 toilet	2021	In Progress
Mambwe (Jumbe)	Construction of Chasera Post Health Centre	2020	In progress
Mambwe (Jumbe)	Construction of a 2 in1 staff house at the Health facility in Mphata VAG	2021	In Progress
Mambwe (Nsefu)	Construction of 1 x 3 classroom block in Chitunda VAG		
Lumezi (Mwanya)	Police post and 2 staff houses (Mwase Waminga)	2020	2022
Lusangazi (Sandwe)	Construction of CFMG/CRB Office (Sandwe VAG)	2021	In progress
Nyimba (Luembe)	Construction of Chikwasha Health Post	2021	2022
Nyimba (Luembe)	Construction of 5 Housing unit ,Chinambi VAG	2021	In Progress
Nyimba (Nyalugwe)	Construction of Maternity Annex, Mchimadzi VAG	2021	In Progress
Nyimba (Nyalugwe)	Construction of 1x3 classroom blocks In Kasansamula VAG	2021	In Progress
Rufunsa (Mpanshya)	Construction of a 1x 2 Classroom block in Chalunda, Nyampande VAG	2021	In Progress
Rufunsa (Mpanshya)	Construction of a 1x 2 Classroom block in Musukambale (Chansanje, Lukwipa Area)	2021	In Progress
Rufunsa (Shikabeta)	Police post, Mweteshi VAG	2021	2022
Rufunsa	CFMG office, Mweteshi VAG	2021	In Progress

(Shikabeta)			
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Table 6b. Infrastructural projects initiated in the current monitoring period (2022; M5).

Chieftdom	Project	Sector	Status
Chitungulu	Construction of Toilets at Nthumbe Health Post	Health	Completed
	Construction of incinerator at Nthumbe Health Post	Health	In progress
	Construction of CRB Offices	Social	In Progress
	Construction of Market Shelter	Social	Completed, Awaiting Handover to the community.
Mwanya	Construction of Rural health post for Lukusuzi area	Education	In Progress
Sandwe	Rehabilitation of 1x3 Classroom Block	Education	In Progress
Nyalugwe	Construction of 2 scout houses at Kasolo Camp	Social	In Progress
	Rehabilitation of the Nyalugwe Rural Health Clinic roof	Health	In Progress
Luembe	Construction of Toilets at Mshalila Primary School, Chisanga VAG	Education	In progress
	Construction of Maternity Annex, Luangazi VAG	Health	In Progress
	Construction of Staff House at Ng'ambwa school, Ng'ambwa VAG	Education	In Progress
Mwape	Rehabilitation of 1x3 Classroom Block	Education	In Progress
Mphanshya	Rehabilitation of DNPW Chinzete Wildlife Camp	Social	Completed

Activity Area 2: Apiculture, Agricultural Improvement, Diversification, and Economic Opportunities

Defined Activities:

- Development of agroforestry demonstration plots to test and demonstrate agricultural multi-cropping, diversification, and intensification.
- Training Lead farmers in conservation farming techniques to support follower farmers through the farmer-to-farmer model.
- Hosting of farmer field schools to teach farmers about climate-smart agriculture
- Development of apiary demonstration plots, and training of beekeeping farmers (mentors) in hive construction, maintenance, and appropriate apiary and beekeeping techniques.
- Distribution of beehives to participating households within the project area.
- Supporting poultry and animal husbandry and facilitating the construction of animal enclosures to prevent the spread of diseases and damage to crops

Additional Activities Undergoing Investigation and Further Consultation:

- Supporting the development of horticulture production
- Assistance with the commercialization of agricultural products and Support in crop aggregation for easy access to markets outside the project area
- Land-based aquaculture

Expected Positive Impacts:

- Increased quantity, diversification, and value of crops
- Improved nutritional value of crops
- More sustainable practices (Conservation farming, Beekeeping, and forest management training)
- Reduced pressure on the ecosystem
- Improved resilience to crop failure and market fluctuations
- Increased employment opportunities (Lead Farmers)
- Improved well-being
- Guaranteed market for honey

Potential Negative Impacts:

- Increased cost of living if the value of crops increases
- Conflict related to unequal distribution of benefits and workload

Support to improved agriculture techniques through Conservation Farming

In 2017, LCFP began its conservation farming activities by supporting the work of the Ministry of Agriculture, specifically through the provision of conservation farming extension services. Agreements were signed in 2017 with the respective territorial Ministry of Agriculture offices within the project zone in the districts listed in Table 7. These signed Agreements were renewed and signed in 2020 (M3) in form of the terms of references.

To be noted that the Lundazi district has not been renewed due to the absence of project areas within the Lundazi district itself. As a direct consequence of changes in district administrative boundary that occurred in 2020 (i.e creation of new districts and therefore modification of previous boundary extents), project areas that were initially located in Lundazi district were relocated to newly created districts named Lumezi district.

Table 7. District agreements with the Department of Agriculture.

Location (Districts)	Initial Agreement Date	Agreement Renewal Date
Mambwe	21/11/2016	08/05/2020
Nyimba	15/03/2017	20/04/2020
Lundazi	21/01/2017	-
Lusangazi	15/04/2020	-
Rufunsa	29/05/2017	21/05/2020
Lumezi	17/04/2020	-

Arising from the FPIC community consultative meeting; forests are one of the main sources of livelihood for the local communities with illegal charcoal production, illegal logging, and unsustainable farming methods such as shifting cultivation (“slash and burn”) being the major activities conducted. These activities coupled with the increased in-migration of farmers from the plateau to the valleys, where soils are still fertile, has increased the demand for opening up new farmland within the project area. Conservation Farming was thus one of the activities identified during the needs assessment process as an intervention that would complement LCFP’s conservation activities. This intervention targeted 1,260 households with conservation farming training and extension services within the first year of implementation.

During the current monitoring period (2022), a total of 127 new lead farmers were enrolled across the 5 new project instance 3 chiefdoms, bringing the total number of lead farmers under climate smart agriculture in 2022 to 478 from 351 lead farmers in 2021. Currently the project also tracks land under Conservation farming for Lead farmers and in 2022 (M5)

alone, the project increased the area under conservation farming by 113 ha, bringing the total area under conservation farming to 408.5 ha since 2020. Table 8 shows summarized numbers of Lead Farmers enrolled per chiefdom. See appendix B for lead farmer registers.

Table 8. Current status of Conservation Lead farmers enrolled in the project zone.

Location (Chiefdom)	Number of Lead Farmers
	2022
Nyalugwe	35
Luembe	42
Sandwe	53
Nsefu	17
Mnkhanya	17
Malama	17
Jumbe	16
Msoro	17
Mwanya	32
Chitungulu	32
Shikabeta	33
Mpanshya	40
Mwape	25
Mpuka	26
Chiawa	25
Mburuma	26
Kazembe	25
Total	478

In 2022 (M5) BCP together with the Ministry of Agriculture/The Zambia Agriculture Research Institute - Msekera Research conducted Climate Smart Agriculture and Farmer Field Schools trainings;

For the Mfuwe sector, the trainings were conducted from the 22/08/2022 to the 29/08/2022, where 84 and 86 Lead farmers were trained under Mambwe (BCP's Mfuwe sector) and Lundazi-Lumezi (BCP's Lundazi sector) districts respectively.

For Rufunsa, the training was conducted from the 17th Aug - 3rd September, 2022 and targeted lead farmers from five chiefdoms, namely; Chiawa of Kafue district, Mpanshya, Shikabeta of Rufunsa district, Mphuka and Mburuma of Luangwa district. 159 farmers trained.

The climate smart agriculture lead farmer training in Nyimba sector was conducted from the 24th August to the 2nd of September 2022, focusing on vegetative propagation, tree nursery establishment and land preparation. A total of 155 farmers were trained.

In total, the project provided training in climate smart agriculture techniques to 484 participants. The main objective of these trainings was to teach farmers the different Climate smart agricultural technologies that are being implemented at the moment. The technologies included agroforestry, Integrated Soil Fertility Management, Conservation Agriculture, Conservation farming and other unique agricultural models like Evergreen Maize for Africa and SILMS. The other objective of the training was to teach farmers the importance of the Farmer Field Schools and how to establish Farmer Fields Schools within their communities (See Appendix B for training reports).

A follower farmer training was conducted under the Rufunsa site for Mpanshya and Shikabeta chiefdoms where 734 follower farmers were trained. See Appendix B for the training report.

To further support conservation farming in the project zone, the project conducted the following additional conservation farming related training in the different project zones.

An Animal Draft Power Tillage Service Providers (ADP TSPs) Technical and Business training, the objective of the training was to provide technical and business training to potential ADP Tillage Service Providers in Nsefu, Jumbe, Msoro, Malama, Mnkhanza, Luembe and Sandwe chiefdoms. The training was conducted from 12/09/2022 to 30/09/2022 and a total of 187 farmers were trained. A similar training was conducted on the 17th of October 2022 with an objective of providing tractor maintenance, tractor safety, technical and business training to potential mechanized tillage service provider (Tithandizeni Cooperative) in Jumbe Chiefdom. The training benefited 10 farmers.

Mfuwe sector also conducted a Mechanized and Animal Draft Power Tillage Service Land Preparation Promotions training in Jumbe and Nsefu chiefdom from 28/10/2022 to 09/12/2022. This training benefited a total of 408 farmers in the project zone.

Bio Fertilizer and liquid inoculant training was conducted in the Rufunsa site. The training was conducted by Alba Agriculture in conjunction with the Ministry of Agriculture extension officers. The training was conducted on the 11th of October 2022 with an aim of sensitizing

BCP farmers on the advantages of bio fertilizer. Ten (10) farmers from Mpanshya were trained.

In order to support farmer field school demonstration plots, a total of 458 conservation farming demonstration packs were distributed to lead farmers in the participating chiefdoms (Table 9), including in the newly added 5 project activity instances in M5. In addition, T-shirts were also distributed.

Table 9. Starter packs distribution during the current monitoring period.

Chiefdom	Number of Starter Packs Distributed
Mbuluma	25
Shikabeta	25
Mphuka	31
Chiawa	25
Mpanshya	39
Mwanya	32
Msoro	17
Nsefu	17
Mnkhanya	17
Malama	17
Chitungulu/Kazembe	60
Sandwe	53
Nyalugwe	35
Mwape	23
Luembe	42
Total	458

Demonstration inputs packs included the following:

- i. Maize seed 5kg
- ii. Soya beans seed 20kg
- iii. Basal dressing fertilizer 2 X 50kgs
- iv. Top dressing fertilizer 1 x 50kgs

Support of Diversification and Economic Opportunities

- Horticulture

Horticulture is important in creating employment, enhancing profits and providing nutritional and economic security for the farmers. In 2022 (M5), the BCP recruited 50 new horticulture farmers in the newly included 3rd instance chiefdoms bringing the total number of farmers to 130. In collaboration with Zambia Agriculture Research Institute - Msekera Research Station (Agroforestry Research Team Staff) the project conducted three (03) horticulture training; See table 10 below and annex B for supporting documentation.

Table 10. Horticulture training conducted during the current monitoring period (M5)

Site	Date	# Males	# Females	Total
Nyimba	22-25 March 2022	22	8	30
Mfuwe/Lundazi	11-16 March 2022	41	9	50
Rufunsa	7 April-17 May 2022	30	20	50
Total		93	37	130

Both the Rufunsa and Mfuwe training included farmers from the new 3rd instance chiefdoms namely, Kazembe, Mburuma, Mphuka and Chiawa (see Appendix B) for supporting documents.

Similar to the situation in M4, all horticulture lead farmers were given demonstration field starter kits containing vegetable seeds and fertilizer. In addition, 60 treadle pumps were also distributed to horticulture farmers in Chitungulu (5), Mwanya (10), Mphuka (10), Mnkhangya (5), Mburuma (10), Kazembe (10), Jumbe (5) and Malama (5).

In 2022, the project through Chitungulu CFMG, supported 6 gardening groups with gardening equipment (treadle pumps) and inputs for their horticulture activities. This initiative benefited 62 individuals in the project zone.

Support for the development of Value Chains in Non- timber forest Products

- Improved Apiculture techniques:

Whilst beekeeping is regarded as a cottage industry in Zambia, that is well suited for individuals and companies with limited resources for investment, its potential to contribute to the reduction of rural poverty cannot be ignored. According to the Central Statistics Office (2010), the incidence of rural poverty in Zambia is about 80 %. If well organized, beekeeping can contribute to poverty reduction as it increases people's income.

Although honey hunting and beekeeping have been practiced for a long time by almost all communities in Zambia, beekeeping is only now being promoted as an alternative livelihood for those people that are involved in charcoal production and illegal game hunting, and for those that want an extra income.

Against this background, LCFP conducted a viability assessment in 2016, on the potential to establish honey production value chains within the LCFP project area. Based on the findings of this viability assessment, an agreement was signed with Bee Sweet, to expand honey production to cover the LCFP project zone. Through this intervention, 12 from a target of 11 honey producer- lead farmers (Mentors), received training from Bee Sweet in hive construction and maintenance and apiary management (Table 11).

In addition, through this intervention, 4050 Kenya top bar beehives were distributed within the LCFP project area to 1,100 beneficiary households (Table 11).

Table 11. Beehive distribution by BeeSweet in the project zone.

District	Chiefdom	Target No. of hives to be distributed	Actual No. distributed	Date of last beehive delivery
Mambwe	Jumbe	2000	1318	Aug-17
	Mnkhanya	1000	388	Feb-18
	Msoro	2000	1515	May-18
	Malama	1000	264	May-18
	Nsefu	1000	375	May-18
	Mwanya	1000	681	May-18
Lumezi	Chitungulu	200	190	May-18
Total		6,200	4,050	

The contract with Bee Sweet ended in April 2020 without renewal. BCP, working with Nature's Nectar, conducted a feasibility study between September and October 2020 for expanding the honey project. This study was used to provide baseline data to the new partner on the current status on the honey project. Following the feasibility study, BCP in March 2021 entered into a Specific Partnership Agreement with Nature's Nectar to manage the beehives and purchase honey from the managed beehives. Currently this Specific Partnership Agreement is working well with Nature's Nectar having already trained 14 Mentors in honey harvesting (Table 12). In 2022 (M5), Nature's Nectar conducted beekeeping training targeting Beekeeping Farmers and Zone Lead farmers and focused on beehive assembly, hive hanging, general beehive management and capturing GPS Coordinates of active beehives. These training were conducted in Nyalugwe (126) and Shikabeta (76).

Table 12. Number of bee mentors trained in the project zone.

District	Chiefdom	No. Mentors trained	Date of Completion of training	Facilitator	Topics
Mambwe	Mwanya	3	Jul-16	BeeSweet	Hive construction and maintenance and apiary management
	Jumbe	2	May-17	BeeSweet	
	Mnkhanya	1	May-17	BeeSweet	
	Msoro	3	Aug-17	BeeSweet	
	Malama	3	Aug-17	BeeSweet	
	Jumbe	3	Jun-21	Nature's Nectar	Honey harvesting
	Mnkhanya	2	Jun-21	Nature's Nectar	
	Msoro	3	Jun-21	Nature's Nectar	
	Nsefu	2	Jun-21	Nature's Nectar	
Lumezi	Mwanya	3	Jun-21	Nature's Nectar	
	Chitungulu	1	Jun-21	Nature's Nectar	
Nyimba	Shikabeta	76	Nov-22	Nature's Nectar	Beehives management, assembling, baiting, and hanging
	Nyalugwe	126	Nov-22	Nature's Nectar	

In the current monitoring period (M5), the project has distributed 5041 new beehives, as shown in Table 14 (refer to Appendix B for supporting documents). Since each beekeeper received 10 beehives, the project has recruited 504 new beekeepers (see Table 13 below). However, at the time of writing of this report, only 399 beekeepers have been registered in the beekeepers' registry (refer to Appendix B), and the remaining beekeepers are yet to be registered. Table 13 (below) provides information on the newly recruited beekeepers based on the number of beehives distributed per chiefdom. This increased the number of beekeepers from 577 in 2021 to 1,081 in 2022. The total number of distributed beehives also increased from 3004 in 2021 to 8,045 in 2022 (refer to Table 14).

Table 13. Beekeepers recruitment during the current monitoring period (M5)

Chiefdom	2021 Beekeepers	New Beekeepers Recruited (2022)	Total
Jumbe	120	70	190
Nsefu	93	35	128
Mpanshya	0	100	100
Shikabeta	0	99	99
Luembe	0	100	100
Nyalugwe	0	100	100
Chitungulu	24	0	24
Mnkhanya	151	0	151
Msoro	99	0	99
Mwanya	90	0	90
TOTAL	577	504	1081

Table 14: Beehive distribution by Nature's Nectar in the project zone.

Chiefdom	Zone	# beehives distributed	Year of delivery
Jumbe	Mphata	354	2021
Shikabeta	Chisulo	650	2021
Mkhanya	Kumbukile	500	2021
Msoro	Lupande	340	2021
	Kawala	330	2021
	Kasamanda	330	2021
Mwanya	Chasera	350	2021
Nsefu	Kawaza	150	2021

Jumbe	Mphata	347	2022
	Chisulo	350	2022
Luembe	Chalubilo	1000	2022
Mpanshya	Nyampande	750	2022
	Chimusanya	250	2022
Nsefu	Kawaza	350	2022
Nyalugwe	Moombe	500	2022
	Muchimadzi	500	2022
Shikabeta	Mwateshi	250	2022
	Lusemfwa	250	2022
	Shimunguwo	244	2022
	Lubalashi	250	2022
TOTAL 2021		3,004	
TOTAL 2022		5,041	
GRAND TOTAL		8,045	

During the current monitoring period (2022; M5), a total of 3,627.17 kg of honey was purchased and generated K 71,852 ZMW of revenue to farmers in LCFP (Table 15). This has contributed to improved and diversified household income for beekeeping farmers across LCFP.

Table 15. A summary of the harvest data for LCFP for 2021 (M4) and 2022 (M5) for the first harvest season of the year.

Location	Total kg Purchased			Total Paid Out (ZMW)		
	2021	2022 (M5)	Total	2021	2022 (M5)	Total
Chitungulu	384.7	333.74	718.44	10,157	6,606	16,763
Jumbe	376.9	671.3	1,048.2	9,951	13,288	23,239
Mnkhanya	96.1	181.22	277.32	2,536	3,625	6,161

Msoro	175.3	154.53	329.83	4,627	3,061	7,688
Mwanya	1,359.7	1927.7	3,287.4	35,895	38,168	74,063
Nsefu	0	358.68	358.68	0	7,104	7,104
Total	2,392.7	3,627.17	6,019.87	63,166	71,852	135,018

Expected Outcomes:

- Reduced encroachment into the forests under protection through REDD+ by local communities harvesting honey.
- Increased production of natural honey.
- Improvements in the income levels of rural households through the sale of honey.
- Reduction in the use of traditional hives and falling of trees due to honey hunting.
- Enhancement of the knowledge base and skills of the honey producer in modern honey production, processing and marketing practices.
- Confidence built amongst honey producers due to having a guaranteed market for the honey produced.

Activity Area 3: Forest Encroachment and Wildlife Monitoring and Poaching Reduction

BCP provides support to the local CRB community scouts and the Department of National Parks and Wildlife (DNPW) to ensure the protection of the project area from encroachment by monitoring wildlife and reducing poaching. This activity is also supported by satellite-based Global Land Analysis and Discovery forest change alerts (GLAD Alerts) from Global Forest Watch and aerial monitoring.

Defined activities:

- Monthly analysis of GLAD Alerts and follow up by Aerial patrols
- Monthly patrols by community scouts along project area boundaries identified as encroachment hotspots.
- Monthly aerial and ground monitoring of wildlife
- Monthly or bi-monthly aerial monitoring of encroachment along project boundaries
- Monthly monitoring of high conservation value species
- Monthly anti-poaching activities including removal of snares, counting carcasses, and apprehending suspected poachers.

Expected Outcomes:

- Quick response to encroachment in the project area
- Rapid resolution of encroachment through the consultation process by local governance structures notably the CRB and chiefs.
- Reduction of encroachment over time as project activities expand and awareness of monitoring activities take hold.
- Greater understanding of wildlife populations and threats to wildlife.
- Adapt management to changing wildlife populations.
- Increase in wildlife populations, especially high conservation value species.
- Employment creation for community scouts.
- Reduction of leakage and project emissions.

Thus far monitoring has been rolled out in most of the project area by participating CRBs. BCP has developed an Impact Monitoring Manual which has a chapter that outlines the interaction of satellite-based alerts and aerial and ground-based monitoring. Potential encroachments received as GLAD Alerts are verified by aerial monitoring and recorded using a GPS. This information is then stored in a database and follow up ground monitoring is then actioned by BCP conservation staff. Scouts are given the location of encroachment, once confirmed on the ground and contact made with encroachers, the Community Engagement Team and CRB then follow up and investigate incidents further.

Monitoring has been rolled out in all of the project areas and results of aerial monitoring are shared with all CRBs at the end of each month. Figure 7 below is an example of a monthly Aerial Patrol map for the Msoro Chiefdom and the associated post-flight report is shown in Appendix B, as an example.

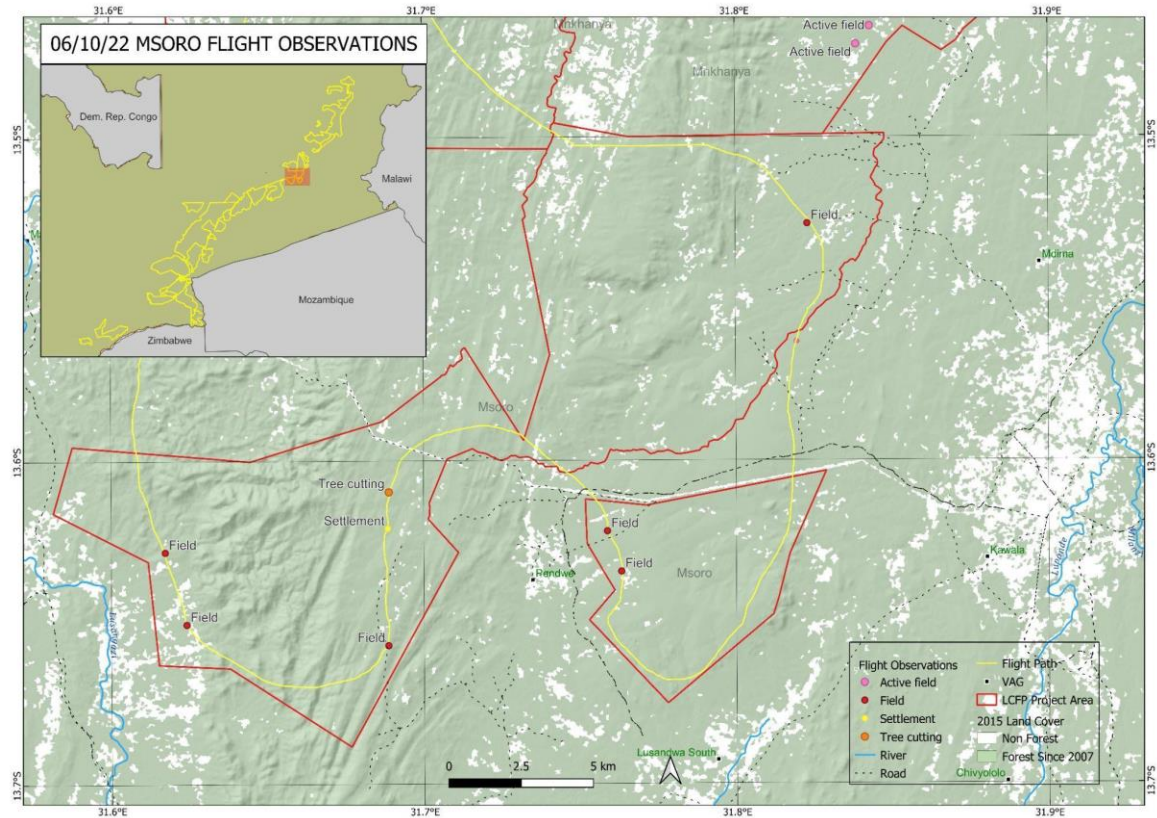


Figure 7. Aerial monitoring map of Msoro for the month of October 2022.

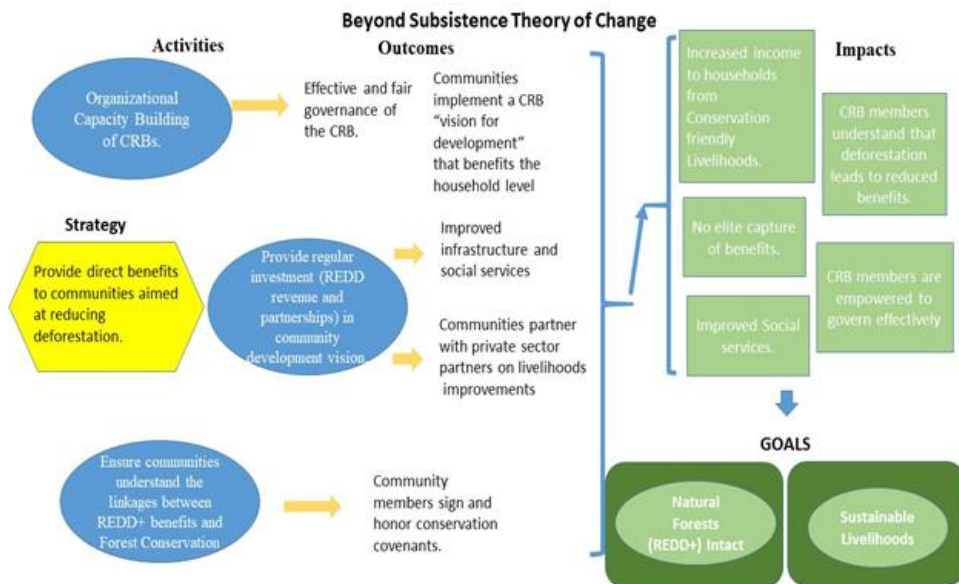


Figure 8. BCP Community Theory of Change

Our Theory of Change (Figure 8) around the provision of livelihoods support is that if we can help communities make long term investments in a shared vision for development and demonstrate that those investments are intrinsically linked to the successful long term conservation of forests, then livelihoods in those communities will improve demonstrably and be sustainable, while natural forests will remain intact.

Key risks to providing benefits to the communities include the following:

- Risk of elite capture of benefits, either from REDD+ or from livelihood activities themselves.
- Communities may not grasp the connection between performance (REDD+ forest conservation) and the benefits they receive.
- Households may not benefit enough from REDD+ to halt deforestation.
- Households may “cheat” and take benefits while continuing to deforest.

To address some of these risks BCP is implementing forest and biodiversity protection measures, in addition. Since these risks are complex and difficult to deal with we also forward a Beyond Subsistence Livelihoods approach that:

- Builds the capacity of CRBs to manage financial resources, govern themselves effectively and build a shared vision for development.
- Provides demonstrable, regular benefits that are used to invest in the shared vision for development that benefit the household level.
- Ensures communities understand linkages between receiving REDD+ benefits and long term forest conservation.

This combination of community-based natural resource governance and sustainable livelihood development will result in long term conservation and sustainable development impacts for these communities.

Our Beyond Subsistence Livelihoods activities are focused on achieving outcomes that will set the stage for achieving development impacts. CRBs need a vision for development implemented through effective and fair governance of the institution. Investments should lead to improved infrastructure and social services (education and health). The private sector should invest in these communities to build mutually beneficial partnerships. This approach requires that all members of the CRB “sign on” to and honor conservation covenants that spell out the responsibilities that go along with the benefits that come from REDD+.

If we can support our community partners to achieve these outcomes, then these impacts will flow from our work:

- Increased income at the household level from forest conservation-friendly livelihoods.
- CRB members will understand that deforestation leads to reduced benefits from REDD+.

- CRB members will be empowered to govern effectively.
- There will be no elite capture of REDD+ benefits.
- Households will benefit from improved social services.

In combination with our efforts to build local capacity and responsibility for forest and biodiversity conservation, these livelihood efforts lead to our ultimate goals of keeping natural forest ecosystems intact and ensuring that community livelihoods are profitable and sustainable.

Expected Positive Impacts:

- Reduction in the cost of living
- Increased level of education in communities
- Increased local capacities
- Increased life expectancy
- Reduced child mortality and disease transmission

Potential Negative Impacts:

- Conflicts related to the equitable distribution of benefits

All of the above-mentioned activities are connected in some way to reducing the non-permanence risk. By ensuring little deforestation takes place in the project area, community members' shares are benefited by the project, hence reducing opportunity cost risks and activity displacement leakage is limited to a minimum.

New Projects Implemented in 2022.

During the current monitoring period (2022; M5), the project completed 3 infrastructure projects that were in progress in M4 and initiated 13 new infrastructure projects of which 3 were completed and the remaining 10 are still ongoing, planned to be completed in 2023 (M6). Table 6 above provides details of infrastructure projects implemented in 2022 with their status of either being ongoing or completed, while table 6a shows projects initiated in the previous monitoring periods but still ongoing or completed in the current monitoring period.

3.2 Deviations

3.2.1 Methodology Deviations

MR.97

Demonstration of Forest Age

The methodology requires (in PDR.22) that a digital GIS-based map of the project accounting areas, including aerial or satellite imagery showing that they are comprised of

forests or native grasslands as of the project start date and 10 years prior to the project start date. The project did not meet this requirement because it chose to map forests in the project area with PALSAR radar data and the earliest available data for that sensor is from 2007. However, this is justified on several grounds. The project areas are virgin forests that were previously remote and are inaccessible. Thus, there is no history of large disturbances in the area that would have been followed by regeneration between 2005 and 2007. An accurate 2007 forest map of these areas should be virtually indistinguishable from a 2005 map. Additionally, using the 2007 PALSAR imagery results in a much more accurate map of the forests that existed 10 years prior to the start date than would be possible with optical imagery. The optical imagery best suited to mapping the project area is generally from April-June. It is at this time of year, that relatively cloud-free imagery can be obtained when the forests still have leaves and are not dotted with burn scars. Even so, the passing of just a couple of weeks during this period, which is immediately after the rains, results in significant changes in the appearance of the forest in optical imagery. The project initially tried mapping forests with optical imagery but found significant changes between years that were the result of seasonality rather than real changes in forest areas. These effects are much reduced when using PALSAR radar data. This is because, in addition to being cloud penetrating, L band radar has a relatively long bandwidth and thus does not generally interact with the leaves of trees, but does interact with large branches and stems¹. Finally, VM0009 does not require a remote sensing analysis of an area, but rather to visually show that it is unconverted as of the project start date and 10 years prior. Such an approach is highly subjective since unconverted grasslands and farmlands may at times appear similar. With the PALSAR based analysis, all non-forest areas, whether natural or not, were excluded. Thus, using the PALSAR data improved the accuracy of the analysis and did not negatively impact the conservativeness of the quantification of GHG emission reductions or removals. The deviation only affects the monitoring and measuring part of the methodology.

Leakage Plot Protocol

There was a minor deviation of the visual approach for leakage monitoring (Appendix R, VM0009). Instead of Forest Inventory Team members estimating degradation, they were required to geo-reference every felled tree in the permanent plots and the total number of felled trees were compared against the average density of trees/ha in the project accounting area, again avoiding biased estimations/erroneous counting of trees in 1ha plots. This value was then placed in a degradation category as per VM0009. This approach removed all potential bias and greatly improves the robustness and repeatability of degradation monitoring. This deviation does not affect the conservativeness of any emissions reductions as it is in fact an improvement of the prescribed procedure and removes observer bias that

¹ Mitchard, E. T. A., *et al.* "Using satellite radar backscatter to predict above - ground woody biomass: A consistent relationship across four different African landscapes." *Geophysical Research Letters* 36.23 (2009)

could result in degradation in the leakage area being underestimated. The deviation only affects the monitoring and measuring part of the methodology.

Since a section of 19,887 ha of the new second project activity instance areas overlapped with the existing activity shifting leakage area (originally 319,179 ha), the new leakage area comprises 299,291 ha. This is not strictly regarded as a methodological deviation, as VM0009 allows for existing leakage areas to be converted to future project activity instance areas.

Biomass estimation

The project uses two different allometric equations to estimate above-ground biomass. One was developed from data provided by Prof. Chudimayo from Zambian Miombo (see Appendix G) and was tested on trees up to 55 cm DBH. It is used to estimate the biomass of trees up to 63 cm dbh, with the biomass of trees between 55 cm and 63 cm dbh assumed to be equal to a 55 cm dbh tree. Then, a second equation developed by Mugasha et al. in Tanzanian Miombo (see Appendix G) is used to estimate the biomass of trees between 63 cm dbh and 110 cm dbh with trees with a greater than 110 cm dbh assumed to have biomass equal to a 110 cm dbh tree. This parsing is necessary because the methodology requires that the directly measured biomass of the largest tree in the representative sample of destructively harvested trees be greater than the biomass predicted by the equation and that the value of the first-order derivative of the largest tree in the representative sample and inventory not differ by more than 10%. The first selected allometric equation does underpredict the biomass of the largest tree in the representative sample, but the difference of the first-order derivative values was greater than 10%. The second allometric equation doesn't pass the first test (it slightly over predicts the biomass of the largest tree in the sample). However, compared to the Zambian equation, it predicts lower biomass for every size tree. Therefore, the equations were applied as described above to enable the project to make an estimate of the biomass of larger trees that fell outside the range of the size of trees used to develop the equations. Only 33 trees out of 3589 trees in the inventory had a dbh above 55 cm and only 3 had a dbh above 110 cm. Applying the equations in this way is conservative and erases the possibility of the equation over predicting biomass of trees outside the size classes for which it was tested. This deviation only relates to how biomass is measured for monitoring purposes and doesn't affect any other part of the methodology. It results in a more conservative estimate of biomass and emissions savings. Therefore, it meets the criteria for an acceptable methodological deviation.

Soil Emission Calculations

The VM0009 baseline soil emissions model assumes that SOC in deforested areas will deplete to the same value as the proxy area after 5 years. Project SOC emissions are supposed to be tracked directly through field measurements in encroached areas. However, in the project area, the main drivers of deforestation (smallholder farmers) deliberately

choose high fertility areas for conversion with higher than average SOC values. Following the methodology, it would appear that encroachment causes an increase in SOC, which is not consistent with what is actually occurring to soil stocks. To avoid this bias, we conservatively assumed that once encroached, the average SOC stocking per ha in the encroached area becomes the proxy area SOC stocking. This methodological deviation is, therefore, more conservative in calculating emissions reductions from SOC. It does not relate to any other part of the methodology, except for calculating soil carbon stocks in the encroached area. See the master biomass summary spreadsheets in Appendix G for this conservative calculation.

Carbon Stock Figures- New Project Activity Instances

The project has opted to use the first instance accounting area carbon stock value of 236.2 tCO₂e/ha in the other 2 project activity instances accounting areas, and has recognized this as a methodology deviation.

Miombo woodlands across Zambia can be found in various stages of disturbance from human activities, mostly due to selective wood harvesting for charcoal production, leading to changes in canopy structure and species diversity. Often, these extractive activities modify the ecosystem to an extent that makes it vulnerable to further degradation. Studies such as Handavu et al. (2021)² and Bulusu et al. (2021)³ have demonstrated that these disturbances significantly affect plant density and height, which is likely to affect carbon storage of these woodlands. Furthermore, as explained by Bouvet et al. (2018)⁴, while efforts to estimate carbon stocks in African woodlands have advanced in recent years, there are still challenges and uncertainties associated with accuracy. Improving the accuracy of carbon stock estimation requires continued refinement of remote sensing techniques, validation with ground-based measurements, development of region-specific allometric equations, and enhanced data availability and quality.

Field studies conducted in Miombo woodlands across the Sub Saharan Africa region have estimated carbon stocks values ranging between 167 tCO₂e/ha and 845 tCO₂e/ha (Handavu et al. (2021)) indicating a high variability that results in high uncertainty when trying to adopt to a different location. As such, the LCFP project resorted to the use of its own forest inventory data, rather than reliance on literature. The carbon stocks for the project are calculated using locally derived allometric equations as mentioned in section 4.3 of the Monitoring Report.

² Handavu, F., Syampungani, S., Gudeta, W., Sileshi, W., & Paxie, W. C. Chirwa (2021) Aboveground and belowground tree biomass and carbon stocks in the miombo woodlands of the Copperbelt in Zambia, *Carbon Management*, 12:3, 307-321, <https://doi.org/10.1080/17583004.2021.1926330>

³ Bulusu, M.; Martius, C.; Clendenning, J. Carbon Stocks in Miombo Woodlands: Evidence from Over 50 Years. *Preprints* 2021, 2021030029. <https://doi.org/10.20944/preprints202103.0029.v1>

⁴ Bouvet, A. et al. An above-ground biomass map of African savannahs and woodlands at 25 m resolution derived from ALOS PALSAR. *Remote sensing of environment* 206, 156–173 (2018).

The LCFP project applied Miombo biomass carbon stock values of 236.2 tCO₂e/ha across the 3 project activity instances estimated for a single stratum of intact forest. As explained in the LCFP PD (page 195), biomass plots are re-measured on a rotating basis, with 20% measured per year, such that the project's overall carbon stock estimates are updated every 5 years after all plots (100 %) have been re-measured. As per the current monitoring report (Section 4.3 of this MR), the upcoming carbon stock dataset would be updated for the 2023 crediting period (M6) as the biomass plot monitoring rotation started in 2019. The forest inventory raw data for instances 2 and 3 to date (Appendix G) shows that the carbon stocks in PAI#2 (551.38 tCO₂e/ha) and PAI#3 (388.26 tCO₂e/ha) are actually higher than the PA#1 value of 236.2 tCO₂e/ha). The project, therefore, conservatively considered the use of the lower value in all the instances for this monitoring period. Moreover, most of the studies conducted in miombo woodlands reported carbon stocks that are higher than those used by the LCFP project in the current monitoring period. For instance, Munishi et al. (2010)⁵ calculated Miombo carbon stocks values of 385 tCO₂e/ha in the southern highlands of Tanzania, and Missanjo et al. (2015)⁶ obtained carbon stocks of 845±16 tCO₂e/ha in Chongoni, Malawi.

These statistics, therefore, confirm that the locally calculated and VCS validated and four times verified carbon stock value of 236.2 tCO₂e/ha is a conservative stand-in value that can be used until the scheduled carbon stock master dataset update in the next monitoring period.

3.2.2 Project Description Deviations

Parameters $t^{[i-1]}$, $t^{[m]}$, $t^{[m-1]}$, $U_B^{[m]}$, $U_{EM}^{[M]}$ were mistakenly excluded from section 3.3.2 of the validated PD but they are monitored during the crediting period and therefore this omission does not impact on the applicability of the methodology, additionality or appropriateness of the baseline scenario. These parameters are included in section 4.2 of this MR and will be included in the PD when it gets to be updated at the 10 years' baseline re-evaluation.

Section 3.3.3 of the validated Project Description, only highlights Sentinel-2 or Landsat imagery as data sources for the project's visual interpretation forest disturbance analysis, and does not mention the 4.77m spatial resolution Planet-NICFI mosaics dataset. It also lacks clarity on the accuracy assessment procedures for the manual forest disturbance analysis. Recognizing the significance of incorporating the Planet-NICFI mosaics dataset into the visual interpretation forest disturbance analysis and accuracy assessment procedure in the current monitoring period, modifications have been made in Section 5.2 of the

⁵ Munishi PKT, Shirima, DD (2010). Measuring and Valuing the Carbon Stocks in Chome Nature Reserve, Eastern Arc Mountains, Report submitted to Eastern Arc Endowment Fund for Forest and Beekeeping Division, Dar es salaam Tanzania. p. 17

⁶ Edward Missanjo* and Gift Kamanga-Thole Estimation of Biomass and Carbon stock for Miombo Woodland in Dzalanyama Forest reserve, Malawi. (2010) Research Journal of Agriculture and Forestry Sciences, ISSN 2320-6063 Vol. 3(3), 7-12, March (2015) Res. J. Agriculture and Forestry

Monitoring Report to incorporate both. This improvement is very important because the 4.77m spatial resolution of the Planet-NICFI dataset enables very reliable accuracy assessment of the manually digitized converted areas. The project has also developed an SOP to guide the manual digitization of converted areas (Appendix E) which also includes an accuracy assessment procedure. It's important to note that these alterations do not affect the methodology's applicability, nor do they compromise the baseline scenario's validity. In fact, the inclusion of the Planet-NICFI mosaics dataset as reference data for accuracy assessment has strengthened the accuracy and correctness of the analysis without diminishing the conservative approach in quantifying greenhouse gas emission reductions. To ensure completeness and transparency, both Planet-NICFI mosaic datasets and the accuracy assessment procedure will be integrated into the monitoring plan section of the Project Description at the next baseline re-evaluation, underscoring the commitment to continuous improvement and accuracy in MRV system and analysis processes.

Early CDM Update

VM0009 requires a baseline reevaluation every 10 years, with the initial one slated for 2025 as per the validated project PD in section 2.1.13. However, the inclusion of new project activity instances has resulted in the initial reference area not meeting the methodology requirement of having a forested area at any point in time during the historic reference period equal to or more than the project accounting area. Instead of delineating a reference area for the new instances only, the project has opted to delineate a new reference area that is representative of the whole project area (all the 3 project activity instances) and consequently execute a CDM update, one year ahead of schedule. Despite the early action, this approach is deemed conservative. The project recognizes this as a PD deviation.

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Reference Area Selection

A reference area has been identified (see Figure 9) for the F-U3 baseline type that is similar to the project area as per the methodology requirements and had at least as much forest area as the project accounting areas at the start of the historical analysis in 1986.

Just like the initial reference area, the selection and delineation of the revised reference area for LCFP was done using ward boundaries data from the Electoral Commission of Zambia (ECZ) and Central Statistics of Zambia (CSO)⁷. Wards are delineated by the Electoral Commission of Zambia (ECZ) and the Central Statistics of Zambia (CSO) for the purpose of the population and housing census and elections. Administrative Wards units were used because they provide a similarity measure, i.e. population density that can be used to relate the reference area to the project area. Figure 9 below shows the new reference area and the LCFP project area (Instances 1, 2 and 3).

⁷ http://riskprofilesundrr.org/layers/geonode:popward/metadata_detail

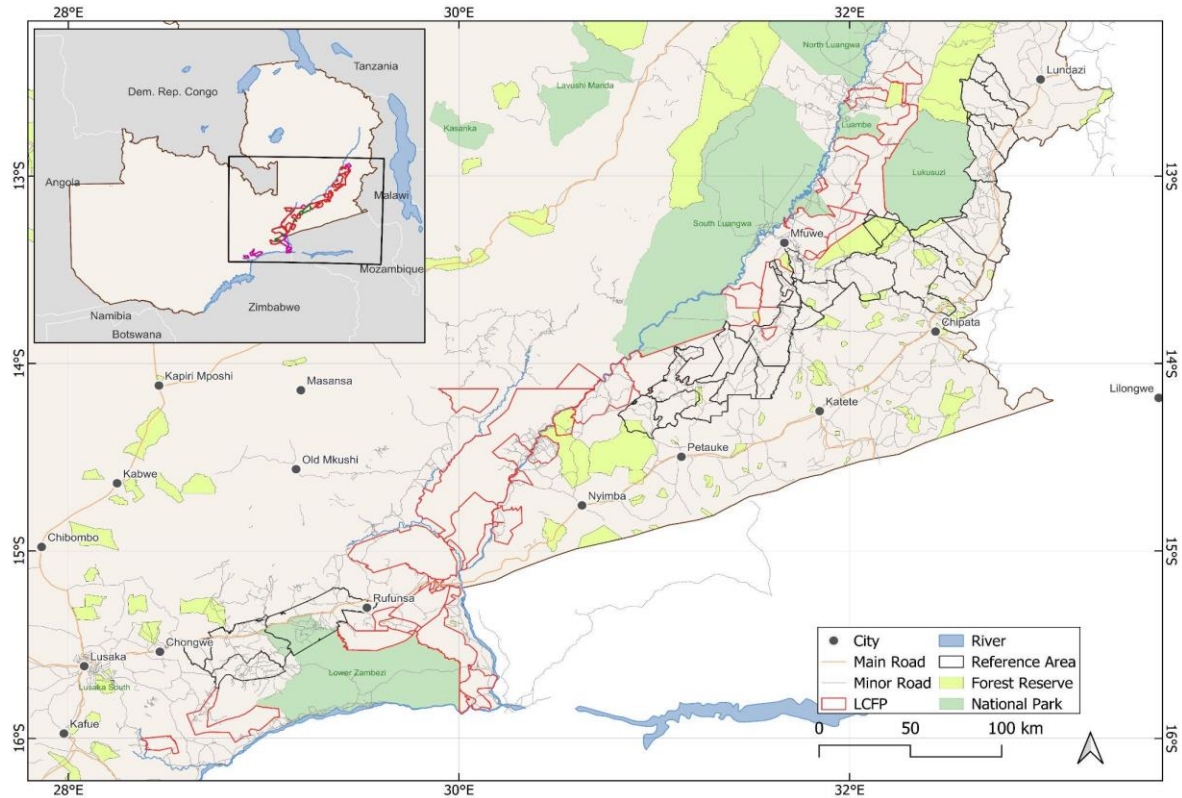


Figure 9: Reference and Project Area (Instance 1,2 and 3)

In order to come up with the reference area, administrative wards that overlap or are adjacent to the project area administrative wards and have bio-physical characteristics that represent the baseline scenario were selected. The resultant selection was 28 administrative wards in total. Project areas and potential project expansion areas were then clipped out from the 28 selected reference area wards. All areas located beyond 6.5 km from the roads and consisting of slopes steeper than 45% were excluded. The remaining area results in the reference area.

The new reference area consists of a total of 28 reference administrative wards (11 wards overlap with the initial reference area wards, and 17 are new) with an overall extent of 1,151,327 ha. Out of 5,910 image interpretation points (See figure 12), 5,080 points were interpreted as forest in 1986. This is equivalent to 86% of the points, which would mean that approximately 989,634.7 ha of reference area in 1986 was forested. When compared with the extent of the accounting area in M5 i.e. 659,081 ha, the forested area in the reference area in 1986 is 1.5 times greater than the accounting area, thus the new reference area meets the VM0009 criteria in terms of having, at one point in the historic reference period, a forest area that is equal or greater than the accounting area forest size.

A significant portion of the new reference area is situated within protected areas such as National Parks, National and Local Forest reserves, including Game Management Areas (GMAs), where most of the project area is located. Even though, on paper, these areas are

designated as Protected Areas based on the IUCN Protected Area categories and Zambia Law, in reality they are open-access and experience little to no law enforcement due to lack of institutional capacity.

Reference & Project Area Similarity Analysis

- Geographical Proximity Factors

Ten of the 28 reference area administrative wards overlap with the project area administrative wards and the remaining 18 are adjacent. This indicates that the reference area is geographically contiguous with the project area, implying that the agents of conversion in the reference area are affected by similar cultural and socioeconomic factors as the agents of conversion in the project area, and therefore the social economic status of both the project and reference area is considered to be similar.

The same inference can be drawn regarding market access. Given the geographical proximity of the reference area to the project area (with 10 wards overlapping and 18 being adjacent), the same transportation routes/roads passing through the project areas also traverse the reference area, linking the same major towns. Thus the reference area exhibits significant similarity in terms of market access as the project area. Figure 9 shows the roads and main towns around the project and reference areas.

- Road Density

Road density similarity analysis between the project area and the reference area was conducted using (1) a historic roads layer (1980-2010) accessed from NASA's socioeconomic data application centre (SEDAC)⁸ to assess the road density in the reference area at the start of the reference period, and (2) an updated roads layer, with more recent roads (2021) mapped and used to delineate project accounting areas, to assess the road density in the project area at the start of the project crediting period.

Note that the SEDAC dataset used for the analysis includes additional roads that were manually digitized using 1986 Landsat 5 image in QGIS.

The updated roads layer within the project area have been shared with the VVB for verification, as many of the roads in the project area have been manually digitized to delineate the accounting area. The vast majority of the roads present in the project area are not delineated on publicly available maps. The same was done with the SEDAC roads dataset used for analysis as it also includes a few manually digitized roads.

The SEDAC historic roads dataset (merged with a few manually digitized roads) was used to obtain the road density at the beginning of the historic reference period in the reference area. The assumption is that the reference area had fewer roads in 1986 compared to 2010, therefore using 2010 roads to represent the 1986 scenario in the reference area is

⁸ <https://sedac.ciesin.columbia.edu/data/set/groads-global-roads-open-access-v1>

conservative. The digitized 2021 data set was used to get the current road density scenario in the project area.

To calculate the road density, the historic SEDAN roads dataset (merged with 1986 manually digitized roads) and the updated 2021 roads dataset were clipped using the reference and project area polygons respectively. Length for the resultant roads dataset was calculated in QGIS and divided by the respective areas sizes, i.e. reference and project areas, to get the road density in table below.

Table 16: Road density - Reference Area (1986) and Project areas(2021)

Period	Area	Road Distance (km)	Area Size (km2)	Road Density (km/km2)
SEDAN : 1980-2010 Road Dataset (Merged with additional 1986 manually digitized roads)	Reference	972	11,513.27	0.084
2021 Road Dataset	Project	1377	12,362.58	0.1114
	Difference			0.027

LCFP's independent deforestation agents and drivers' study⁹ (Appendix K), identified the main driver variables explaining the quantity and location of deforestation as population density and accessibility to forests, i.e., distance to roads, distance to settlements, and slope, making road density a proxy to forest access and consequently, to threat level. The results in the table above indicate that the project areas in 2021 with a road density of 0.11 is more threatened than the reference area at the beginning of the Historic Period (1986) with a road density of 0.08. Therefore, the reference area delineated is considered a conservative choice for a VM0009 reference area in this regard.

- Accessibility and Mobility of the Agents of Deforestation.

LCFP's independent agents and drivers study established that between 2007 and 2017, all deforestation in Eastern and Lusaka province happened on slopes less than 45 degrees and

⁹ https://pdf.usaid.gov/pdf_docs/PA00THNB.pdf

within 6.5 km from the roads. Therefore, the project opted to use these 2 parameters to identify areas that are accessible by the agents of conversion in both the project area and the reference area. In terms of the mobility and accessibility of the agents of conversion, the project area has been limited to those areas within 6.5 km from roads and slopes of less than 45% (to generate the accounting area - where the CDM is applied). The entire reference area is also limited to within 6.5 km of roads, and all areas with slopes greater than 45% were excluded.

Considering that the same distance from roads and slope delimitation was used in the project area to get the accounting area (where the CDM is applied) and when delineating the reference area, the reference area can be considered similar to the project area as far as accessibility and mobility of the agents of conversion is concerned.

- Population Density

LCCP's independent agents and drivers' study looks at the drivers of deforestation across the entire Eastern and Lusaka Provinces, and so includes both the project area and the reference area. In this study, population density was "...identified as the driver variable that possesses a clear relationship with historical deforestation, and thus, can explain the quantities of deforestation in the historical reference period in [Lusaka and Eastern Provinces]." Population density in the reference area wards in 1986, at the beginning of the historic reference period, was examined in relation to population density in 2021 in the project area's reference wards.

The analysis used ward level population data from 2000 and 2010 census reports, district level 1990-2010 and 2010-2022 average annual population growth rates, and 1980-1990 Provincial annual average population growth rates published by the Zambian Central Statistics Office¹⁰. With these data, population density figures were estimated for the individual reference wards in 1986 (beginning of the historic reference period) and project area wards in 2021. Lastly, the analysis also used population wards spatial data¹¹. The resultant wards population density data for the reference and project wards is shown in the table 17 and 18.

Because district level data was not available in the 1990 census report, provincial annual growth rates were used instead of district level to get the 1986 population numbers. In addition, because the Chamtowa, Kachama, Nyamanongo and Ukwimi wards were introduced in 2010 only, their 1986 population numbers were estimated based on the 2010 population numbers and population growth rates between 1980-1990 and 1990-2010. (See supporting documentation and the population estimation dataset in Appendix O).

¹⁰ <https://www.zamstats.gov.zm/2022-census/reports/>

¹¹ http://riskprofilesundrr.org/layers/geonode:popward/metadata_detail

Table 17 :Reference Area Administrative Wards Population Density in 1986

Ward	Province	District	Area Size	Pop1986	PopDensity1986
Bunda Bunda	Lusaka	Rufunsa	1497	3959	2.64
Chamtowa	Eastern	Lumezi	477	5655	11.86
Chibande	Eastern	Lumezi	2224	3677	1.65
Chikowa	Eastern	Mambwe	332	886	2.67
Chipangali	Eastern	Chipangali	695	8564	12.32
Chipapa	Eastern	Mambwe	535	1605	3.00
Chisangu	Eastern	Lusangazi	243	7661	31.53
Diwa	Eastern	Lumezi	995	8785	8.83
Jumbe	Eastern	Mambwe	734	3078	4.19
Kachama	Eastern	Lumezi	283	3990	14.10
Kakumbi	Eastern	Mambwe	510	3528	6.92
Kamimba	Eastern	Lumezi	471	5724	12.15
Kazembe	Eastern	Lumezi	1264	2592	2.05
Lusangazi	Eastern	Lusangazi	1310	1491	1.14

Mateyo Mzeka	Eastern	Lusangazi	352	4998	14.20
Mawanda	Eastern	Lusangazi	1794	5132	2.86
Mnkhanya	Eastern	Mambwe	168	7199	42.85
Mphomwa	Eastern	Mambwe	662	655	0.99
Msoro	Eastern	Mambwe	156	583	3.74
Ncheke	Eastern	Mambwe	344	246	0.71
Ndima	Eastern	Mambwe	1009	3727	3.69
Nthope	Eastern	Chipangali	1117	8493	7.60
Nyamanongo	Lusaka	Rufunsa	1232	1189	0.97
Rufunsa	Lusaka	Rufunsa	2970	5960	2.01
Singozi	Eastern	Lusangazi	296	5292	17.88
Ukwimi	Eastern	Lusangazi	419	5087	12.14
Wachitangachi	Eastern	Lumezi	255	3146	12.34
Mwachilele	Eastern	Rufunsa	174	1740	10.00
AVERAGE					8.82

Table 18 :Project Area Administrative Wards Population Density in 2021

Ward	Province	District	Area Size	Pop2021	PopDensity2021
Chamilala	Eastern	Nyimba	1380	6112	4.43
Chibande	Eastern	Lumezi	2224	15230	6.85
Chinambi	Eastern	Nyimba	1580	14183	8.98
Chinsimbwe	Eastern	Nyimba	562	1319	2.35
Chisomo	Central	Serenje	3299	3306	1.00
Chiyaba	Lusaka	Kafue	1630	10166	6.24
Dzalo	Lusaka	Luangwa	45	7289	161.98
Jumbe	Eastern	Mambwe	734	8474	11.54
Kabowo	Lusaka	Luangwa	638	292	0.46
Kakumbi	Eastern	Mambwe	510	24386	47.82
Kambale	Lusaka	Kafue	806	8723	10.82
Katipa	Eastern	Nyimba	968	1509	1.56
Katondwe	Lusaka	Luangwa	407	3593	8.83
Kaunga	Lusaka	Luangwa	65	3995	61.46

Kazembe	Eastern	Lumezi	1264	14552	11.51
Luangwa	Eastern	Nyimba	3454	3528	1.02
Lukusuzi	Eastern	Lumezi	1403	3497	2.49
Lumimba	Eastern	Lumezi	2202	14085	6.40
Lunya	Lusaka	Luangwa	47	895	19.04
Lusangazi	Eastern	Lusangazi	1310	3559	2.72
Malama	Eastern	Mambwe	271	711	2.62
Mandombe	Lusaka	Luangwa	32	1043	32.60
Mankanda	Lusaka	Rufunsa	1225	6222	5.08
Mankhokwe	Lusaka	Luangwa	58	958	16.51
Mawanda	Eastern	Lusangazi	1794	18964	10.57
Mburuma	Lusaka	Luangwa	173	4113	23.78
Mkaliva	Lusaka	Luangwa	1884	1241	0.66
Mnkhanya	Eastern	Mambwe	168	11757	69.98
Mphuka	Lusaka	Luangwa	225	1665	7.40
Ncheke	Eastern	Mambwe	344	930	2.70

Ndimba	Eastern	Mambwe	1009	16196	16.05
Nsefu	Eastern	Mambwe	609	13290	21.82
Phwazi	Lusaka	Luangwa	157	1315	8.38
Rufunsa	Lusaka	Rufunsa	2970	24206	8.15
Shikabeta	Lusaka	Rufunsa	1931	3008	1.56
AVERAGE					17.30

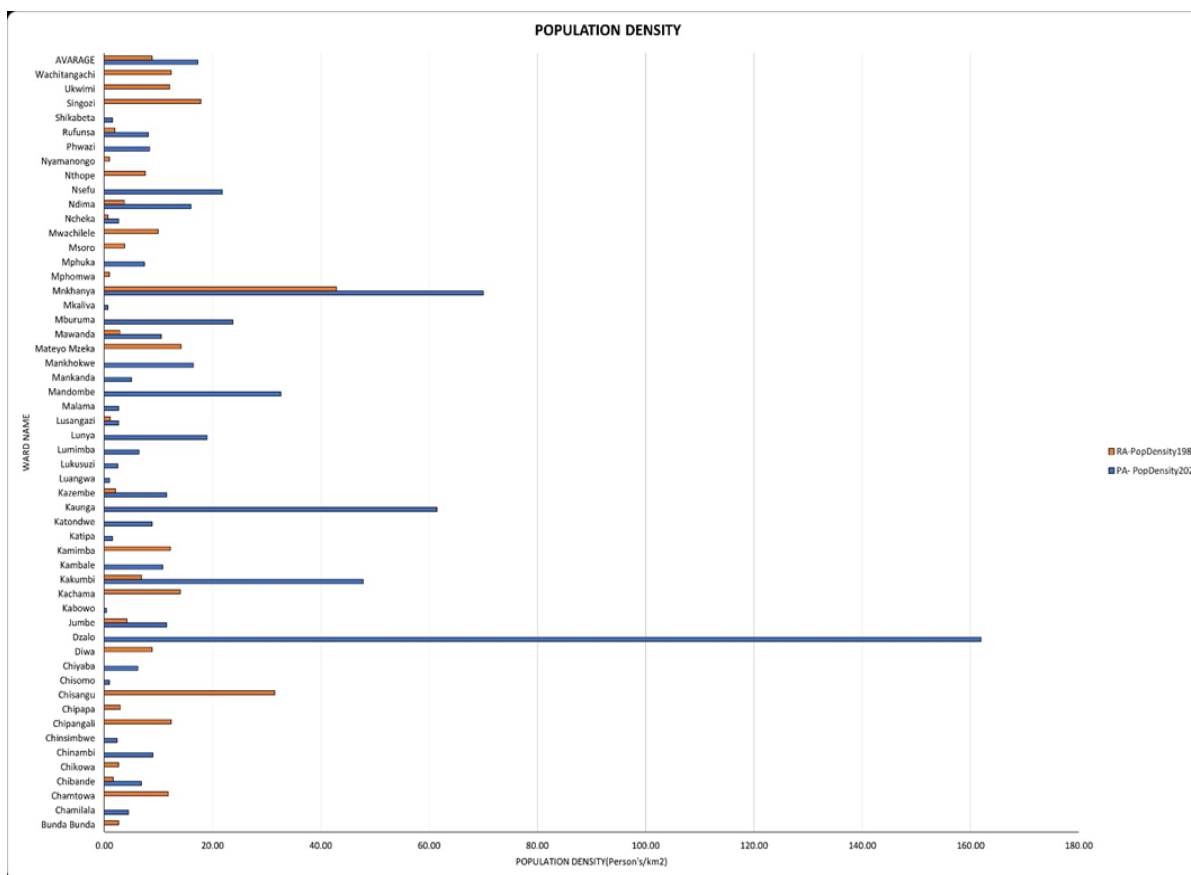


Figure 10: Population Density - Project Area Administrative Wards vs Reference Area Administrative Wards

The similarity is sufficient in that the 1986 average population density in the reference area wards was estimated to be 8.82 persons per km², while the 2021 population density for project area wards is estimated at 17.3 persons per km². This indicates that there are more people per square kilometer in 2021 in the project area wards than there were in the reference area wards in 1986 at the beginning of the Reference Period.

Given that the independent agents and drivers of deforestation study for the project identified population density as the driver variable that possesses a clear relationship with historical deforestation, population density is therefore considered as a suitable proxy for deforestation threat. In this respect, the results above indicate that the project area faces higher levels of deforestation threat in 2021 compared to the reference area at the beginning of the reference period (1986). Therefore, the choice of the reference area for the project is considered conservative in this regard.

- Landscape configuration.

In terms of landscape configuration similarity, there is some disparity between the reference area and the project accounting areas in terms of elevation. Utilizing NASA SRTM Digital Elevation 30m data, the mean elevation was estimated for both regions in Google Earth Engine. The reference area is predominantly on the plateau with a mean elevation of 926.85 meters above sea level, whereas the project accounting area is primarily in the Luangwa Valley, with a mean elevation of 728.54 meters above sea level. However, insights from focus group discussions conducted during the projects independent drivers and agents' study suggest that the baseline agents of conversion are migrating from the plateau, indicating a preference for the landscape conducive to agricultural activities, charcoal production, and firewood collection, which are major drivers of deforestation across the region. Moreover, deforestation is predominantly influenced by slope rather than absolute elevation. With the reference area and the accounting area having mean slopes of 7.1% and 9.5%, respectively. It can therefore be concluded that on average, there is a minimal difference in susceptibility to deforestation between the reference area and the project area, underscoring the conservative choice of the reference area. Furthermore, to standardize the comparison, all areas with slopes exceeding 45% were excluded from both the project accounting and reference areas. Consequently, in terms of landscape configuration, the reference area and the project area can be deemed similar. See slope and elevation in figure 11.

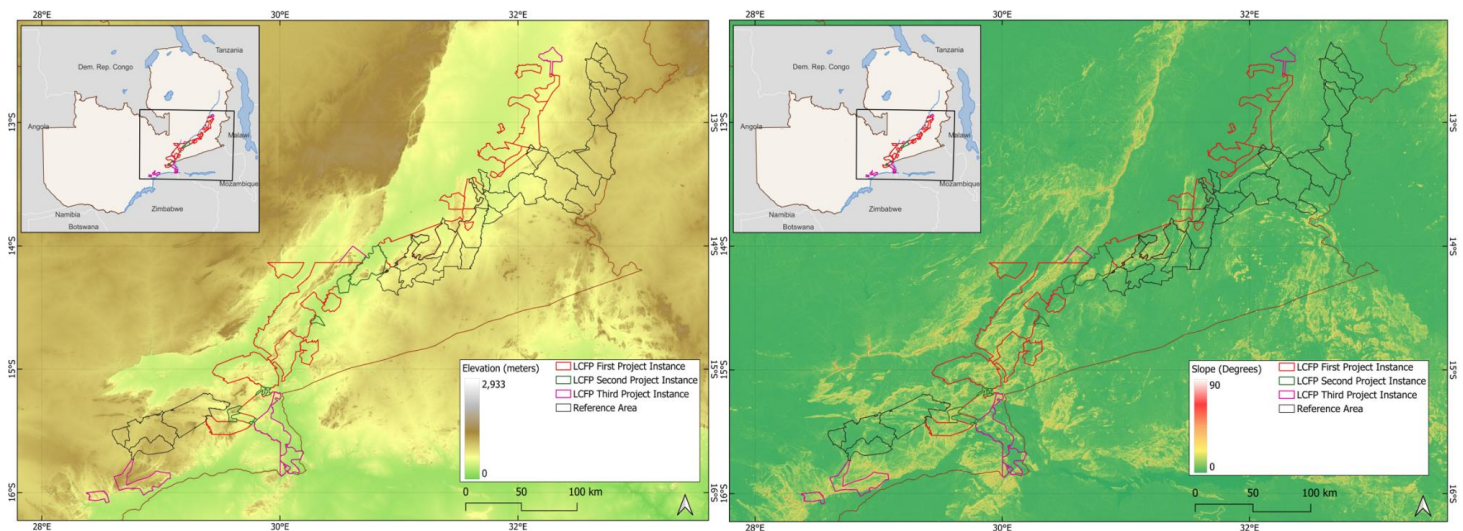


Figure 11. Project and Reference Area Configuration (Elevation and Slope)

Number of CDM Sample Points

Given the overlap of the new reference area with the original and the necessity to utilize the data already collected for the initial reference area, the project has chosen to estimate the

necessary number of CDM interpretation points for the new reference area by extrapolating from the CDM point density of the initial reference area, ensuring consistency.

The new reference area covers a total extent of 1,151,327 ha and based on the original reference area point density, i.e. 0.004673662 points per hectare, approximately 5,443 points were estimated to be dispersed into the new reference area. However, the project opted to allocate more points than estimated. In total, the project randomly allocated 5910 points in the new reference area. The resulting logistic function for the second baseline included 50,523 forest state observations during the historic reference period.

Given that the initial portion of the reference area already had 2,600 points installed, only 3,310 points were installed in the new segment of the new reference area. This random distribution was carried out using QGIS to ensure an equitable representation across the entire area.

To further maintain consistency in our sampling methodology, the project adhered to the original sample point numbering system. Consequently, while the initial sample points ranged from 0 to 2,599, the new sample points were designated within the range of 3,000 to 5,909. This systematic approach facilitates the organization and analysis of data.

In summary, with 5,910 points distributed within the 1,151,327 ha of the new reference area, building upon the previous analysis, the new CDM results from a comprehensive executed sampling approach aimed at capturing the diverse characteristics of the reference area.

Reference Period and Historic Imagery to Parameterize Cumulative Deforestation Model

PDR.48

The start of the reference period is May, 1986 and the end is May, 2021.

PDR.49

Table 19: List of available historical imagery for the reference area

Sensor	Path/Row or Tile Name	Date
Landsat 5	169/069	5/4/1986
Landsat 5	170/070	5/11/1986
Landsat 5	171/070	5/18/1986
Landsat 5	171/071	5/18/1986
Landsat 5	169/069	5/20/1986
Landsat 5	170/069	5/11/1986

Landsat 5	169/068	5/4/1986
Landsat 5	170/070	5/9/1991
Landsat 5	169/070	4/16/1991
Landsat 5	170/069	5/9/1991
Landsat 5	170/070	5/25/1991
Landsat 5	169/068	5/18/1991
Landsat 5	171/070	4/14/1991
Landsat 5	171/071	4/14/1991
Landsat 5	169/069	6/8/1993
Landsat 5	169/068	5/23/1993
Landsat 5	169/070	5/23/1993
Landsat 5	171/071	5/21/1993
Landsat 5	169/069	5/24/1999
Landsat 5	169/068	5/24/1999
Landsat 5	169/070	5/24/1999
Landsat 5	170/069	5/15/1999
Landsat 5	171/071	5/6/1999
Landsat 5	170/070	5/4/2001
Landsat 5	170/069	5/4/2001
Landsat 5	169/069	8/17/2001
Landsat 5	171/071	8/15/2001
Landsat 5	171/070	8/15/2001
Landsat 5	169/069	4/28/2007
Landsat 5	169/068	4/28/2007

Landsat 5	170/069	5/21/2007
Landsat 5	170/070	5/21/2007
Landsat 5	171/070	5/12/2007
Landsat 5	171/071	5/12/2007
Landsat 5	170/070	5/26/2009
Landsat 5	169/068	5/19/2009
Landsat 5	169/069	5/19/2009
Landsat 5	169/070	5/19/2009
Landsat 5	170/068	5/26/2009
Landsat 5	170/069	4/8/2009
Landsat 5	171/070	5/17/2009
Landsat 5	171/071	5/17/2009
Sentinel-2	T36LTK	5/7/2016
Sentinel-2	T36LUK	5/7/2016
Sentinel-2	T36LVL	5/7/2016
Sentinel-2	T36LVM	5/7/2016
Sentinel-2	T36LWM	5/7/2016
Sentinel-2	T36LUL	5/7/2016
Sentinel-2	T35LPC	4/20/2016
Sentinel-2	T35LPD	4/20/2016
Sentinel-2	T35LQC	4/20/2016
Sentinel-2	T35LQD	4/20/2016
Sentinel-2	T36LTK	5/22/2017
Sentinel-2	T36LUK	5/22/2017

Sentinel-2	T36LVL	5/22/2017
Sentinel-2	T36LVM	5/22/2017
Sentinel-2	T36LWM	5/22/2017
Sentinel-2	T36LUL	5/12/2017
Sentinel-2	T35LPC	5/5/2017
Sentinel-2	T35LPD	5/5/2017
Sentinel-2	T35LQC	5/25/2017
Sentinel-2	T35LQD	5/25/2017
NICFI	planet_medres_normalized_analytic_2020-12_mosaic	2020-12
NICFI	planet_medres_normalized_analytic_2021-05_mosaic	2021-05

PDR.50

No specific timeline of events is known for the agents of conversion

PDR.51

Deforestation in Eastern Province started in the southern side of the province near settlements along the Great East Road. As population has expanded, deforestation advanced towards the Luangwa valley, passing through the reference areas on the way towards the project area. No specific events regarding this movement were identified, so just like in the initial baseline estimation, the project simply selected a reference period based on the best historical imagery that covered the new reference areas. Landsat 5 imagery was used for older periods (1986-2009), Sentinel-2 imagery was used for the middle period (2016-2017) and NICFI images used for the last 2 years (2020-2021) of the historic reference period. The May, 1986 Landsat scenes provided the earliest suitable imagery. The inclusion of NICFI images, with a spatial resolution of 4.77 meters, made it easier to interpret the land use compared to Landsat imagery and Sentinel 2, so this helped to improve the accuracy of the analysis.

PDR.52-58

The selected imagery was chosen specifically to minimize clouds and was thus almost entirely cloud free, resulting in 100% double coverage, meaning that all parts of the reference area were covered by the selected imagery at least twice during the historical reference

period. Figure 12 below shows the reference area, Location of CDM interpretation sample points/Double coverage points (100%).

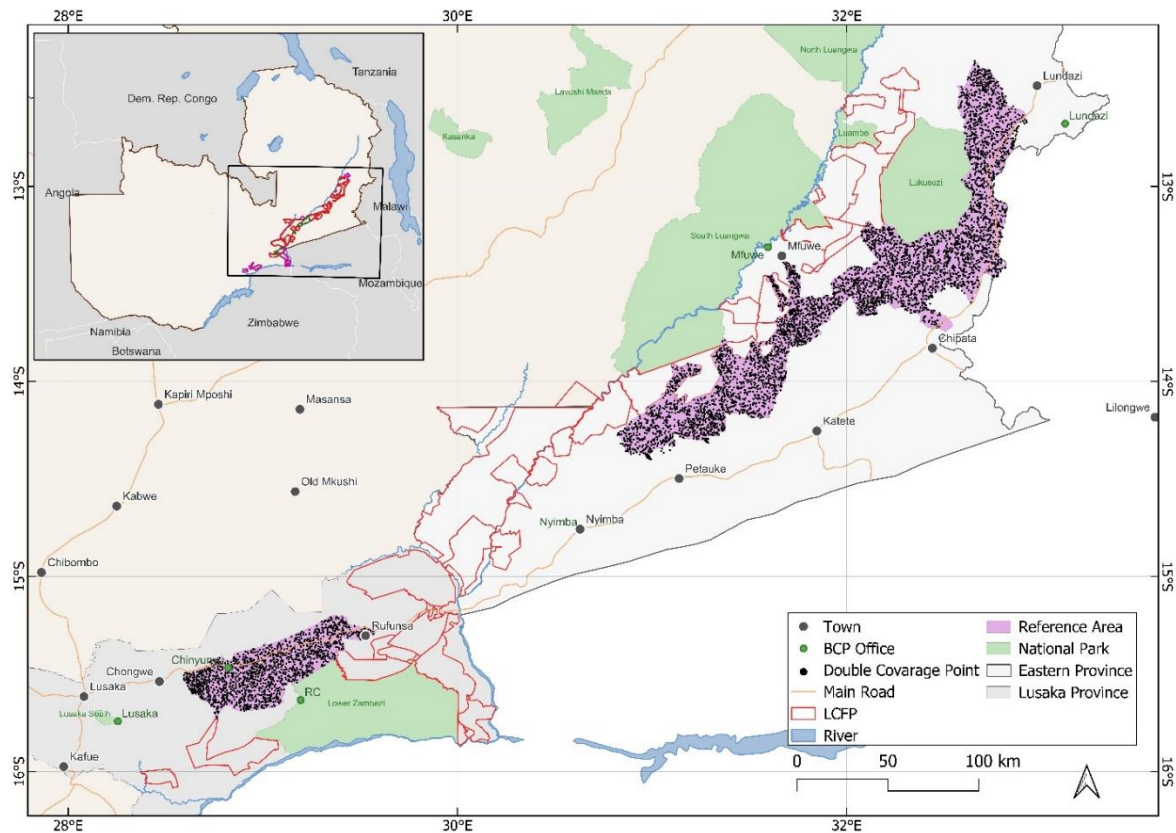


Figure 12. Map of reference area and Location of the CDM interpretation sample points/Double Coverage (100%)

To develop the point interpretation sampling frame, 5910 random points were generated using this QGIS random points in polygons function. These are also shown in Figure 12 above

Figure 13 below shows the temporal location of the images used to develop the CDM for the first baseline for the Luangwa Community Forests Project, while Figure 14 shows the temporal location of the images used to develop the CDM for the baseline re-evaluation. The temporal locations are, on average, located evenly across the reference period thereby fulfilling the stationarity requirement.

The analysis of conversion in the interpolation points was conducted in Earth Engine (script is available to share with VVB). This made it very easy to examine each point systematically and rapidly flip from one image date to the next. As shown in Table 19, all of the imagery used was 30-meter resolution or less.

The registration for the imagery was visually checked before starting the analysis. All of the images were well registered with less than 10% RMSE with the exception of the Landsat 5 image from 1999. The shift in that image was considered while interpreting the images.

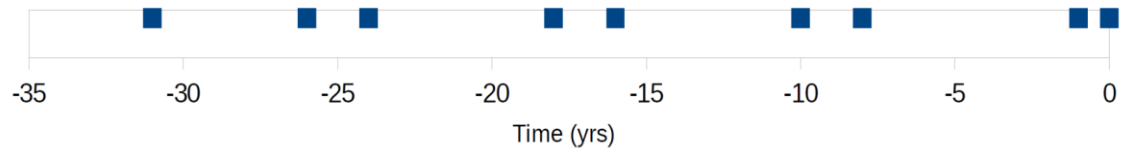


Figure 13. Initial Line Plot to Demonstrate Approximate Stationarity of Historical Imagery

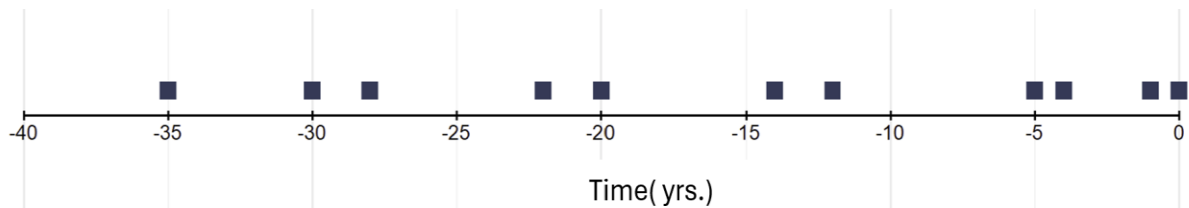


Figure 14. Updated CDM Line Plot to Demonstrate Approximate Stationarity of Historical Imagery

Figure 14 above shows that the time series is stationary (evenly distributed through the timeline).

Point Interpretation and fitting the Logistic Function

PDR.59-62

For this CDM updated, the project has used the same covariates and parameters as those in the validated PD therefore because, VM0009 allows for using parameters (θ) other than time to improve the fit of the logistic function. However, the project only tested a single model with time and found that it provided a good fit. Therefore, no other parameters were considered.

PDR.63

The point interpretation analysis was conducted using a script in Earth Engine to display the historical imagery and make it easy for the user to navigate to each of the 5,910 points (script

available for VVB review). A protocol was developed to guide the 3 interpreters who conducted the work. The protocol was developed by a BCP employee who was an expert in remote sensing in Miombo forests in Zambia and very familiar with interpreting a variety of imagery types. The protocol describes, with illustrations:

- How to discern conversion based on changes in pixels, the shapes of changes, surrounding context, and using supplementary higher resolution imagery.
- Being aware of seasonal changes that occur in vegetation, which was most prominent in June imagery (though the historical imagery was specifically selected to avoid this issue as much as possible)
- Defined deforestation as a clearing of 0.5 ha in order to reduce errors associated with image speckle or degradation that did not lead to deforestation. A 5 x 5 grid of Landsat pixel boundaries was overlaid on the imagery and the protocol required that the interpretation pixel (the center pixel) be deforested and have an orthogonal connection to at least 5 other pixels to be considered deforested.

PDR.64-65

A 5% ($5,910 \times 0.05 = 296$) random sample was drawn from the forest state interpretations by the 3 interpreters and compared to a forest state classification performed by another interpreter who is more familiar with the study area. Table 20 reports the classification accuracy as well as the Kappa statistic for each of the image dates. The accuracies returned were suitable and well within the criteria stipulated by the methodology. The data used to determine the errors can be found in Appendix O

Table 20. Results of an independent Updated CDM error assessment.

Year	Richard Burns				
	Agreement	Kappa	SE	95% conf	
1986	97.64%	0.91	0.20	0.89	0.94
1991	98.99%	0.99	0.13	0.97	1.00
1993	98.31%	0.98	0.17	0.96	1.00
1999	97.97%	0.97	0.19	0.94	0.99
2001	96.94%	0.95	0.23	0.92	0.98
2007	95.95%	0.93	0.28	0.90	0.96

2009	96.62%	0.95	0.22	0.93	0.98
2016	96.30%	0.95	0.25	0.92	0.98
2017	96.28%	0.95	0.25	0.93	0.98
2020	95.95%	0.95	0.26	0.92	0.98
2021	96.28%	0.96	0.28	0.93	0.99

On average, there was 97% agreement. No systematic errors were detected, but following guidance from VM0009 v3.0, it was decided to set all values after observed conversion to conversion, so the data was recorded with a formula in LibreOffice.

PDR.66-67

The logistic function (Figure 15) was fit using an R script. The results and standard errors are shown in table 21. Additionally, the standard deviation of observed conversion $\sigma^{\wedge}_{EM}$ was estimated using equation F.13 from VM0009 v3.0. $\sigma^{\wedge}_{EM}$ was found to be 0.442783

$$\hat{\sigma}_{EM} = \sqrt{\left[\sum_{i \in J} w_i o_i \right] \left[1 - \sum_{i \in J} w_i o_i \right]}$$

Equation F.13

Table 21: Initial CDM function fitting results from R

Coefficient	Estimate	Standard Error
A	0.2188	0.02932
B	0.0003271	0.000006976

Table 22: Updated Logistic function fitting results from R

Coefficient	Estimate	Standard Error
A	0.3530	0.01548
B	0.0003166	0.000003562

PDR.74-75

No covariates to conversion were used as in the initial CDM.

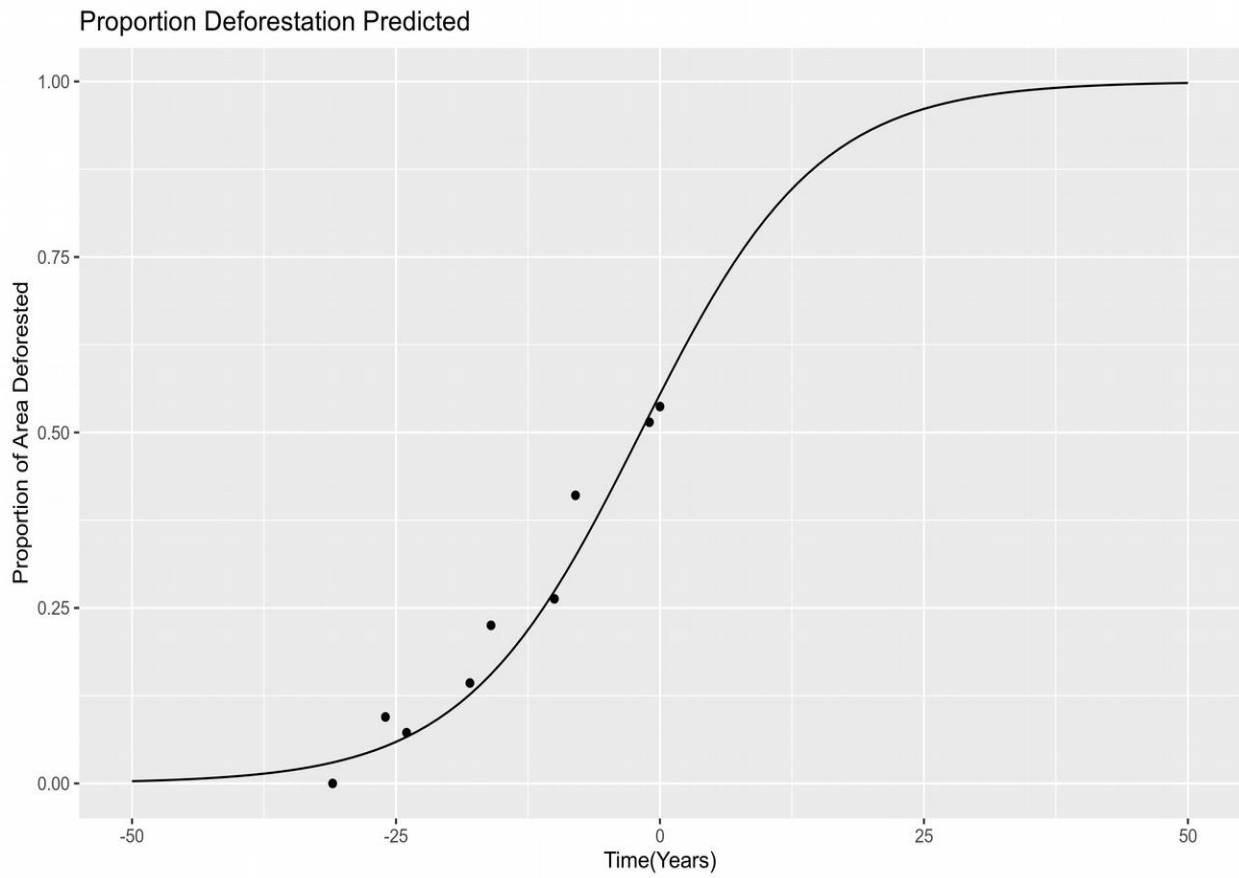


Figure 15: Initial Logistic Function

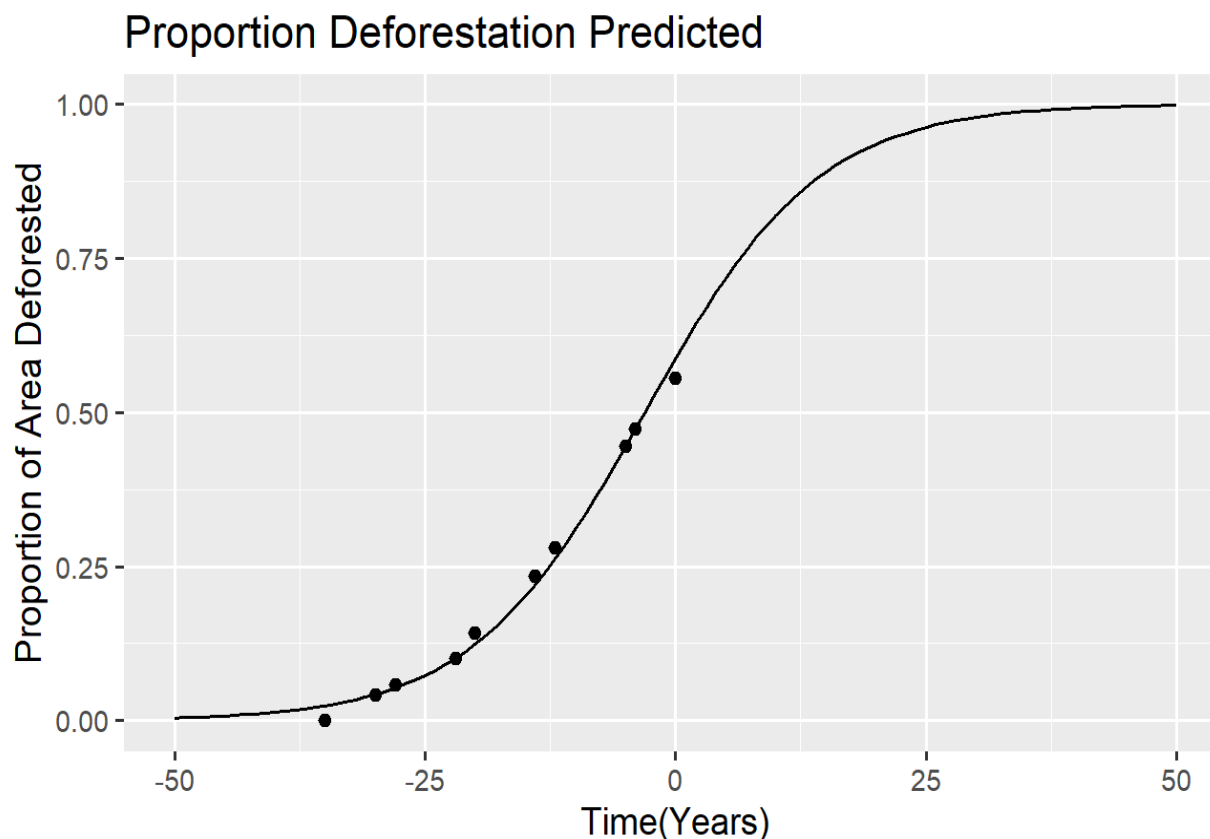


Figure 16 : Updated Logistic function used in CDM

3.3 Grouped Projects

The second project activity instances verified in M4 comprised a total project area of 92,990 ha and in the current monitoring period (M5), a third project activity instance has been included comprising of 199,621 ha of project area, of which 183,507 ha consist of forest according to the Zambian definition of forest (Table 23 and 24). Within this forested area, 122,856 ha is calculated as a project accounting area (Table 24) and subject to verification in this monitoring period.

Table 23. Summary of Project Area - REDD+ Project Zones on Private Land, Second activity instances.

Project Activity Instance	Private Game Ranch	Chiefdom	Project area (ha)	Forest area (ha)	Accounting area (ha)
2	Nyakolwe	Luembe	57,706	32,013	29,856
2	Nyamvu	Luembe	15,238	7,176	7,172
2	Kazumba	Mwape	4,775	2,732	2,724

Table 24. Summary of Project Area - REDD+ Project Zones under communal tenure, Second and third project activity instances.

Project Activity Instance	CRB	GMA	Chiefdom	Project area (ha)	Forest area (ha)	Accounting area (ha)
2	Mpanshya corridor	Rufunsa	Mpanshya	15,271	13,430	10,978
3	Kazembe	Upper Lumimba	Kazembe	18,515	13,241	13,248
3	Mwape	Chisomo	Mwape	20,500	18,845	5,149
3	Mburuma	Rufunsa	Mburuma	66,520	62,537	34,146
3	Mphuka	Rufunsa	Mphuka	25,881	24,541	24,375
3	Chiawa	Rufunsa	Rufunsa	68,205	64,343	45,938

Following the guidance on grouped projects under the VCS, all the new project activity instances shall meet the eligibility criteria for inclusion as explained in the PD, section 2.1.20 as shown below in table 25.

Table 25. Eligibility criteria under which the second and third project instance is incorporated into the project.

Criteria	Justification	Proof (Relevant Documents)
Must be within the chiefdoms already identified as potential future activity instances in the PD	All the project instances are within the project boundaries identified in the PD (page 36-40)	Map showing project boundaries see Figure 9 & 10 and Tables 6 -8 in the PD.
The project type for future project instances must be avoided unplanned deforestation. Accordingly, the end land use in the baseline scenario must be non-forest.	The baseline scenario and the demonstration of additionally are determined for the entirety of the geographic project area within which project activity instances are developed.	PD, Figure 10, page 40.
The baseline drivers and agents of deforestation in future project instances must be of the same type as the current project instances - clearing of forested land for expansion of small-holder farming and, or charcoal production.	All instance 2 and 3 areas share the same agents and drivers study as that of the first instance project areas and therefore share the same conclusion Made	Drivers and agents study by UNIQUE (Appendix K)
Future project accounting areas must have met the Zambian definition of forest for at least 10 years prior to the project start date.	All second instance project areas meet the definition of forest and accounting areas were calculated according to the same method as First Activity Instance areas.	See the classification method described in the PD.
Some point within 2km of the perimeter of future accounting areas must have been converted from forest to farming.	Perimeters of the newly added areas under the Third project activity instance did satisfy this requirement without exception.	See r_U value in NER calculations in Appendix G

Future project accounting areas must be located in a tropical ecosystem and must not contain peat soil.	No peatland was detected in the third project activity instance areas	Spatial data for accounting areas does not overlap with available peatland data (See Appendix I)
For each new accounting area, a reference area meeting the requirements in terms of size and similarity to the accounting area as per section 6.8.1 of VM0009 v3.0 must be delineated.	First instance project areas share a common reference area as the Second and Third instances.	Map and spatial data for the common reference areas
For any new reference areas that may be required, there is sufficient historical imagery meeting the requirements of section 6.8.4 of VM0009 v3.0.	Third and Second instance project areas share a common reference area as the first project activity instances	Map and spatial data for the common reference areas
Project activities for new project instances are the same as those described in this PD.	NA	NA
The project proponents can access any new leakage or proxy areas that may be required for monitoring related to new project instances as per sections 8.3.2.1 and 6.4 of VM0009 v3.0.	Instance 3 areas share common leakage and proxy areas with the 1 st instance project areas	Map showing accessibility to the common leakage and proxy areas – see Figure 9.
Same Governance and Land Tenure and Social Economic Systems	All second and third instance activity areas are private and communal land tenure systems.	Title Deed from private game ranch owners and community forest management Agreements signed between government and Community resources boards (CFMGs)

Since a section of 19,887 ha of the second project activity instance areas overlapped with the existing activity shifting leakage area (originally 319,179 ha), the leakage area comprises

299,291 ha. This is not strictly regarded as a methodological deviation, as VM0009 allows for portions of existing leakage areas to be converted to future project activity instance areas.

MR.7 - 8

The project activity start date for the initial project area instance is the same as the project start date: May 1st, 2015. Therefore, the number of days since the project start date for the initial project area instance is 0.

The start date of the second project activity instance is 11th August 2021. The number of days since the project start date is 2,294 days.

The start date of the third project activity instance is 1st January 2022, The number of days since project start is 2,437 days.

4 DATA AND PARAMETERS

4.1 Data and Parameters Available at Validation

Data / Parameter	α
Data unit	Unitless
Description	Combined effects of β and θ at the start of the historic reference period
Source of data	Reference area and historic reference period
Value applied	0.3530
Justification of choice of data or description of measurement methods and procedures applied	Time and place in which the logistic model is fit. Analysis undertaken as described in section 3.2.1 of the project document. The Analysis was conducted by BioCarbon Partners Carbon Team at Baseline evaluation. See Appendix O for more information Including the LCFP CDM Point Interpretation SOP
Purpose of Data	Calculation of baseline emissions
Comments	This parameter has been changed from the initial due to the CDM update in the current monitoring period (M5)

Data / Parameter	β
Data unit	Unitless
Description	Effect of time on the cumulative proportion of conversion over time

Source of data	Reference area and historic reference period
Value applied	0.0003166
Justification of choice of data or description of measurement methods and procedures applied	Time and place in which the logistic model is fit. Analysis undertaken as described in section 3.2.1 of the project document. The Analysis was conducted by BioCarbon Partners carbon Team at Baseline evaluation. See Appendix O for more information Including the LCFP CDM Point Interpretation SOP
Purpose of Data	Calculation of baseline emissions
Comments	This parameter has been changed from the initial due to the CDM update in the current monitoring period (M5)

Data / Parameter	γ
Data unit	Days
Description	Time shift from beginning of historic reference period to project start date
Source of data	Historic reference period
Value applied	10,589
Justification of choice of data or description of measurement methods and procedures applied	Time in which the logistic model is fit. Analysis undertaken as described in section 3.2.1 of the project document. The Analysis was conducted by BioCarbon Partners carbon Team at Baseline evaluation. See Appendix O for more information Including the LCFP CDM Point Interpretation SOP
Purpose of Data	Calculation of baseline emissions
Comments	

Data / Parameter	λ_{soc}
Data unit	Proportion (unitless)
Description	Exponential soil carbon decay parameter
Source of data	VCS AFOLU default for tropical regions
Value applied	0.2
Justification of choice of data or description of measurement methods and procedures applied	Conservative default based on IPCC value.

Purpose of Data	Calculation of baseline emissions
Comments	

Data / Parameter	σ_{EM}
Data unit	Standard deviation (unitless)
Description	The estimated standard deviation of the state observations used to fit the logistic function
Source of data	Remote sensing image interpretation. Analysis undertaken as described in section 3.2.1 of the project document.
Value applied	0.442783
Justification of choice of data or description of measurement methods and procedures applied	Time and place in which the logistic model is fit. Analysis undertaken as described in section 3.2.1 of the project document. The Analysis was conducted by BCPs Carbon Team at Baseline evaluation. See Appendix O for more information Including the LCFP CDM Point Interpretation SOP
Purpose of Data	Calculation of baseline emissions
Comments	

Data / Parameter	B
Data unit	Set
Description	The set of all selected carbon pools in biomass. Is a subset of C
Source of data	PDD
Value applied	2
Justification of choice of data or description of measurement methods and procedures applied	AGOT and BGOT, See section 1.1.1 of the project document for justification
Purpose of Data	Calculation of baseline emissions
Comments	

Data / Parameter	C
Data unit	Set
Description	The set of all selected carbon pools

Source of data	Monitoring records
Value applied	3
Justification of choice of data or description of measurement methods and procedures applied	AGOT, BGOT, SOC, See section 1.1.1 of the project document for justification
Purpose of Data	Calculation of baseline emissions
Comments	

Data / Parameter	<i>J</i>
Data unit	Set
Description	The set of all observations of conversion. When superscripted with a monitoring period, the conversion observations are taken for leakage analysis
Source of data	Remote sensing image interpretation or field observations in the leakage area
Value applied	13,194
Justification of choice of data or description of measurement methods and procedures applied	Analysis undertaken as described in section 3.2.1 of the project document. Leakage area monitored by BCP's forest monitoring teams as per Leakage Monitoring SOP in Appendix R
Purpose of Data	Calculation of baseline emissions
Comments	

Data / Parameter	<i>M</i>
Data unit	Set
Description	The set of all monitoring periods
Source of data	Monitoring records
Value applied	27
Justification of choice of data or description of measurement methods and procedures applied	The project crediting period is 30 years and after the first verification, the project uses annual verification. Thus there will be 27 verification periods.
Purpose of Data	Calculation of baseline emissions

Comments	
Data / Parameter	A_{PAA}
Data unit	ha
Description	Cumulative size of all three activity instance project accounting areas. The accounting areas of the first, second and third project instances are 485,495 ha, 50,730 ha, and 122,856 ha respectively.
Source of data	GIS analysis prior to sampling
Value applied	659,081
Justification of choice of data or description of measurement methods and procedures applied	Area within the project area that is on slopes of less than 45%, less than 6.5 km from a road, and was forest at least 10 years before the project start date. Described in more detail in section 3.2.1 of the project document. Spatial Analysis conducted by BCPs Carbon Accounting team.
Purpose of Data	Calculation of baseline emissions
Comments	

Data / Parameter	A_{PX}
Data unit	ha
Description	Area of proxy area
Source of data	GIS analysis prior to sampling
Value applied	3,840
Justification of choice of data or description of measurement methods and procedures applied	Areas converted from forest to non-forest between 2010 and 2017 falling within 2 km of buffer around roads passing through communities favorable to having proxy plots on their lands. Described in more detail in section 3.2.1 of the project document. Spatial Analysis conducted by BCPs Accounting team.
Purpose of Data	Calculation of baseline emissions
Comments	

Data / Parameter	n_d
Data unit	unitless
Description	Number of spatial points in the reference area
Source of data	Remote sensing image interpretation

Value applied	5910
Justification of choice of data or description of measurement methods and procedures applied	Value deemed sufficient as per VM0009 to adequately quantify forest state in the reference area. Described in more detail in section 3.2.1 of the project document. Remote sensing Image interpretation and Analysis was conducted by BCP's Carbon Team at Baseline evaluation. See Appendix O for more information.
Purpose of Data	Calculation of baseline emissions
Comments	

Data / Parameter	o_i
Data unit	Binary
Description	State observation for the sample point in the reference area
Source of data	Remote sensing image interpretation
Value applied	0 or 1
Justification of choice of data or description of measurement methods and procedures applied	Data collected as per the methodology. Described in more detail in section 3.2.1 of the project document. Remote sensing Image interpretation and Analysis was conducted by BCP's Carbon Team at Baseline evaluation. See Appendix O for more information.
Purpose of Data	Calculation of baseline emissions
Comments	

Data / Parameter	P_{LME}
Data unit	Proportion (unitless)
Description	Portion of leakage related to market
Source of data	Analysis of drivers of deforestation (Appendix K)
Value applied	0
Justification of choice of data or description of measurement methods and procedures applied	Few wood products are taken from the project area in the baseline scenario and most farming in the baseline scenario is subsistence.
Purpose of Data	Calculation of baseline emissions

Comments	
Data / Parameter	q
Data unit	Days
Description	Lag between start of degradation and conversion
Source of data	Consultation with agents of deforestation in near project area in Lusaka province – 28 farmers were questioned to determine how long they usually wait after clearing an area before ploughing it.
Value applied	620
Justification of choice of data or description of measurement methods and procedures applied	Based on observations in the reference area, hence applicable to the project area.
Purpose of Data	Calculation of baseline emissions
Comments	

Data / Parameter	r_{RS}
Data unit	Unitless
Description	Expansion factor for above-ground biomass to below-ground biomass (root/shoot ratio)
Source of data	Chidumayo, E. N. 2013. Estimating tree biomass and changes in root biomass following clear-cutting of Brachystegia-Julbernardia (Miombo) woodland in central Zambia. Environmental Conservation. 42(1), pp. 54 - 63.
Value applied	0.54
Justification of choice of data or description of measurement methods and procedures applied	The root-shoot ratio was determined within a study site that overlaps one of the same provinces the project is in. See Appendix P for more information including documentation pertaining to measurements and procedures that were applied.
Purpose of Data	Calculation of baseline emissions
Comments	

Data / Parameter	r_U
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Data unit	Proportion (unitless)
Description	Onset proportion of conversion immediately adjacent to project area
Source of data	Remote Sensing Deforestation analysis (Described in section 1.1.2 of the PD) /GIS analysis and image interpretation
Value applied	0.084 for PAI1, 0.163 for PAI2, and 0.054 for PAI3
Justification of choice of data or description of measurement methods and procedures applied	Result of remote sensing analysis described in more detail in section 3.2.1 of the project document. For project activity instances 1 and 3 (PAI1 and PAI3) this calculation is based on the percentage of threatened perimeter close to recent deforestation(Detailed Steps for calculating the threatened boundary, converted boundary and rU value is in section 3.2.1 of the validated PD)In summary; Getting Threatened perimeter - Removal of the accounting area perimeter that border with areas not accessible to the agents and drivers of deforestation (Interior boundaries for the accounting areas, natural non-forest areas such as wide rivers, grassland and slopes greater than 45%, protected area) - The remaining perimeter is then rasterized Getting Converted perimeter - Raster buffer of 5 pixels was made for areas mapped as deforested in the PALSAR deforestation analysis. - Then band math was used to identify the threatened boundary pixels that overlapped with the converted areas. Calculating the Ru Value r^U value was then calculated as the ratio of the threatened boundary pixels that overlapped with conversion or the buffer compared to the total threatened boundary pixels. For project activity instances 2 (PAI2) this calculation is based on the ratio of conversion and accounting area minus conversion area. Steps for the Ru value calculation included mapping of deforested area using PALSAR deforestation analysis, calculating the size and then getting the ratio of converted area to the difference between the accounting area and the converted area. See the calculations provided in the ER calculation spreadsheets, Appendix G. GIS analysis and image interpretation conducted by BCPs Carbon team.

Purpose of Data	Calculation of baseline emissions
Comments	

Data / Parameter	t_i
Data unit	Days
Description	The point in time of the observation made at point i
Source of data	Remote sensing image interpretation
Value applied	1-12810
Justification of choice of data or description of measurement methods and procedures applied	The historical reference period stretches from May 4 th , 1986 to May 30 th 2021, which is 12810 days in total. Remote sensing image interpretation and data analysis conducted by BCPs Carbon team.
Purpose of Data	Calculation of baseline emissions
Comments	

Data / Parameter	t_{PL}
Data unit	Days
Description	Length of project crediting period
Source of data	PD
Value applied	10957
Justification of choice of data or description of measurement methods and procedures applied	The project crediting period is from May 1 st , 2015 to April 30 th , 2045 which is equivalent to 10957 days.
Purpose of Data	Calculation of baseline emissions
Comments	

Data / Parameter	w_i
Data unit	Unitless
Description	Weight applied to the ith sample point in the reference area
Source of data	Remote sensing image interpretation
Value applied	Varies (see Appendix O)
Justification of choice	Calculated as specified in VM0009. Remote sensing image

of data or description of measurement methods and procedures applied	interpretation and data analysis conducted by BCP's Carbon team. See Appendix O for more information.
Purpose of Data	Calculation of baseline emissions
Comments	

Data / Parameter	$x_i y_i$
Data unit	Geographic coordinates
Description	Latitude and Longitude of the i^{th} sample point
Source of data	Remote sensing image interpretation
Value applied	Varies (see Appendix O)
Justification of choice of data or description of measurement methods and procedures applied	
Purpose of Data	Calculation of baseline emissions
Comments	

4.2 Data and Parameters Monitored

MR.85-87

Data / Parameter	$A_{P1}^{[m=0]}$
Data unit	ha
Description	Area of project accounting area stratum 1 prior to first verification event
Source of data	GIS analysis prior to sampling
Description of measurement methods and procedures to be applied	GIS analysis removed areas in the project area that were non-forest prior to the project start date, on slopes greater than 45% or in areas farther than 6.5 km from a road.
Frequency of monitoring/recording	First monitoring period / 10 years
Value monitored	485,495

Monitoring equipment	N/A
QA/QC procedures to be applied	Double checking GIS analysis
Purpose of the data	Calculation of baseline emissions
Calculation method	GIS area calculation
Comments	

Data / Parameter	$c_B^{[m]}$
Data unit	tCO ₂ e / ha
Description	Baseline carbon stocks at the end of the current monitoring period
Source of data	Proxy area sampling
Description of measurement methods and procedures to be applied	Proxy area sampling as per the protocol Appendix P.
Frequency of monitoring/recording	20% of plots measured annually
Value monitored	15.6
Monitoring equipment	See Appendix P. Standard vegetation quantification equipment.
QA/QC procedures to be applied	See Appendix P and description in monitoring report.
Purpose of the data	Calculation of baseline emissions
Calculation method	Using standard vegetation ecology plot sampling techniques and subsequent quantification of biomass using allometric equations.
Comments	

Data / Parameter	$C_{BBGB}^{[m]}$
Data unit	tCO ₂ e

Description	Carbon not decayed in BGB at the end of the current monitoring period
Source of data	Equation F.32 from VM0009
Description of measurement methods and procedures to be applied	Ongoing forest monitoring as per Appendix P
Frequency of monitoring/recording	Based on peer reviewed literature and prescribed equation in VM0009
Value monitored	-67,841
Monitoring equipment	See Appendix P
QA/QC procedures to be applied	See Appendix P
Purpose of the data	Calculation of baseline emissions
Calculation method	Applied equation F.32 in VM0009 3.0
Comments	

Data / Parameter	$C_{B\ SOC}^{[m]}$
Data unit	tCO ₂ e
Description	Carbon not decayed in SOC at the end of the current monitoring period
Source of data	Project accounting area sampling and application of equation F.33 from VM0009 3.0
Description of measurement methods and procedures to be applied	Accounting area sampling as per the protocol in Appendix P and application of equation F.33, VM0009 V3.0
Frequency of monitoring/recording	Every monitoring period
Value monitored	- 849,070
Monitoring equipment	See Appendix P standard soil sampling procedures and equipment.

QA/QC procedures to be applied	See Appendix P
Purpose of the data	Calculation of baseline emissions
Calculation method	Application of equation F.33 VM0009 V3.0 using SOC values from the project accounting area.
Comments	

Data / Parameter	$C_{Bb}^{[m]}$
Data unit	tCO ₂ e / ha
Description	Baseline scenario average carbon stock in selected carbon pools
Source of data	Proxy area sampling
Description of measurement methods and procedures to be applied	Proxy area sampling as per the protocol Appendix P.
Frequency of monitoring/recording	20% of plots measured annually
Value monitored	15.6
Monitoring equipment	See Appendix P. Standard vegetation quantification equipment.
QA/QC procedures to be applied	See Appendix P and description in monitoring report.
Purpose of the data	Calculation of baseline emissions
Calculation method	Using standard vegetation ecology plot sampling techniques and subsequent quantification of biomass using allometric equations.
Comments	

Data / Parameter	$C_{BBM}^{[m]}$
Data unit	tCO ₂ e / ha
Description	Baseline carbon stocks in biomass at the end of the current

	monitoring period
Source of data	Proxy area sampling
Description of measurement methods and procedures to be applied	Proxy area sampling as per the protocol in Appendix P
Frequency of monitoring/recording	At first verification and baseline revision
Value monitored	15.6
Monitoring equipment	See Appendix P
QA/QC procedures to be applied	Review of monitoring records as described in Appendix P and re-sampling of 20% of plots
Purpose of the data	Calculation of baseline emissions
Calculation method	Using standard vegetation ecology plot sampling techniques and subsequent quantification of biomass using allometric equations. Averages calculated as per equation B.9 VM0009 V3.0
Comments	

Data / Parameter	$C_{BSOC}^{[m]}$
Data unit	tCO ₂ e / ha
Description	Baseline soil carbon stocks at the end of the current monitoring period
Source of data	Project accounting area sampling
Description of measurement methods and procedures to be applied	Project accounting area sampling as per the protocol in Appendix P
Frequency of monitoring/recording	Once at first verification
Value monitored	210.6
Monitoring equipment	See Appendix P, standard soil sampling equipment
QA/QC procedures to	Review of monitoring records at next baseline re-evaluation

be applied	
Purpose of the data	Calculation of baseline emissions
Calculation method	Quantification of SOC done according to Annex B.2.6, VM0009 3.0
Comments	

Data / Parameter	$c_p^{[m]}$
Data unit	tCO ₂ e / ha
Description	Project carbon stocks at the end of the current monitoring period
Source of data	Project accounting area sampling
Description of measurement methods and procedures to be applied	Plot monitoring as per the protocol in Appendix P
Frequency of monitoring/recording	20% of plots monitored annually
Value monitored	236.2 in intact forest and 15.2 in encroached forest
Monitoring equipment	See Appendix P, standard vegetation sampling equipment
QA/QC procedures to be applied	Review of monitoring records, see Appendix G.
Purpose of the data	Calculation of baseline emissions
Calculation method	Standard plot sampling techniques to quantify vegetation and application of allometric equations. Averages calculated using equation B.9 VM0009 V3.0.
Comments	A methodology deviation was applied in this monitoring period in that instance one intact forest carbon stock value was applied in both instance two and three. The justification being that instance one values are lower than instance two and three ,therefore using instance one figures is a conservative approach until the consolidation of the datasets in the next monitoring period as per the project's monitoring plan in the validated PD (see section 3.2.1 of the MR for details of this Deviation)

Data / Parameter	$c_p^{[m-1]}$
Data unit	tCO ₂ e / ha
Description	Project carbon stocks at the beginning of the current monitoring period
Source of data	Project accounting area sampling
Description of measurement methods and procedures to be applied	Project accounting area sampling as per the protocol in Appendix P
Frequency of monitoring/recording	20% of plots monitored annually
Value monitored	236.2 in intact forest and 15.2 in encroached forest
Monitoring equipment	Review of monitoring records, see Appendix G.
QA/QC procedures to be applied	Review of monitoring records, re-sampling of 20% of plots.
Purpose of the data	Calculation of baseline emissions
Calculation method	Standard plot sampling techniques to quantify vegetation and application of allometric equations. Averages calculated using equation B.9 VM0009 V3.0.
Comments	A methodology deviation was applied in this monitoring period in that instance one intact forest carbon stock value were applied in both instance two and three. The justification being that instance one values are lower than instance two and three values ,therefore using instance one figures is a conservative approach, until the consolidation of the biomass datasets in the next monitoring period as per the monitoring plan(every 5 years) in the validated PD (see section 3.2.1 of the MR for details of this Deviation)

Data / Parameter	$c_p^{[m=0]}$
Data unit	tCO ₂ e / ha
Description	Project carbon stocks prior to first verification event
Source of data	Project accounting area sampling

Description of measurement methods and procedures to be applied	Project accounting area sampling as per the protocol in Appendix P
Frequency of monitoring/recording	First monitoring period
Value monitored	236.2
Monitoring equipment	Appendix P describes the plot based monitoring procedure.
QA/QC procedures to be applied	Review of monitoring records, see Appendix G.
Purpose of the data	Calculation of baseline emissions
Calculation method	Standard plot sampling techniques to quantify vegetation and application of allometric equations. Averages calculated using equation B.9 VM0009 V3.0.
Comments	

Data / Parameter	$C_{P1BM}^{[m=0]}$
Data unit	tCO ₂ e / ha
Description	Project carbon stocks in stratum 1 prior to first verification event
Source of data	Project accounting area sampling
Description of measurement methods and procedures to be applied	Project accounting area sampling as per the protocol in Appendix P
Frequency of monitoring/recording	First monitoring period
Value monitored	236.2
Monitoring equipment	Appendix P describes the plot based monitoring procedure.
QA/QC procedures to be applied	Review of monitoring records, see Appendix G.
Purpose of the data	Calculation of baseline emissions

Calculation method	Standard plot sampling techniques to quantify vegetation and application of allometric equations. Averages calculated using equation B.9 VM0009 V3.0.
Comments	

Data / Parameter	$C_{PBM}^{[m=0]}$
Data unit	tCO2e
Description	Project carbon stocks in biomass prior to first verification event
Source of data	Project accounting area sampling
Description of measurement methods and procedures to be applied	Project accounting area sampling as per the protocol in Appendix P
Frequency of monitoring/recording	First monitoring period
Value monitored	114,668,494
Monitoring equipment	Appendix P describes the plot based monitoring procedure.
QA/QC procedures to be applied	Review of monitoring records, see Appendix G.
Purpose of the data	Calculation of baseline emissions
Calculation method	Standard plot sampling techniques to quantify vegetation and application of allometric equations. Averages calculated using equation B.9 VM0009 V3.0.
Comments	

Data / Parameter	$C_{Pb}^{[m]}$
Data unit	tCO2e / ha
Description	Average carbon in biomass in the project accounting area
Source of data	Project accounting area sampling
Description of measurement	Project accounting area sampling as per the protocol in Appendix P

methods and procedures to be applied	
Frequency of monitoring/recording	20% of biomass plots annually
Value monitored	236.2
Monitoring equipment	Appendix P describes the plot based monitoring equipment.
QA/QC procedures to be applied	Review of monitoring records, see Appendix G.
Purpose of the data	Calculation of baseline emissions
Calculation method	Standard plot sampling techniques to quantify vegetation and application of allometric equations. Averages calculated using equation B.9 VM0009 V3.0.
Comments	A methodology deviation was applied in this monitoring period in that instance one intact forest carbon stock value was applied in both instance two and three. The justification being that instance one values lower than instance two and three ,therefore using instance one figures is a conservative approach until the consolidation of the datasets in the next monitoring period as per the monitoring plan in the validated PD (see section 3.2.1 of the MR for details of this Deviation)

Data / Parameter	$c_{P1b}^{[m]}$
Data unit	tCO ₂ e / ha
Description	Average carbon in biomass for project accounting area stratum 1
Source of data	Project accounting area sampling
Description of measurement methods and procedures to be applied	Project accounting area sampling as per the protocol in Appendix P
Frequency of monitoring/recording	20% of biomass plots annually
Value monitored	236.2
Monitoring equipment	Appendix P describes the plot based monitoring equipment.

QA/QC procedures to be applied	Review of monitoring records, see Appendix G.
Purpose of the data	Calculation of baseline emissions
Calculation method	Standard plot sampling techniques to quantify vegetation and application of allometric equations. Averages calculated using equation B.9 VM0009 V3.0.
Comments	A methodology deviation was applied in this monitoring period in that instance one carbon stock figures were applied in both instance two and three. The justification being that instance one values lower than instance two and three ,therefore using instance one figures is a conservative approach until the consolidation of the datasets in the next monitoring period as per the monitoring plan in the validated PD (see section 3.2.1 of the MR for details of this Deviation)

Data / Parameter	$c_{PSOC}^{[m=0]}$
Data unit	tCO ₂ e / ha
Description	Project soil carbon stocks prior to first verification event
Source of data	Project accounting area sampling
Description of measurement methods and procedures to be applied	Project accounting area sampling as per the protocol in Appendix P
Frequency of monitoring/recording	First monitoring period
Value monitored	235.2
Monitoring equipment	Appendix P, standard soil sampling equipment
QA/QC procedures to be applied	Quantification of SOC done according to Annex B.2.6, VM0009 3.0 at soil lab.
Purpose of the data	Calculation of baseline emissions
Calculation method	Quantification of SOC done according to Annex B.2.6, VM0009 3.0 at soil lab.
Comments	

Data / Parameter	$E_{\Delta GER}^{[m]}$
Data unit	tCO2e
Description	GERs for the current monitoring period
Source of data	Area measurements
Description of measurement methods and procedures to be applied	Accounting area plots monitoring as per Appendix P.
Frequency of monitoring/recording	Every monitoring period
Value monitored	2,163,019
Monitoring equipment	See Appendix P, standard plot sampling equipment
QA/QC procedures to be applied	Review of GER calculations
Purpose of the data	Calculation of project emissions
Calculation method	Calculated with equation F.53 as per section 8.4.1 of VM0009 V3.0
Comments	

Data / Parameter	$E_{\Delta GER}^{[i]}$
Data unit	tCO2e
Description	GERs for monitoring period i
Source of data	Area measurements
Description of measurement methods and procedures to be applied	Accounting area plots monitoring as per Appendix P.

Frequency of monitoring/recording	Prior monitoring period
Value monitored	102,251,032
Monitoring equipment	See Appendix P, standard plot sampling equipment
QA/QC procedures to be applied	Review of GER calculations
Purpose of the data	Calculation of project emissions
Calculation method	Calculated with equation F.53 as per section 8.4.1 of VM0009 V3.0
Comments	

Data / Parameter	$E_{\Delta NER}^{[i]}$
Data unit	tCO ₂ e
Description	NERs for monitoring period i
Source of data	Area measurements
Description of measurement methods and procedures to be applied	Accounting area plots monitoring as per Appendix P.
Frequency of monitoring/recording	Prior monitoring period
Value monitored	91,697,667
Monitoring equipment	See Appendix P, standard plot sampling equipment
QA/QC procedures to be applied	Review of NER calculations
Purpose of the data	Calculation of project emissions
Calculation method	Calculated with VCS AFOLU Non-Permanence Risk Tool and equation F.55 as per section 8.4.3 of VM0009 V3.0
Comments	

Data / Parameter	$E_B^{[m]}$
Data unit	tCO ₂ e
Description	Cumulative baseline emissions at the end of the current monitoring period
Source of data	Proxy area measurements
Description of measurement methods and procedures to be applied	Proxy area plots monitoring as per Appendix P and equation F.16.
Frequency of monitoring/recording	Every monitoring period
Value monitored	11,699,622
Monitoring equipment	See Appendix P, standard plot sampling equipment
QA/QC procedures to be applied	Annual monitoring of proxy area. See Appendix P for error correction procedures.
Purpose of the data	Calculation of baseline emissions
Calculation method	Equation F.16 from VM0009 V3.0
Comments	

Data / Parameter	$E_B^{[m-1]}$
Data unit	tCO ₂ e
Description	Cumulative baseline emissions at the beginning of the current monitoring period
Source of data	Proxy area measurements
Description of measurement methods and procedures to be applied	Proxy area plots monitoring as per Appendix P and equation F.16.
Frequency of monitoring/recording	Prior monitoring period

Value monitored	9,432,108
Monitoring equipment	See Appendix P, standard plot sampling equipment
QA/QC procedures to be applied	Annual monitoring of proxy area. See Appendix P for error correction procedures.
Purpose of the data	Calculation of baseline emissions
Calculation method	Equation F.16 from VM0009 V3.0
Comments	

Data / Parameter	$E_{BA}^{[m]}$
Data unit	tCO ₂ e
Description	Change in baseline emissions
Source of data	Proxy area measurements
Description of measurement methods and procedures to be applied	Proxy area plots monitoring as per Appendix P and equation F.15.
Frequency of monitoring/recording	Every monitoring period
Value monitored	2,267,514
Monitoring equipment	See Appendix P, standard plot sampling equipment
QA/QC procedures to be applied	Annual monitoring of proxy area. See Appendix P for error correction procedures.
Purpose of the data	Calculation of baseline emissions
Calculation method	Equation F.15 from VM0009 V3.0
Comments	

Data / Parameter	$E_{BABGB}^{[i]}$
Data unit	tCO ₂ e
Description	Change in baseline emissions from below-ground biomass

	during monitoring period i
Source of data	Monitoring the proxy area
Description of measurement methods and procedures to be applied	Proxy area plots monitoring as per Appendix P
Frequency of monitoring/recording	Every monitoring period
Value monitored	4,010,031
Monitoring equipment	See Appendix P, standard plot sampling equipment
QA/QC procedures to be applied	Annual monitoring of proxy area. See Appendix P for error correction procedures.
Purpose of the data	Calculation of baseline emissions
Calculation method	Calculation of above ground biomass using standard vegetation sampling techniques and application of an acceptable root: shoot ratio and application of equation F.10 of VM0009 V3.0, the decay emissions model
Comments	

Data / Parameter	$E_{BASOC}^{[m]}$
Data unit	tCO ₂ e
Description	Baseline change in emissions from soil carbon
Source of data	Measurement in the project area
Description of measurement methods and procedures to be applied	Soil sampling in project accounting area as per Appendix P
Frequency of monitoring/recording	At first verification but values updated at every monitoring period
Value monitored	232,817
Monitoring equipment	See Appendix P, standard plot sampling equipment

QA/QC procedures to be applied	Annual monitoring of proxy area. See Appendix P for error correction procedures.
Purpose of the data	Calculation of baseline emissions
Calculation method	Calculation of SOC using standard soil sampling techniques and application of equation F.7 and F.29 of VM0009 V3.0, the SOC decay emissions model
Comments	

Data / Parameter	$E_{BASOC}^{[i]}$
Data unit	tCO ₂ e
Description	Baseline emissions from soil carbon in monitoring period i
Source of data	Measurement in the project area
Description of measurement methods and procedures to be applied	Soil sampling in project accounting area as per Appendix P
Frequency of monitoring/recording	At first verification but values updated at every monitoring period
Value monitored	1,180,491
Monitoring equipment	See Appendix P, standard plot sampling equipment
QA/QC procedures to be applied	Annual monitoring of proxy area. See Appendix P for error correction procedures.
Purpose of the data	Calculation of baseline emissions
Calculation method	Calculation of SOC using standard soil sampling techniques and application of equation F.7 and F.29 of VM0009 V3.0, the SOC decay emissions model
Comments	

Data / Parameter	$E_{BBGB}^{[m]}$
Data unit	tCO ₂ e

Description	Cumulative baseline emissions from below-ground biomass at the end of the current monitoring period
Source of data	Measurement in the proxy area
Description of measurement methods and procedures to be applied	Standard vegetation sampling techniques to quantify above ground biomass, application of a root-shoot ratio and a decay emissions model.
Frequency of monitoring/recording	Every monitoring period
Value monitored	3,942,220
Monitoring equipment	See Appendix P, standard plot sampling equipment
QA/QC procedures to be applied	Annual monitoring of proxy area. See Appendix P for error correction procedures.
Purpose of the data	Calculation of baseline emissions
Calculation method	Calculation of above ground biomass using standard vegetation sampling techniques and application of an acceptable root to shoot ratio and application of equation F.10 of VM0009 V3.0, the decay emissions model
Comments	

Data / Parameter	$E_{B\ BGB}^{[m-1]}$
Data unit	tCO ₂ e
Description	Cumulative baseline emissions from below-ground biomass at the beginning of the current monitoring period
Source of data	Measurement in the proxy area
Description of measurement methods and procedures to be applied	Standard vegetation sampling techniques to quantify above ground biomass, application of a root-shoot ratio and a decay emissions model.
Frequency of monitoring/recording	Every monitoring period

Value monitored	3,203,934
Monitoring equipment	See Appendix P, standard plot sampling equipment
QA/QC procedures to be applied	Annual monitoring of proxy area. See Appendix P for error correction procedures.
Purpose of the data	Calculation of baseline emissions
Calculation method	Calculation of above ground biomass using standard vegetation sampling techniques and application of an acceptable root-to-shoot ratio and application of equation F.10 of VM0009 V3.0, the decay emissions model
Comments	

Data / Parameter	$E_{BBM}^{[m]}$
Data unit	tCO ₂ e
Description	Cumulative baseline emissions from biomass at the end of the current monitoring period
Source of data	Measurement in the proxy area
Description of measurement methods and procedures to be applied	See Appendix E
Frequency of monitoring/recording	Every monitoring period
Value monitored	11,368,203
Monitoring equipment	See Appendix P, standard plot sampling equipment
QA/QC procedures to be applied	Annual monitoring of proxy area. See Appendix P for error correction procedures.
Purpose of the data	Calculation of baseline emissions
Calculation method	Standard vegetation sampling techniques, quantification of deforestation in the reference area and application of equation F.4 and F.24, VM0009 V3.0
Comments	

Data / Parameter	$E_{BSOC}^{[m]}$
Data unit	tCO ₂ e
Description	Cumulative baseline emissions from soil carbon at the end of the current monitoring period
Source of data	Measurement in the project area
Description of measurement methods and procedures to be applied	Standard soil sampling techniques in the proxy area, laboratory analysis and application of the soil decay model.
Frequency of monitoring/recording	At first verification but values updated at every monitoring period
Value monitored	1,135,047
Monitoring equipment	See Appendix P, standard plot sampling equipment
QA/QC procedures to be applied	See Appendix P for error correction procedures.
Purpose of the data	Calculation of baseline emissions
Calculation method	Quantification of soil carbon and application of equation F.9 and F.29, VM0009, V3.0
Comments	

Data / Parameter	$E_{BSOC}^{[m-1]}$
Data unit	tCO ₂ e
Description	Cumulative baseline emissions from soil carbon at the beginning of the current monitoring period
Source of data	Measurement in the project area
Description of measurement methods and procedures to be applied	Standard soil sampling techniques in the proxy area, laboratory analysis and application of the soil decay model.
Frequency of monitoring/recording	At first verification but values updated at every monitoring period

Value monitored	947,674
Monitoring equipment	See Appendix P, standard plot sampling equipment
QA/QC procedures to be applied	See Appendix P for error correction procedures.
Purpose of the data	Calculation of baseline emissions
Calculation method	Quantification of soil carbon and application of equation F.9 and F.29, VM0009, V3.0
Comments	

Data / Parameter	$E_{BA}^{[m]}$
Data unit	tCO ₂ e
Description	Cumulative emissions allocated to the buffer account at the end of the current monitoring period
Source of data	VCS Non-Permanence Risk Report, See Appendix H
Description of measurement methods and procedures to be applied	See Appendix H – standard VCS procedure used.
Frequency of monitoring/recording	Non-permanence risk revised at every baseline re-assessment.
Value monitored	-1,164,534
Monitoring equipment	None
QA/QC procedures to be applied	Annual review of risk criteria and disturbance monitoring
Purpose of the data	Calculation of project emissions
Calculation method	Procedure followed in VCS non-permanence risk tool and emissions allocated to buffer account subtracted from GERs.
Comments	

Data / Parameter	$E_L^{[m]}$
Data unit	tCO2e
Description	Cumulative emissions from leakage at the end of the current monitoring period
Source of data	Measurements in the leakage area(s)
Description of measurement methods and procedures to be applied	Plot based sampling of leakage area and estimation of degradation.
Frequency of monitoring/recording	Every monitoring period
Value monitored	0
Monitoring equipment	None
QA/QC procedures to be applied	See Appendix R, standard vegetation sampling equipment
Purpose of the data	Calculation of leakage emissions
Calculation method	Application of equation F.44 and F.45 to field data collected.
Comments	See Appendix R, all evidence of tree-felling by humans are geo- referenced and plots photographed

Data / Parameter	$E_L^{[m-1]}$
Data unit	tCO2e
Description	Cumulative emissions from leakage at the beginning of the current monitoring period
Source of data	Measurements in the leakage area(s)
Description of measurement methods and procedures to be applied	Plot based sampling of leakage area and estimation of degradation.
Frequency of monitoring/recording	Every monitoring period
Value monitored	0

Monitoring equipment	None
QA/QC procedures to be applied	See Appendix R, standard vegetation sampling equipment
Purpose of the data	Calculation of leakage emissions
Calculation method	Application of equation F.44 and F.45 to field data collected.
Comments	See Appendix R, all evidence of tree-felling by humans are geo- referenced and plots photographed

Data / Parameter	$E_{LA}^{[m]}$
Data unit	tCO ₂ e
Description	Change in emissions due to leakage
Source of data	Measurements in the leakage area(s)
Description of measurement methods and procedures to be applied	Plot based sampling of leakage area and estimation of degradation.
Frequency of monitoring/recording	Every monitoring period
Value monitored	0
Monitoring equipment	None
QA/QC procedures to be applied	See Appendix R, standard vegetation sampling equipment
Purpose of the data	Calculation of leakage emissions
Calculation method	Application of equation F.44 and F.45 to field data collected.
Comments	See Appendix R, all evidence of tree-felling by humans are geo- referenced and plots photographed

Data / Parameter	$E_{LASF}^{[m]}$
Data unit	tCO ₂ e
Description	Cumulative emissions from activity-shifting leakage in forested strata at the end of the current monitoring period

Source of data	Measurements in the leakage area(s)
Description of measurement methods and procedures to be applied	Plot based sampling of leakage area and estimation of degradation.
Frequency of monitoring/recording	Every monitoring period
Value monitored	0
Monitoring equipment	None
QA/QC procedures to be applied	See Appendix R, standard vegetation sampling equipment
Purpose of the data	Calculation of leakage emissions
Calculation method	Application of equation F.44 and F.45 to field data collected.
Comments	See Appendix R, all evidence of tree-felling by humans are geo- referenced and plots photographed

Data / Parameter	$E_{LME}^{[m]}$
Data unit	tCO ₂ e
Description	Cumulative emissions from market leakage at the end of the current monitoring period
Source of data	N/A
Description of measurement methods and procedures to be applied	Land-use by deforestation agents is subsistence and therefore all market leakage is captured by monitoring in activity shifting leakage areas.
Frequency of monitoring/recording	N/A
Value monitored	0
Monitoring equipment	None
QA/QC procedures to be applied	N/A

Purpose of the data	Calculation of leakage emissions
Calculation method	N/A
Comments	

Data / Parameter	$E_{P\Delta}^{[m]}$
Data unit	tCO ₂ e
Description	Change in project emissions
Source of data	Monitoring records for Forest Fire, Burning, logging, wood products, and natural disturbance events
Description of measurement methods and procedures to be applied	Standard vegetation sampling techniques and remote sensing to determine the extent of fires and other disturbances.
Frequency of monitoring/recording	Every monitoring period
Value monitored	-104,495
Monitoring equipment	Standard vegetation sampling equipment and satellite imagery.
QA/QC procedures to be applied	Remote sensing, patrols and field measurements
Purpose of the data	Calculation of project emissions
Calculation method	Monitoring data from areas of Forest Fire, Burning, logging, wood products, and natural disturbance events and equation F.41 VM0009 V3.0.
Comments	

Data / Parameter	$E_{P\Delta BRN}^{[m]}$
Data unit	tCO ₂ e
Description	Cumulative project emissions due to burning at the end of the current monitoring period
Source of data	Monitoring plots in the project accounting area

Description of measurement methods and procedures to be applied	Application of equation F.42 using burning monitoring data as input.
Frequency of monitoring/recording	Every monitoring period
Value monitored	0
Monitoring equipment	Standard vegetation sampling equipment
QA/QC procedures to be applied	Review of monitoring records
Purpose of the data	Calculation of project emissions
Calculation method	Monitoring data from areas of Burning and equation F.42 VM0009 V3.0.
Comments	Biomass burning in the project accounting area is not a project activity

Data / Parameter	$E_U^{[m]}$
Data unit	tCO ₂ e
Description	Cumulative confidence deduction at the end of the current monitoring period
Source of data	N/A
Description of measurement methods and procedures to be applied	Application of equation F.57 which is a linear combination of weighted standard errors of estimates from baseline emission models and carbon stock measurements.
Frequency of monitoring/recording	Every monitoring period
Value monitored	0
Monitoring equipment	N/A
QA/QC procedures to be applied	Review of calculations
Purpose of the data	Calculation of project emissions

Calculation method	Equation F.57 from VM0009 V3.0
Comments	

Data / Parameter	$P_{LDEG}^{[m]}$
Data unit	Proportion (unitless)
Description	Portion of leakage due to degradation at the end of the current monitoring period
Source of data	Monitoring in the leakage area
Description of measurement methods and procedures to be applied	Plot based sampling of leakage area and estimation of degradation.
Frequency of monitoring/recording	Every monitoring period
Value monitored	0.08
Monitoring equipment	See Appendix R, standard vegetation sampling equipment
QA/QC procedures to be applied	See Appendix R, all evidence of tree-felling by humans are geo-referenced and plots photographed
Purpose of the data	Calculation of leakage
Calculation method	Calculated using equation B.9, VM0009 V3.0. Minimum sample size determined using equation B.31, VM0009 V3.0
Comments	

Data / Parameter	$P_{LDEG}^{[m=0]}$
Data unit	Proportion (unitless)
Description	Portion of leakage due to degradation prior to first verification event
Source of data	Monitoring in the leakage area
Description of measurement methods and procedures to be applied	Plot based sampling of leakage area and estimation of degradation.

applied	
Frequency of monitoring/recording	First monitoring period
Value monitored	0.0815
Monitoring equipment	See Appendix R, standard vegetation sampling equipment
QA/QC procedures to be applied	See Appendix R, all evidence of tree-felling by humans are geo-referenced and plots photographed
Purpose of the data	Calculation of leakage
Calculation method	Calculated using equation B.9, VM0009 V3.0. Minimum sample size determined using equation B.31, VM0009 V3.0
Comments	

Data / Parameter	t
Data unit	Days
Description	Time since project start date
Source of data	Calendar/Monitoring records
Description of measurement methods and procedures to be applied	Number of days since project start (Project start date is May 1st, 2015 and the current monitoring period ended December 31st, 2022.)
Frequency of monitoring/recording	Every Monitoring period
Value Monitored	2,800
Monitoring equipment	N/A
QA/QC procedures to be applied	N/A
Purpose of Data	Calculation of baseline emissions
Calculation method	Deduct project start date from end of date of current monitoring
Comments	

Data / Parameter	$t^{[i-1]}$
Data unit	Days

Description	Time from the project start date to beginning of monitoring period i
Source of data	Monitoring records
Description of measurement methods and procedures to be applied	NA
Frequency of monitoring/recording	Prior monitoring period
Value monitored	0
Monitoring equipment	NA
QA/QC procedures to be applied	NA
Purpose of the data	Calculation of baseline emissions
Calculation method	Count of days in project crediting period
Comments	

Data / Parameter	$t^{[m]}$
Data unit	Days
Description	Time from project start date to end of current monitoring period
Source of data	Monitoring records
Description of measurement methods and procedures to be applied	NA
Frequency of monitoring/recording	Every monitoring period
Value monitored	PAI1 = 2,800 ; PAI2 = 507 ; PAI3 = 364
Monitoring equipment	NA
QA/QC procedures to be applied	NA

Purpose of the data	Calculation of baseline emissions
Calculation method	Deduct project start date from end of date of current monitoring
Comments	

Data / Parameter	$t^{[m-1]}$
Data unit	Days
Description	Time from project start date to beginning of current monitoring period
Source of data	Monitoring records
Description of measurement methods and procedures to be applied	NA
Frequency of monitoring/recording	Prior monitoring period
Value monitored	PAI1= 2,436 ; PAI2 = 143 ; PAI3 = 0
Monitoring equipment	NA
QA/QC procedures to be applied	NA
Purpose of the data	Calculation of baseline emissions
Calculation method	Deduct project start date from end of date of previous monitoring
Comments	

Data / Parameter	$U_B^{[m]}$
Data unit	tCO2e
Description	Total uncertainty in proxy area carbon stock estimate
Source of data	Biomass and soil carbon inventory data, Appendix G
Description of measurement	Calculation of the standard error of the total of the baseline emissions model, proxy area carbon stocks and accounting

methods and procedures to be applied	area carbon stocks
Frequency of monitoring/recording	Every monitoring period
Value monitored	15,727
Monitoring equipment	NA
QA/QC procedures to be applied	Review of monitoring records
Purpose of the data	Calculation of baseline emissions
Calculation method	Equation B.10 VM0009 v3.0
Comments	

Data / Parameter	$U_{EM}^{[M]}$
Data unit	tCO ₂ e
Description	Total uncertainty in Baseline Emissions Models
Source of data	Forest State Interpretation, Appendix O
Description of measurement methods and procedures to be applied	Calculation of the standard error of the total of the baseline emissions model, proxy area carbon stocks and accounting area carbon stocks
Frequency of monitoring/recording	Every monitoring period
Value monitored	14,616
Monitoring equipment	NA
QA/QC procedures to be applied	Review of monitoring records
Purpose of the data	Calculation of baseline emissions
Calculation method	Equation B.14 VM0009 v3.0
Comments	

Data / Parameter	$U_P^{[m]}$
Data unit	tCO ₂ e
Description	Total uncertainty in project accounting area carbon stock estimate
Source of data	Biomass and soil carbon inventory data, Appendix G
Description of measurement methods and procedures to be applied	Calculation of the standard error of the total of the baseline emissions model, proxy area carbon stocks and accounting area carbon stocks
Frequency of monitoring/recording	Every monitoring period
Value monitored	25,071,680
Monitoring equipment	NA
QA/QC procedures to be applied	Review of monitoring records
Purpose of the data	Calculation of baseline emissions
Calculation method	Equation B.10 VM0009 v3.0
Comments	

4.3 Monitoring Plan

Table 26. Summary of monitoring activities

Monitoring Activity	Timing	Methods
Trespassing and boundary transgressions	Monthly patrols backed up with aerial surveys at least once per dry season.	Scouts patrol the project area on an ongoing basis, it is their core duty
Leakage monitoring	Annually	Forest Inventory team following VM0009 and re-measuring 100% of leakage plots each year as per protocol in Appendix R.
Plot measurements	Annually	Forest Inventory Team

		following VM0009, re-measuring 20% of plots in project accounting and project proxy areas as per protocol in Appendix P. Since all the plots (100%) were measured a year prior to project validation (i.e. 2018), the 20% per year rotation started in 2019. An additional 20% were measured in 2020 and 2021. Therefore, a 100% rotation will be completed in 2023.
Disturbance monitoring	One a year using satellite imagery from April-May	Remote sensing using Sentinel-2 or Landsat imagery. Followed up with aerial surveys and, or scout patrols.

PDR.122

The principal means of monitoring project emissions is with inventory plots in project accounting areas where above-ground biomass is measured by recording tree dbh. Allometric equations are used to convert the measures to above and below-ground biomass. Soil samples are also collected during the first monitoring period to establish the soil carbon stocks for the project accounting area. VM0009 requires that all such plots must be measured during the first monitoring period and then 20% measured during each subsequent monitoring period. After mapping the project accounting area described in section 3.2.1 of the project document, the project used GIS software to generate 122 plots randomly within the project accounting area polygon. One of these plots became encroached, thus biomass monitoring plots used for biomass stocks accounting in the forested area for M1 to M5 monitoring periods are 121 (see **MR.33**). The sample size was estimated based on the variance in biomass inventory plot data from BCP's other REDD project, the Lower Zambezi REDD project.

The incorporation of the second project activity instance in M4 necessitated the inclusion of additional biomass plots in the new areas in the current monitoring period(M5). Employing the plot density from the precedent project activity, instance 1, 13 additional biomass plots were randomly allocated to the newly implemented instance 2 areas. Consequently, the project's total number of biomass plots has increased to 134. 100% of the new project activity instance two(2) biomass plots were monitored in the current monitoring period(M5) and

project activity instance three(3) biomass plots will be installed and monitored in the next monitoring period(M6).It is worth noting that both project activity Instance two(2) plot data monitored in the current monitoring period(M5) and instance 3 plot biomass data to be monitored in the next monitoring period (M6), will only be incorporated in the project biomass stocks accounting in the next monitoring period (M6).This is because based on the project's monitoring plan as per the PD, the carbon stock dataset is updated every after 5 years when the annual 20% biomass plot monitoring comes to a full cycle. The next carbon stock dataset update is scheduled for 2023 (M6) as the 1st 20% biomass plot monitoring rotation was conducted in 2019.

MR.93

Plot monitoring is conducted in plot locations shown in figure 18, according to the BCP field sampling manual (Appendix P). Optimum plot size and minimum tree diameter were determined under BCP's other project, the Lower Zambezi REDD Project. Circular plots (12m radius) are used due to their simplicity, ease of layout and reduced measurement error. A randomly generated plot location is navigated-to using a handheld GPS or Android Smartphone. When the plot location is reached, the center is permanently marked with a steel rebar driven into the ground.

Trees are identified to species level as far as possible. The use of a generic allometric equation to calculate AGOT mass fortunately negated mistakes or uncertainties with tree species identification.

As explained above, plots are re-measured on a rotating basis, with 20% re-measured per year, however stock estimates are only updated every 5 years after all plots have been re-measured.

In addition to the inventory plots, human and natural disturbances in the project accounting area are mapped using visual interpretation of Sentinel-2 and if needed Landsat data. This is done once a year using the most cloud-free imagery from April-May. This represents the end of the rainy season and the beginning of the dry season. This is generally the best time of the year for mapping woodlands in Zambia as there is less cloud cover, trees generally still have leaves, and the understory grass has started to dry out providing better contrast between grassy and wooded areas. The woodlands in the project area are fire-adapted and fires generally don't cause a massive disturbance. April-May imagery helps avoid mistakenly mapping fire scars as a disturbance when the fire likely only burned the understory grass. With April-May imagery, only permanently damaging fires will be mapped as a disturbance.

When significant disturbances are detected in an area not covered by an inventory plot, a polygon will be drawn around the area and the per hectare emissions factors determined from the plot inventory work will be applied to the area to calculate project emissions.

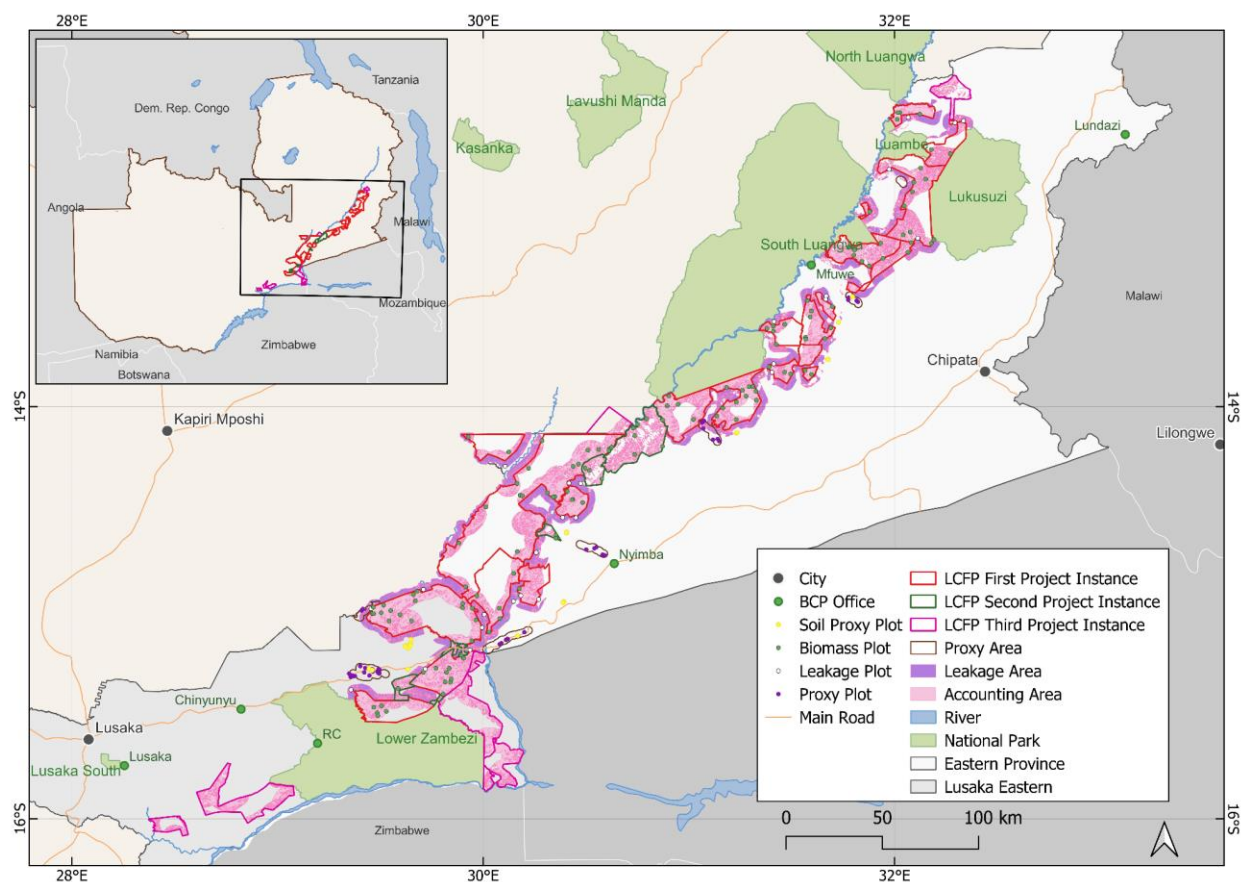


Figure 17. Accounting, Proxy, and Leakage Plots (MR.91)

PDR.123

Proxy plots are used to estimate residual biomass after conversion and the end point for soil carbon decay. For estimating residual biomass, after mapping the proxy areas as described in section 3.2.1 of the project document, the project generated a random sample of 45 points within the proxy areas. Variability in proxy plots is high since some farmers leave large isolated trees in their farms and invariability, some proxy plots will include these trees. Based on experience from the previous LZRP REDD project, BCP settled on a sample size of just 45 points (Table 16). This is because while the variability with proxy plots is high relative to the proxy plot average, that variability is low relative to the accounting area carbon stocks and thus represents a relatively small portion of the overall uncertainty in baseline emissions.

The same biomass and soil sampling protocols used for accounting areas (Appendix P) were used for proxy areas as well.

Proxy soil measurements were conservatively excluded in M1 and M2. From M3 onwards, proxy soil sampling was conducted in areas that represented the end point of conversion and the data was analyzed for SOC accounting. See Appendix C for details.

PDR.124

Leakage plots in the leakage area are used to estimate activity-shifting leakage from the project accounting areas. The project generated a random sample of 25 points within the leakage area where 1 ha leakage plots could be established. Equation B.31 as per VM0009, suggested the minimum number of plots required was 22.

The leakage plot protocol is described in Appendix R.

MR.92

GPS coordinates for all of the accounting area, proxy, and leakage plots can be found in Appendix I.

MR.98

All accounting area, proxy, and leakage plots were measured during monitoring period 1. The project is verified on an annual basis, therefore, measures all leakage plots each year and 20% of the accounting area and proxy plots each year.

Organizational Structure, responsibilities, and competencies

BioCarbon Partners has a team of experts responsible for carbon stock monitoring and standards compliance. As per VM0009, carbon stock monitoring is done using a combination of permanent plots in the project accounting area, proxy area, and leakage area, and remote sensing. Scout teams are used to confirm encroachment and deforestation in the project area as part of the community engagement process to resolve encroachment (Figure 18). The carbon accounting department oversees all forest-related data collection, processing, storage, GIS and remote sensing, and electronic data capture systems with a central database (Figure 18).

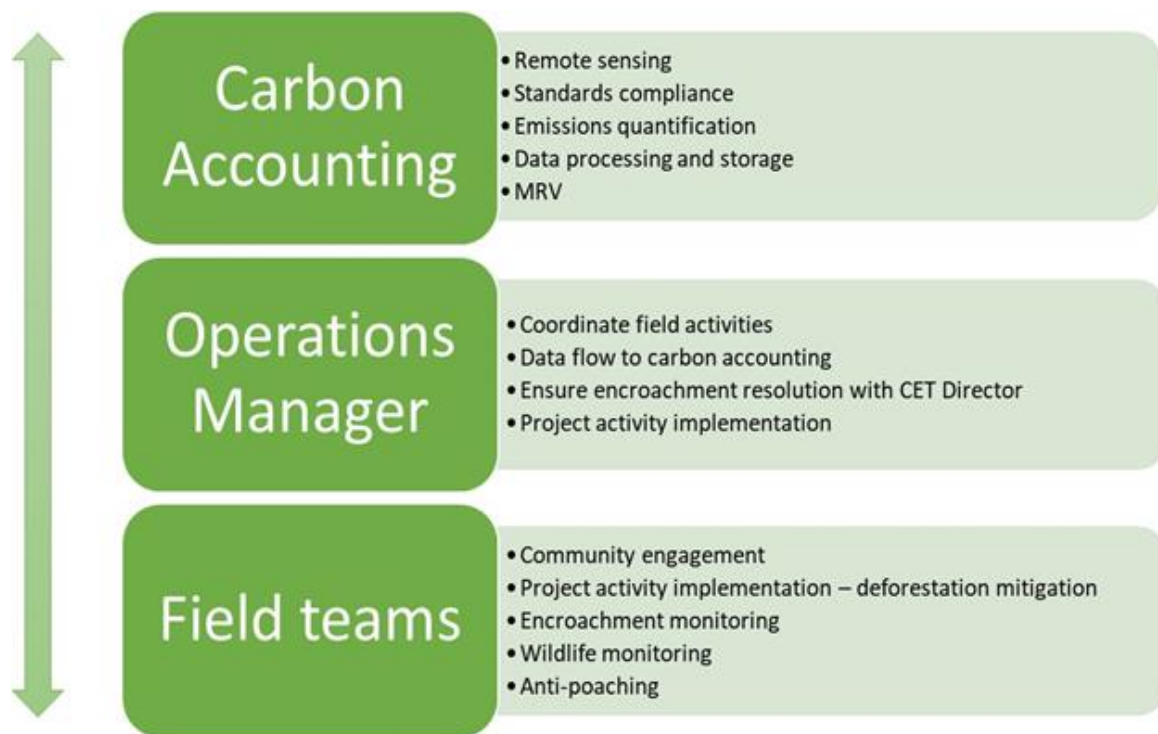


Figure 18. BCP Carbon Monitoring Process.

Both the carbon accounting manager and remote sensing and GIS specialist have post graduate degrees with a strong focus on African ecology and collectively have more than 20 years' experience in biological inventories, remote sensing, and GIS.

Data Management

The process followed for field data collection is outlined in Figure 19. It is a feedback system that relies heavily on training, review and good communication. All training is followed by trial runs of the sampling procedure and then actual data collection under supervision. Data is continuously submitted from the field to the carbon accounting department, and feedback is provided if any corrective actions are required. Remote Sensing data does not have a field component, but all data is also stored on Google Drive. Once data has been error checked and cleaned it is safely stored online using Google Drive and then transferred to an online relational database.

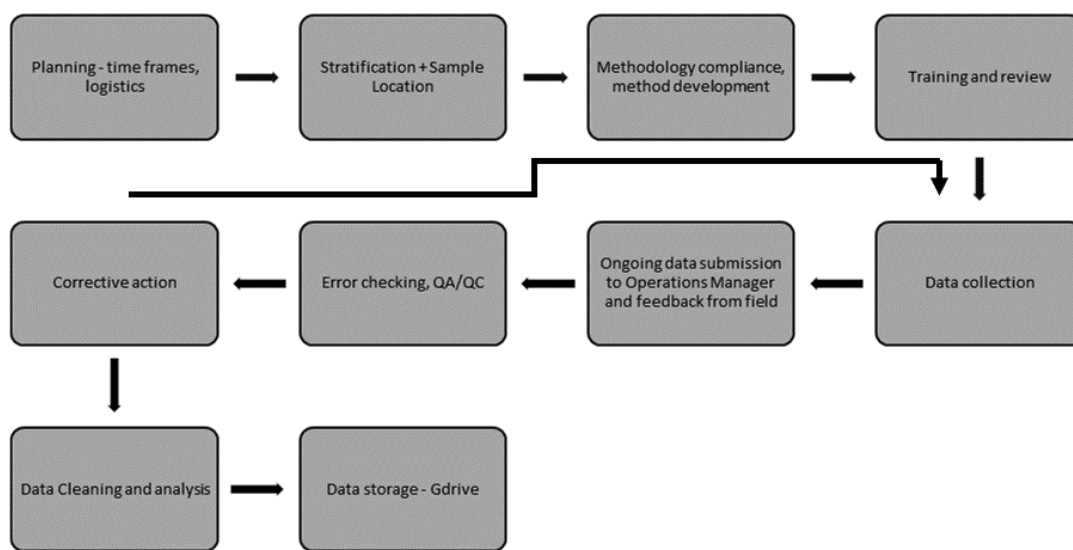


Figure 19. BioCarbon Partners data collection, QA/QC and storage process.

MR.90

Quality Assurance and Quality Control

All data submitted to the central database is subjected to quality control by appropriate staff. In 2022, the Carbon Department was composed of the Senior Carbon Manager, a Carbon Accountant, a Senior Carbon Specialist, a Carbon Specialist, a GIS technician and a Carbon Specialist/Biodiversity Manager. Together, the team has over 50 years of cumulative GIS & Remote Sensing experience and contributed to 13 successful verification of BCP REDD+ projects. The majority of field data is collected using digital tools on smartphones and are either directly loaded to the database or saved on Google Drive where it is checked and then transferred to the database.

A set of Standard Operating Procedures and Field Manuals have been developed for all the various aspects of the carbon inventory. As indicated above, BCP has strong systems in place to ensure all carbon-related data is collected according to the required quality standards. Quality Control is embedded in the organizational structure of the Carbon Accounting department (Figure 19).

Continuous feedback from the field, coupled with field visits from senior staff, ensure that errors are identified in a timely fashion and corrected (Figure 19). Spot checks and internal audits are conducted to make sure all data collected is of the highest standard. The new digital data capture system furthermore ensures that errors are identified and rectified immediately, unlike paper-based systems where data often takes weeks to enter and error check. Spot errors are dealt with as per Appendix P.

With regards to GIS and RS, similar procedures are followed although this role resides mainly within the senior management's responsibility.

MR.88

Three forest monitoring teams were used to collect the biomass and SOC data. Each team was led by an experienced member of what was once a single team formed for the assessment of biomass and SOC data in the Lower Zambezi Redd+ Project. This initial team was trained over a period of 12 days in February and September of 2012. They received a refresher 2-day training in December of 2012, June of 2014, and May of 2016. This core team has been collecting data annually for the other REDD project since 2012 and are thus highly experienced.

In order to speed up the carbon inventory work, a second forest monitoring team was formed. Each of the new teams was led by a highly experienced member of team 1 and all new team members received training before starting field work. At the beginning of and half-way through each year, refresher training is conducted with the forest monitoring teams.

MR.99

The project uses two allometric equations to estimate above-ground biomass. For the majority of tree stems, a generic miombo equation developed by Chidumayo (2013) at a study site very close to the project area was used:

$$\text{Wood Biomass (kg)} = 0.0446 * \text{DBH}^{2.765}$$

It was based on a destructive sampling of 113 trees of 19 different species (all of which are found in the project accounting area) with a DBH range of 2 to 39 cm.

The other equation used by the project is another general miombo equation developed by Mugasha et al, 2013 (see Appendix P for publication copy):

$$\text{Wood Biomass (kg)} = 0.1027 * \text{DBH}^{2.4798}$$

It was based on destructive sampling in 4 different sites in Tanzania of 167 sample trees of 60 tree species with a DBH range from 1.1 to 110 cm.

MR.100

Having been developed from sample trees in Zambia near the project area, the Chidumayo equation was the more local of the two equations. It also gave higher biomass estimates for all DBHs than the Mugasha equation. Therefore, it was the preferred equation. Unfortunately, it was not validated for trees above 55 cm. Therefore, it was applied to all trees with a DBH of 55 cm or less. It was also applied to trees less than 63 cm DBH, but the DBH of these trees was conservatively changed to be 55 cm since the equation had not been tested on

trees larger than 55 cm DBH. This deviation is explained in more detail in section 2.2. The equation was used to estimate 80% of the project area carbon stocks.

For trees between 63 and 110 cm DBH, the Mugasha equation, which was validated for trees up to 110 cm DBH was applied. It was also applied to 3 trees with diameters greater than 110 cm DBH, by changing the DBH of these trees to 110 cm. This deviation is explained in more detail in section 2.2. The equation was used to estimate 20% of the project area carbon stocks.

MR.101

Both equations were used in the same way as for the first monitoring period.

MR.102

As per VM0009, there was no need to validate the Chidumayo equation with destructive harvesting because the equation came from existing published literature and was developed for miombo forests very similar to the project area. However, BCP did conduct some relevant destructive harvesting as part of developing its first REDD+ project in Lusaka province. Additionally, BCP has the destructive harvesting data from Chidumayo's study of miombo biomass allometry in Lusaka province. So, these two datasets were used to validate the Mugasha et al. equation, which comes from further away.

Also, the samples used to develop both equations did not include trees as large as the largest trees in the inventory data and the equations do not pass the first-order derivative test required by VM0009. Therefore, as described in **MR.100** when the equations were applied to a tree with a DBH greater than the range used to develop the equation, the DBH was changed to match the maximum DBH for which the equation was validated. This deviation is described in more detail in section 2.2.

MR.103

Chidumayo (2013) developed the equation through destructive harvesting of study plots located between latitudes 15.2167° and 15.2833°S and longitudes 29.1500° and 29.2333°E, which was miombo woodland in Lusaka province. The sample consisted of 113 trees spanning 19 species, all of which occur in the project and proxy areas.

The Mugasha equation was also developed for miombo from 4 different sites in Tanzania. Out of 156 trees identified at the species level in the Mugasha et al. Dataset, 87 (56%) were species found in the accounting area biomass plots for this project. Out of 165 sample trees identified at the genus level, 141 (85%) were from a genus found in the accounting area inventory plots. Thus, there was a large number of species and genus overlap between the project accounting area and the Mugasha et al. 2013 sample trees, showing that miombo in

Tanzania is similar. Additionally, the equation appears to be conservative relative to the Chidumayo equation and predicts less biomass for any DBH than the Chidumayo equation.

MR.104

The Chidumayo equation did not require validation with destructive harvesting as it was developed from miombo sites in the centre of Lusaka province within 60 km of the project area. Thus, the destructive harvesting data from the Chidumayo study and 7 additional trees destructively harvested by BCP were used to validate the Mugasha equation.

MR.105

Chidumayo (2013) developed the equation through destructive harvesting of study plots located between latitudes 15.2167° and 15.2833°S and longitudes 29.1500° and 29.2333°E, which was miombo woodland in Lusaka province within 60 km of the project accounting area. The diameter at breast height (DBH) range for the Chidumayo model was 2-39 cm and it was developed from a sample size of 113 destructively sampled trees covering 19 species, including most of the common species in the project accounting area. BCP then destructively harvested 7 more trees to bring the size range up to 55 cm DBH.

BCP has the original data from the author and has made it available to the VVB (Appendix G). As part of the work to establish its first REDD+ project, BCP destructively sampled another seven trees of various species to extend the diameter range of the Chidumayo equation to 55 cm (Table 27). The procedures followed for destructive harvesting are described in Appendix P and was developed under the guidance of Professor Chidumayo.

Table 27. Results of allometry validation and diameter range expansion.

Species	DBH (cm)	Total dry mass (kg)	Mass Predicted by equation (kg) (Chidumayo 2013)	Latitude	Longitude
<i>Brachystegia boehmii</i>	49	2750	2143	-15.3672	29.08735
<i>Brachystegia boehmii</i>	48	2838	1957	-15.3654	29.08534
<i>Brachystegia boehmii</i>	42	1236	1346	-15.3627	29.08307
<i>Brachystegia spiciformis</i>	41	1141	1318	-15.3678	29.08696
<i>Brachystegia utilis</i>	43	2212	1462	-15.3627	29.08266

<i>Brachystegia utilis</i>	48	2006	2030	-15.3635	29.08613
<i>Parinari curatellifolia</i>	55	3823	2950	-15.3654	29.08537
Total Mass		16006	13206		

MR.106

The field protocol is available in Appendix P.

MR.107

The field protocol (Appendix P) was based on the methods used by Chidumayo (2013) to develop the original equation. It is a methodical and robust approach that is conservative. The same wet-dry mass ratio as used by Chidumayo (2013) was applied.

MR.108

As shown in Appendix G, the Chidumayo equation predicts 10% less biomass than the cumulative measured biomass of the destructively sampled trees. Thus, it is acceptable under VM0009. The Mugasha equation is also acceptable under this condition, predicting 25% less biomass than the cumulative measured biomass of the destructively sampled trees (Appendix G).

5 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

5.1 Baseline Emissions

MR.95

Table 28. Estimated carbon stock, standard error of the total for each stock, and the sample size for each stratum in the area selected.

Parameter	Project Accounting Area		Proxy Area	
	Forest (AGOT + BGOT)	Soil	Proxy Area (AGOT + BGOT)	Proxy Area Soil*
Plots (P)	121	25	45	25
Total Possible plots ($N_{p,k}$)	57,953,112	57,953,112	339,531	24,032
C	154,861,739	154,938,433	59,760	57,251
U (B.10) per stratum	13,315,735	21,243,360	15,411	3,138

MR.10

Baseline emissions ($E_{(BA)}^{[m]}$) for the current monitoring period (January 1st, 2022 to December 31st, 2022) are summarized in Table 29. Equation F.15 was used to calculate these baseline emissions. Cumulative baseline emissions were calculated using a logistic deforestation model, based upon a point interpretation of forest state in the reference area applying equation F.16 from VM0009 v3.0.

Table 29. Baseline emissions summary for the current monitoring period (see Appendix G for calculations).

Monitoring Period	Carbon Pool	Baseline Emissions (tCO ₂ e)
M5	AGOT	1,448,783
M5	BGOT	738,265
M5	SOC	80,466
Total		2,267,514

MR.11

Total Cumulative baseline emissions from prior monitoring periods were 9,432,108

tCO₂e. Refer to Appendix G for calculations.

MR.12

Cumulative baseline emissions for each selected pool ($E_{BBM}^{[m]}$ and $E_{BSOC}^{[m]}$) and undecayed carbon ($C_{BBGB}^{[m]}$, $C_{BSOC}^{[m]}$) as of the current monitoring period are summarized below.

See Appendix G for calculations.

$$E_{BBM}^{[m]} = 11,368,203$$

tCO₂e

$$E_{BSOC}^{[m]} = 1,180,491$$

tCO₂e

$$C_{BBGB}^{[m]} = -67,811$$

tCO₂e

$$C_{BSOC}^{[m]} = -849,070$$

tCO₂e

MR.15-17

Baseline emissions from biomass for the current monitoring period are based on the average carbon stock in selected carbon pools from AGOT, BGOT given by equation F.17 for the accounting area and F.18 for the proxy area. There is only one stratum with a carbon stock of 236.2 tCO₂e. The proxy area (baseline scenario) carbon stocks are 15.6 tCO₂e.

Cumulative baseline emissions from biomass ($E_{BBM}^{[m]}$) for the current monitoring period were calculated by equation F.24 from VM0009 V3.0. See Appendix G for calculations.

11,368,203 tCO₂e

Current baseline emissions from SOC ($E_{(BASOC)}^{[m]}$) as of the current monitoring period were 232,817

tCO₂e, calculated with equation F.29. See Appendix G for calculations.

Cumulative baseline emissions from biomass ($E_{BBM}^{[m]}$) for all prior monitoring periods were 9,181,154

tCO₂e

See Appendix G for calculations.

MR.26

Carbon stored in non-decayed BGB ($C_{BGB}^{[m]}$) for the current monitoring period was calculated using equation F.32. See Appendix G for calculations.

-67,841 tCO₂e

MR.27

Cumulative baseline emissions from BGB for the current monitoring period ($E_{BGB}^{[m]}$) were calculated using equation F.30. See Appendix G for calculations.

3,942,220

tCO₂e

MR.28

Cumulative baseline emissions from BGB ($E_{BGB}^{[m]}$) for all prior monitoring periods were calculated using equation F.30. See Appendix G for calculations.

3,203,954

tCO₂e

MR.29

Carbon stored in non-decayed SOC for the current monitoring period ($C_{SOC}^{[m]}$) was calculated using equation F.33. See Appendix G for calculations.

-849,070

tCO₂e

5.2 Project Emissions

MR.32

Major disturbances within the project area, typically associated with clearing for cropland preparation, were manually digitized from cloud-free Sentinel-2 images obtained between April and May 2022 (as per the validated PD) for comparative analysis with the previous year's assessment conducted during the same period. Figures 20, 21a, 21b and 22 display maps delineating the boundaries of these areas. The imagery was sourced from Earth

Engine and exported to Google Cloud as map tiles (script provided to VVB). Both Sentinel-2 mosaics from 2021 and 2022 were imported into QGIS and compared to each other to detect changes in the forest cover. Subsequently, any identified forest disturbances were manually delineated as polygons.

To ensure the accuracy of the forest disturbance analysis process, the project implemented a three-tier verification approach. This involved two analysts and one verifier, consisting of a GIS Technician, a Carbon Specialist, and a Carbon Manager, respectively. All three possess extensive expertise in visual image interpretation, enabling them to accurately identify forest loss. Following the BCP annual deforestation Standard Operating Procedure (SOP) detailed in Appendix E, the first analyst (GIS technician) applied Sentinel-2 imagery to identify and delineate converted areas. Subsequently, the second analyst (Carbon Specialist) conducted a comprehensive assessment of the dataset generated by the first analyst, employing the same Sentinel-2 imagery (10m resolution). Finally, both analysts discussed their findings at the completion of the mapping process in order to reconcile any discrepancies. The resulting final dataset revealed a forest cover loss of 425.55 hectares in the current monitoring period (see table 30 for the breakdown of forest disturbed areas stratified by chiefdom).

Table 30. Results of disturbance mapping in the project area

Area Name	Clearing in 2022 (ha)
Chitungulu	0.77
Jumbe	102.34
Mnkhanya	63.19
Mpanshya	4.28
Msoro	106.24
Mwanya	140.32
Sandwe	5.57
Luembe	2.84
Total	425.55

To further enhance the accuracy assessment in the current monitoring period, the project incorporated Planet-NICFI imagery with a spatial resolution of 4.77m. The verifier (Carbon

Manager) conducted an evaluation of the manually digitized polygons (Figures 20, 21, 21 and 22) by comparing the final dataset (10-meter resolution) reported in this monitoring report with a reference dataset generated by the verifier using Planet NICFI imagery (4.77-meter resolution). The outcomes obtained by the verifier were applied to evaluate the accuracy of the deforestation mapping (see Table 30). The sampling design encompassed a comprehensive examination of 100% of the digitized area, as well as the remainder of the project area, to detect any omission (as per Olofsson et al., 2020)¹² or commission errors (as per Ming Lu et al., 2017)¹³. An agreement of equal to or greater than 95% between the two datasets is considered acceptable. The accuracy assessment yielded the following results:

- Total area of final reported dataset (Sentinel-2 imagery) - 425.50 ha
- Total area of reference dataset (Planet-NICFI imagery) - 418.11 ha
- Total Area covered: 430.83 ha
- Commission error: 5.48 ha
- Omission error : 12.72 ha
- Overall agreement(Intersection): 412.63 ha
- Overall accuracy: 95.78 %

Overall accuracy: 95.78 %. The results of the accuracy assessment demonstrated acceptable agreement between the two datasets, and therefore, the reported deforested areas are considered to be accurate. Both the reported final dataset and the reference dataset are provided in Appendix I

MR.33

The disturbed stratum from the M1 to M3 periods represented just 0.63 % of the project accounting area. By random chance, one of the original permanent sample plots fell into the disturbed area and was measured after disturbance. Therefore, in order to have sufficient plots to calculate variance, one more random plot location was selected inside the disturbed area and measured as per Appendix P.

Current emissions from changes in project carbon stocks are calculated with equation F.41 VM0009 v3.0. After restratification (Table 31), the project accounting area stocks at the end of this monitoring period were equal to 309,800,173 tCO₂e (Appendix G) compared to

¹² Olofsson, P., Arévalo, P., Espejo, A.B., Green, C., Lindquist, E., McRoberts, R.E., Sanz, M.J., 557 2020. Mitigating the effects of omission errors on area and area change estimates. *Remote Sensing of Environment* 236, 111492. doi:10.1016/j.rse.2019.111492

¹³ Lu, Ming, Bin Chen, Xiaohan Liao, Tianxiang Yue, Huanyin Yue, Shengming Ren, Xiaowen Li, Zhen Nie, and Bing Xu. 2017. "Forest Types Classification Based on Multi-Source Data Fusion" *Remote Sensing* 9, no. 11: 1153. <https://doi.org/10.3390/rs9111153>

251,989,444 tCO₂e^{14*} at the end of the previous monitoring period (M4). Removing the project carbon stocks for instances added within this monitoring period, and not yet monitored for deforestation, the project area stocks were 251,884,949 tCO₂e. The project emissions for this monitoring period were -104,495 tCO₂e.

Table 31. Project accounting area stocks at the end of the current monitoring period

Parameter	Project Accounting Area			
	Forest Stratum (AGOT + BGOT)	Forest Stratum (Soil)	Disturbed Stratum (AGOT + BGOT)	Disturbed Stratum (Soil)
Plots (P)	121	25	2	25
Total Possible plots (N _{p,k})	57,953,112	57,953,112	322,443	322,443
C	154,806,249	125,272,283	55,491	768,141
U (B.10) per stratum	13,315,619	21,243,318	55,490	42,117

¹⁴ Value is different from that reported in M4 Monitoring report because there was an error in the reported project emissions for M4. Project emissions for M4 were under-estimated by 59 tCO₂e.

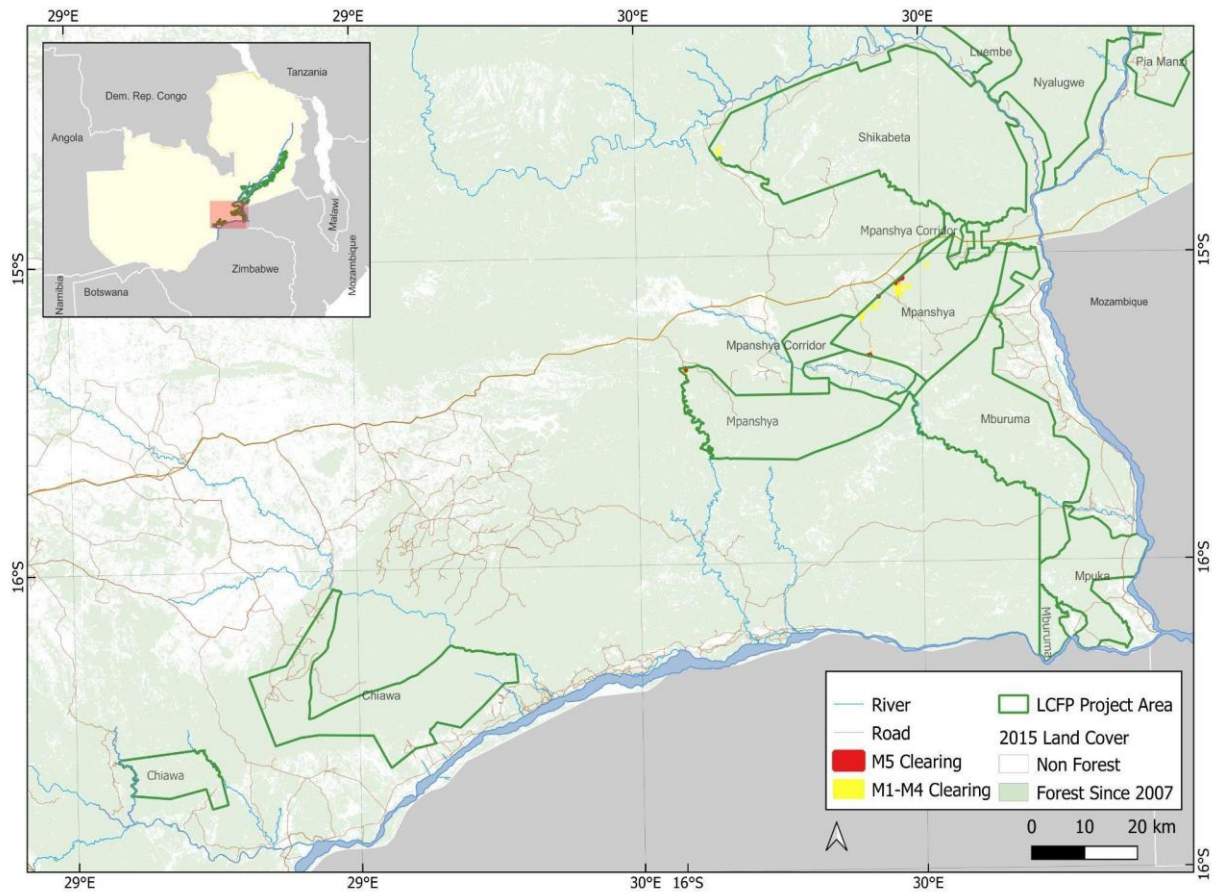


Figure 20. Project accounting area clearings in the Southern part of the project area.

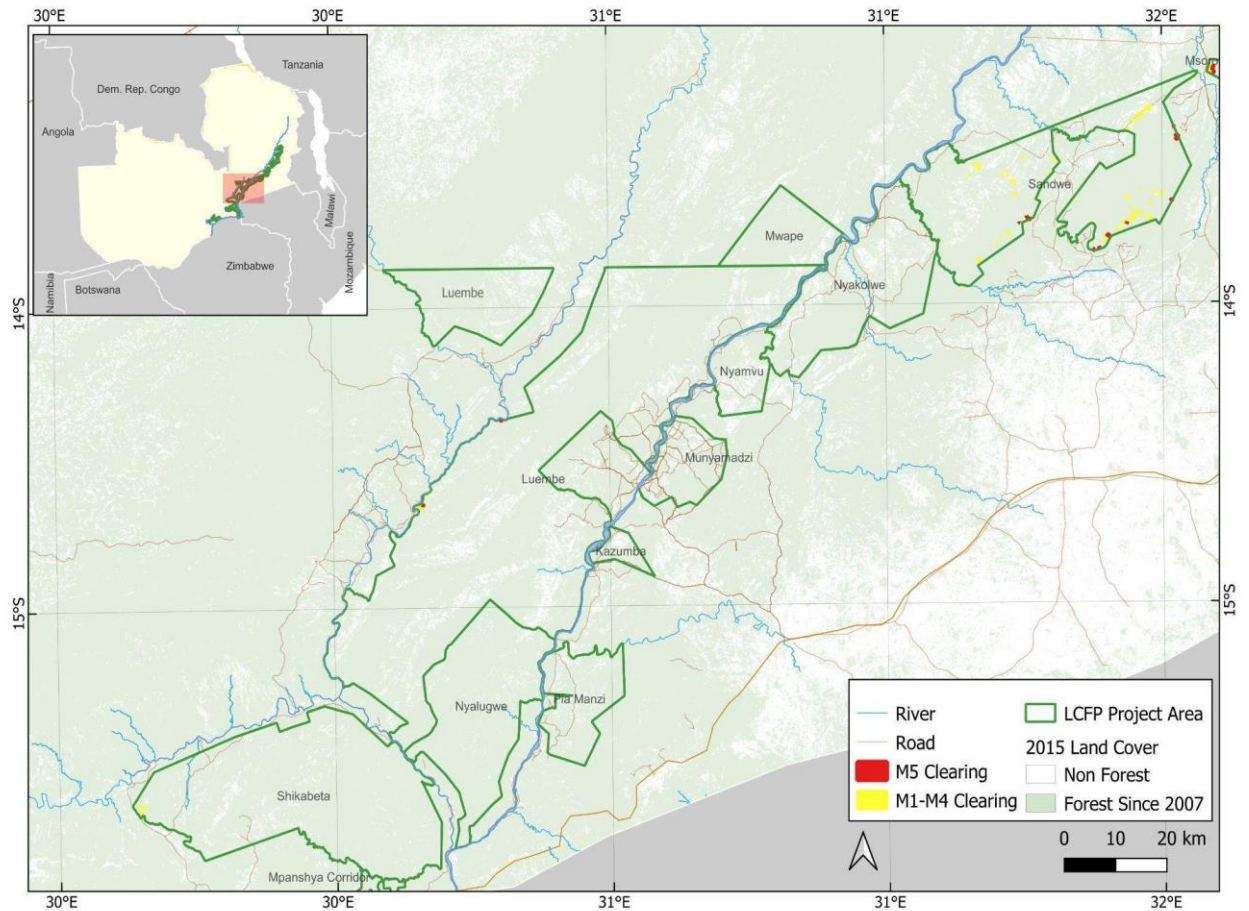


Figure 21a. Project accounting area clearings in the Central part of the project area.

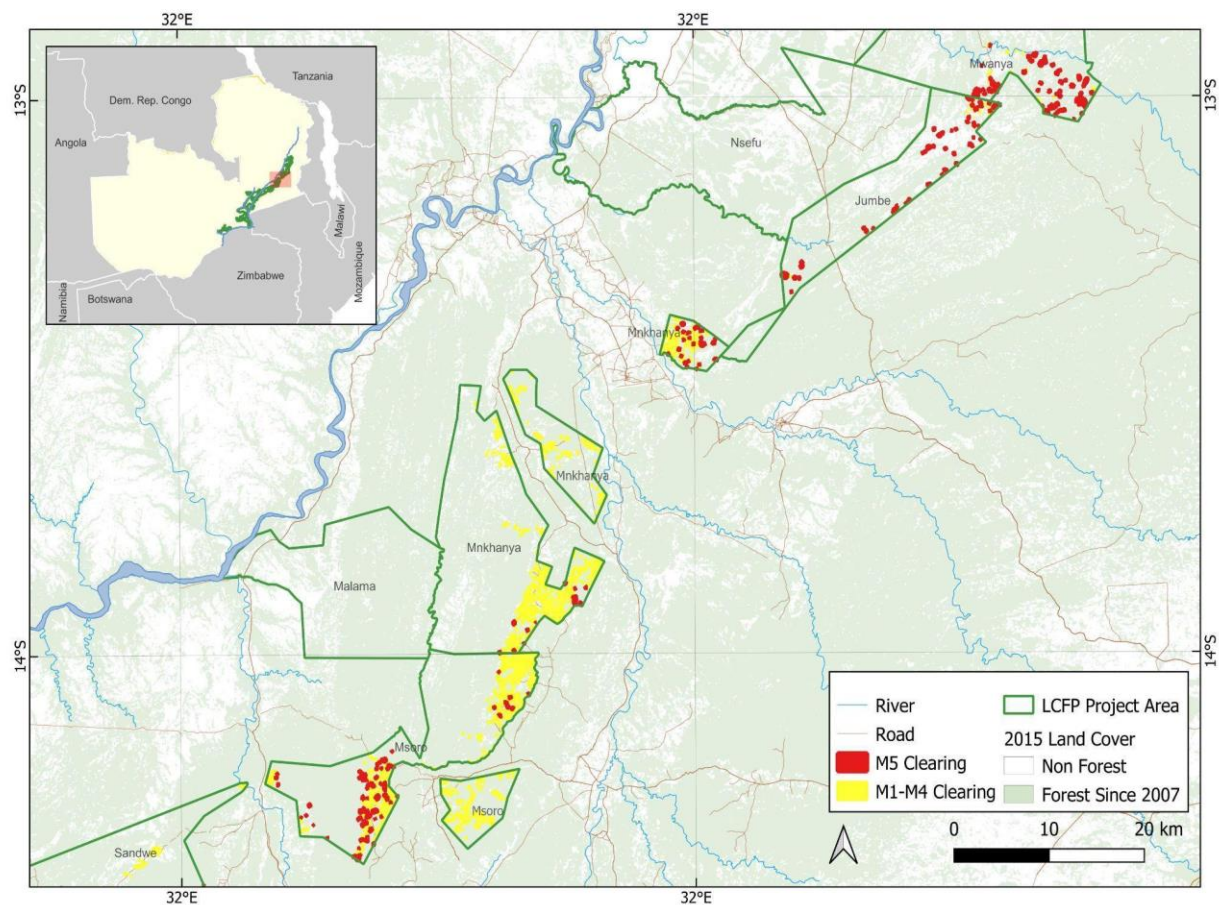


Figure 21b. Project accounting area clearings in the Central part of the project area.

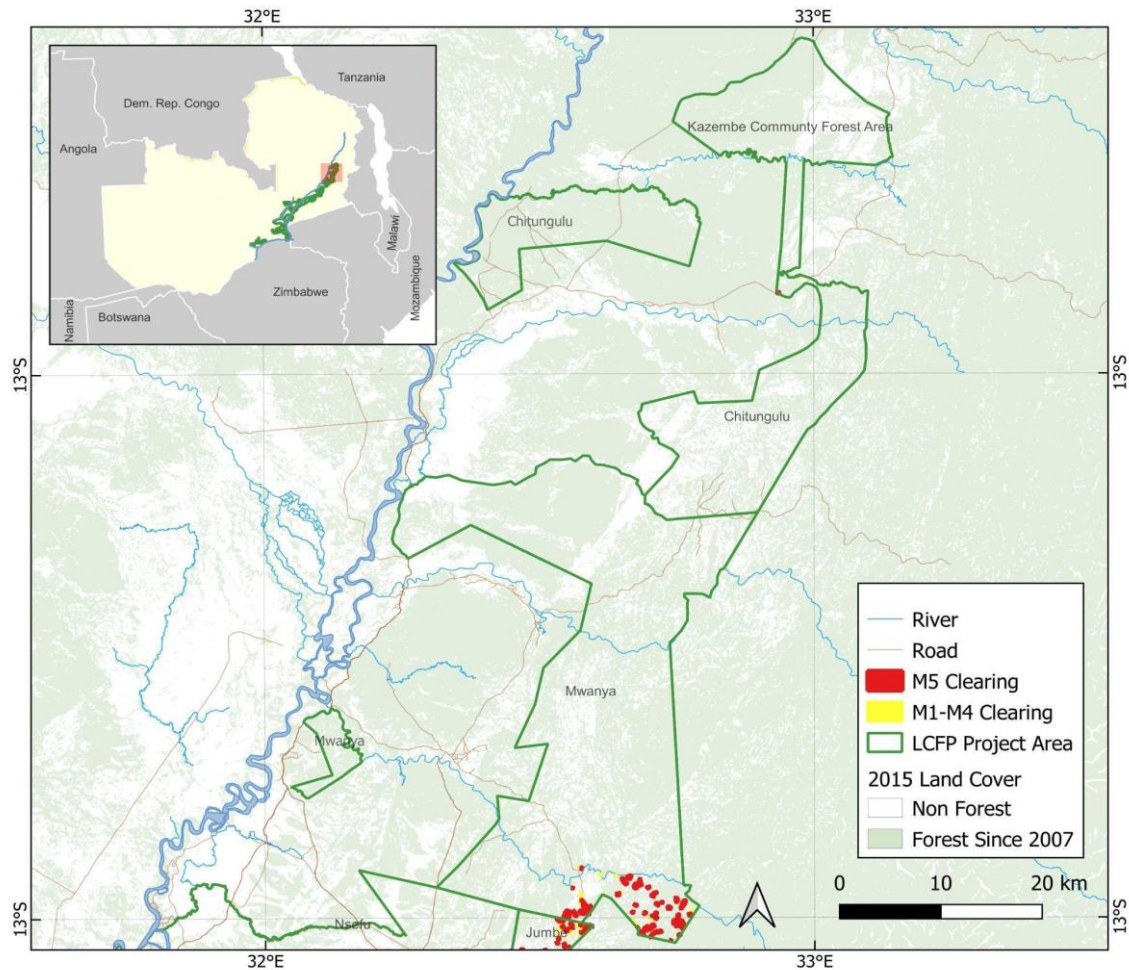


Figure 22. Project accounting area clearings in the Northern part of the project area.

5.3 Leakage

MR.44

The primary leakage mitigation strategies of the project are improving agricultural techniques with conservation farming and agroforestry and promoting alternative livelihoods that don't depend on forest clearing such as beekeeping. See section 4.3 of this report for more detailed descriptions of the leakage mitigation activities employed by this project. Based on the experience of BCP's other REDD+ project in Lusaka Province, which employed similar leakage mitigation activities and has not experienced any leakage in 10 years, it is assumed that these activities will also be effective at preventing leakage from the LCFP project area.

During the fifth monitoring period, a total of 25 plots were sampled using a predefined leakage monitoring strategy. In the previous monitoring period (M4), two (2) initial leakage plots were discarded because they fell within the second project activity instance Project area. As a result, two (2) new replacement plots were monitored. In the current monitoring

period(M5), all the 25 leakage plots were monitored. Observed degradation was calculated from these 25 plots using equation B.9 VM0009, v3.0 (see Appendix C for calculations). The result was a score of 0.08 (Table 32). This value is slightly lower than what was recorded (0.088) for the third monitoring period.

Table 32. Monitoring results for the activity shifting leakage area for the 5th monitoring period.

Plot ID	Stump_Count	Leakage Score	Degradation Factor
Lkg2	41	8.29%	0.2
Lkg20	5	1.01%	0.2
Lkg26	0	0.00%	0
Lkg27	8	1.62%	0.2
Lkg28	5	1.01%	0.2
Lkg29	0	0.00%	0
Lkg3	13	2.63%	0.2
Lkg30	0	0.00%	0
Lkg48	0	0.00%	0
Lkg32	0	0.00%	0
Lkg33	0	0.00%	0
Lkg34	0	0.00%	0.2
Lkg35	0	0.00%	0
Lkg36	0	0.00%	0
Lkg37	51	10.32%	0.2
Lkg38	1	0.20%	0.2
Lkg39	0	0.00%	0
Lkg40	0	0.00%	0
Lkg41	0	0.00%	0
Lkg42	3	0.61%	0.2

Lkg43	3	0.61%	0.2
Lkg47	2	0.40%	0.2
Lkg5	0	0.00%	0
Lkg6	0	0.00%	0
Lkg46	0	0.00%	0
Pldeg	132		0.08

MR.48

Under VM0009 V3.0, cumulative emissions from activity-shifting leakage ($E_{(LASF)}^{[m]}$) for the current monitoring period are 0.

MR.49

Cumulative emissions from activity-shifting ($E_{(LAS)}^{[m]}$) for the prior monitoring periods are 0.

MR.56

The parameter ($p_{(LDEG)}^{[m]}$) for the current monitoring period was estimated using equation B.9. The calculations can be found in Appendix G.

$$p_{(LDEG)}^{[m]} = 0.08$$

MR.57

For the first monitoring period, $p_{(LDEG)}^{[m=0]} = 0.08$

MR.63

BCP is a project proponent representing the private land-owners in this project. BCP doesn't own any land, but is the project proponent in one other REDD+ project in the country involving a private land-owner. The management plans and land-use of land in that REDD+ project have not changed as a result of this project.

The community resource boards (CRBs) are the project proponents for the project areas under community forest management agreements (CFMAs). The project areas are the only lands under their control.

5.4 Net GHG Emission Reductions and Removals

MR.65

GERs (Net GHG emissions or removals in Table 33) for the current monitoring period were calculated using equation F.53 as per section 8.4.1 of VM0009 V3.0 (see Appendix G).

2,163,019 tCO₂e.

Due to insufficient data available to close finding #9 raised by Verra during its accuracy review (see the PRR), a 30% discount is applied to the verified volume.

1,514,113 tCO₂e. (70% of the NER)

MR.66

GERs (Net GHG emissions or removals) for the prior monitoring period were calculated using equation F.53 as per section 8.4.1 of VM0009, v3.0 (see Appendix G).

1,820,354 tCO₂e¹⁵

MR.67

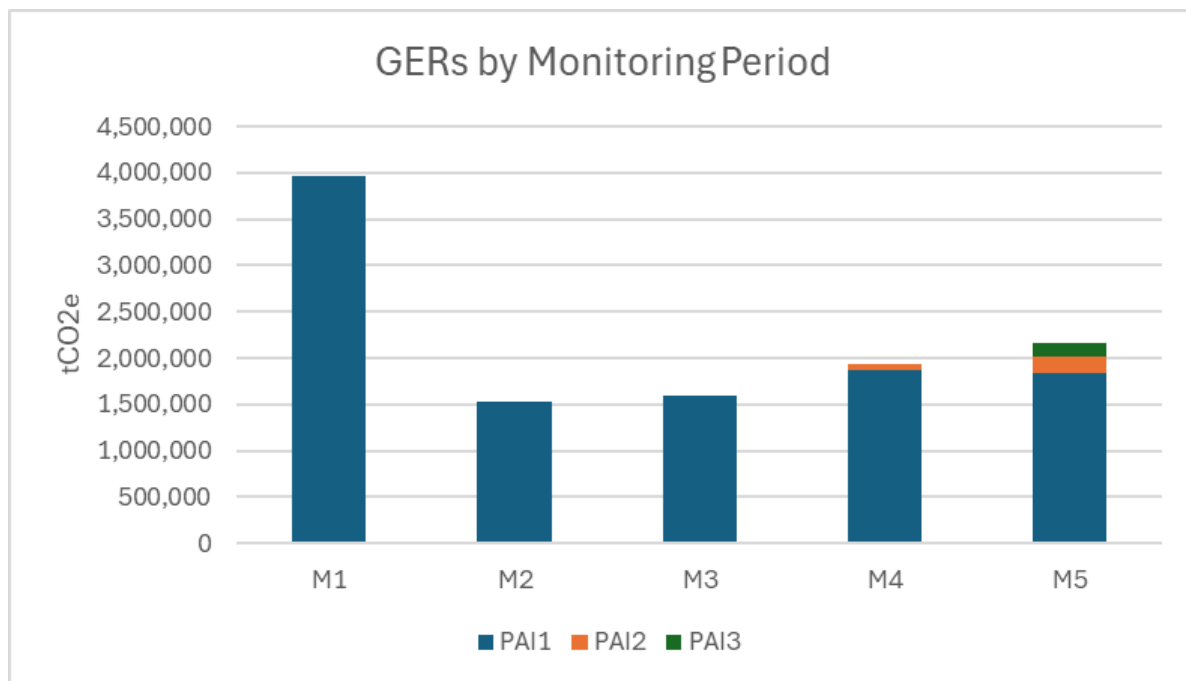


Figure 23. Graph of GERs by monitoring year, and project instance.

¹⁵ Value is different from that reported in M4 Monitoring report because of the CDM update in the current monitoring Period.

Table 33. Net GHG emissions reductions or removals for the current monitoring period

Year	Baseline emissions or removals (tCO ₂ e)	Project emissions or removals (tCO ₂ e)	Leakage emissions (tCO ₂ e)	Net GHG emission reductions or removals (tCO ₂ e)	Buffer pool allocation	VCUs eligible for issuance
2022	2,267,514	-104,495	0	2,163,019	-216,302	1,946,717
Capped	2,267,514	-104,495	0	1,514,113	-151,411	1,362,702
Difference						584,015

MR.68

$U_B^{[m]}$ - Total uncertainty in proxy area carbon stock estimate = 15,727 tCO₂e

$U_{EM}^{[m]}$ - Total uncertainty in Baseline Emissions Models = -13,061 tCO₂e

$U_P^{[m]}$ - Total uncertainty in project accounting area carbon stock estimate = 25,071,680 tCO₂e

$E_u^{[m]}$ - Total uncertainty = -36,126 = 0

There was no confidence deduction as the uncertainty was insignificant.

MR.69

$U_B^{[m]}$ was calculated with equation B.10 VM0009 V3.0

$U_{EM}^{[m]}$ was calculated with equation F.14 VM0009 V3.0

$U_P^{[m]}$ was calculated with equation B.10 VM0009 V3.0

$E_u^{[m]}$ was calculated with equation F.57 VM0009 V3.0

For more details, see Appendix G.

MR.74

NERs (VCUs as per VCS) generated for the current monitoring period were calculated using equation F.55 from VM0009 V3.0 and are reported in Table 33.

MR.75

NERs for the M4 monitoring period were 1,638,318

tCO₂e¹⁶

MR.76

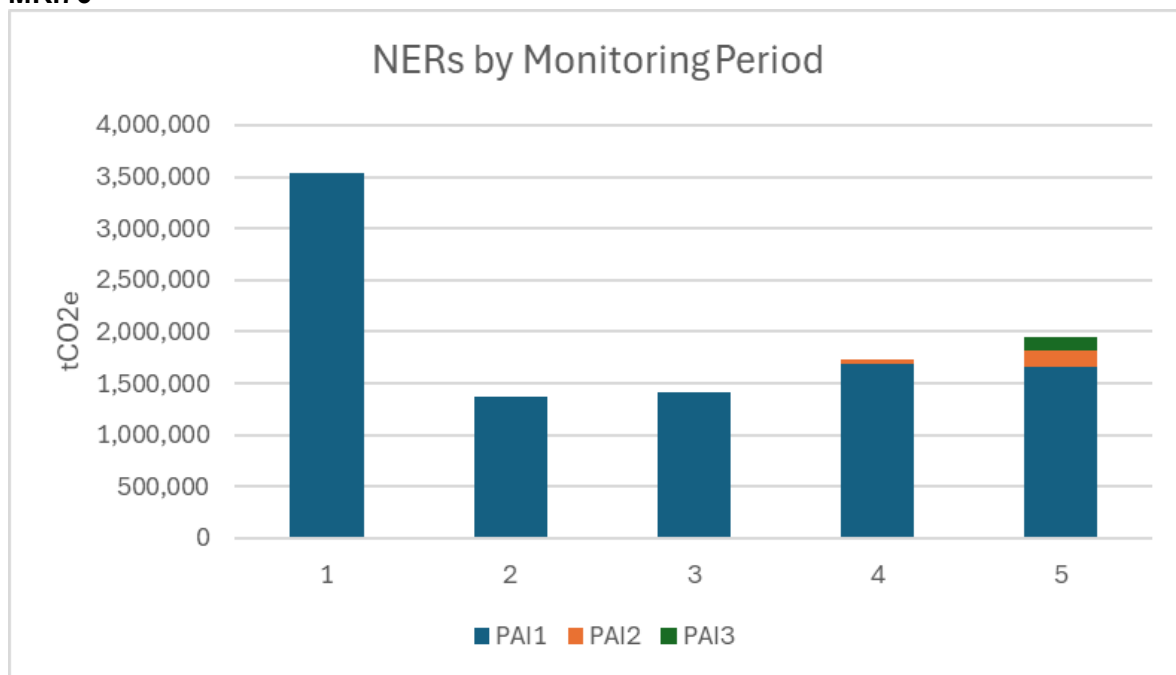


Figure 24. Graph of NERs by monitoring year, and project instance.

MR.77

VCS Non-Permanence Risk Assessment Tool V4.0 (Appendix H) was used to calculate buffer account allocation, calculated as GERs x Non Permanence Risk score (10%) = - 216,302 tCO₂e.

Due to the cap applied to the NERs, the buffer applied is 151,411tCO₂e

MR.78

¹⁶ Value is different from that reported in M4 Monitoring report because of the CDM update in the current monitoring period.

Please see Appendix H for non-permanence risk calculations using the VCS Non-Permanence Risk Assessment Tool V4.0.

MR.82

See Table 33 for quantified NERs by vintage year for the current monitoring period. See Appendix G for calculations.

MR.83

The ex-ante estimated NERs for this monitoring period were 2,267,514 tCO₂e compared to an ex-post estimate of 2,163,019 tCO₂e¹⁷.

MR.84

The difference between ex-ante and ex-post estimates is primarily due to two factors:

1. The inclusion of revised SOC values in monitoring period 3.
2. The inclusion of additional project instances in monitoring periods 4 and 5.

Table 34. Estimated ex-ante and achieved GHG emission reductions for the current monitoring period.

Ex-ante emissions reductions/removals	Achieved emissions reductions/removals	Percent difference	Justification for the difference
2,267,514	2,163,019 ¹⁸	4.6 %	Achieved emissions reductions were less than those estimated ex-ante, as project emissions were underestimated ex-ante.

¹⁷ Due to insufficient data available to close finding #9 raised by Verra during its accuracy review (see the PRR), a 30% discount is applied to this verified volume

¹⁸ Due to insufficient data available to close finding #9 raised by Verra during its accuracy review (see the PRR), a 30% discount is applied to this verified volume

APPENDICES: AVAILABLE UPON REQUEST

Appendix A: Community Forest Management Agreements and Communication Agreement. Confidential

Appendix B: Project Implementation Status Support Documents. Public

Appendix C: Proxy Soil Sampling and Analysis. Confidential

Appendix E: BCP Community Engagement SOPs and Monitoring Systems. Confidential

Appendix G: Emissions Reductions Calculations. Confidential

Appendix H: VCS Non-permanence risk assessment. Public

Appendix I: Spatial Data. Confidential

Appendix K: Drivers and Agents of Deforestation Study Report

Appendix O: Cumulative Deforestation Model. Confidential

Appendix P: Biomass and Soil Sampling Protocol. Confidential

Appendix R: Leakage Monitoring Protocol. Confidential

ANNEX 1: LIST OF VM0009 V3.0 MR REQUIREMENTS APPLICABLE TO F-U3 BASELINE TYPE

MR#	Category	Applicable to Project
MR.1	Spatial Project Boundaries	Yes
MR.2	Temporal Project Boundaries	Yes
MR.3	Temporal Project Boundaries	Yes
MR.4	Grouped Projects	Yes
MR.5	Grouped Projects	Yes
MR.6	Project Accounting Areas	Yes
MR.7	Determining t_{PAI}	Yes
MR.8	Determining t_{PAI}	Yes
MR.9	Determining x_{PAI}	No. No covariates to conversion were used.
MR.10	Baseline Emissions	Yes
MR.11	Baseline Emissions	Yes
MR.12	Baseline Emissions	Yes
MR.15	Applying the Spatial Algorithm	Yes, but there is only one stratum
MR.16	Applying the Spatial Algorithm	Yes, but there is only one stratum
MR.17	Applying the Spatial Algorithm	Yes, but there is only one stratum
MR.21	Carbon Not Decayed in DW	No. All biomass is burned in baseline scenarios
MR.22	Carbon Not Decayed in DW	No. All biomass is burned in baseline scenarios
MR.23	Carbon Not Decayed in DW	No. All biomass is burned in baseline scenarios
MR.24	Carbon Not Decayed in DW	No. All biomass is burned in baseline scenarios
MR.25	Carbon Not Decayed in DW	No. All biomass is burned in baseline scenarios
MR.26	Carbon Not Decayed in BGB	Yes
MR.27	Carbon Not Decayed in BGB	Yes
MR.28	Carbon Not Decayed in BGB	Yes
MR.29	Carbon Not Decayed in SOC	Yes
MR.30	Carbon Stored in Wood Products	No. No timber harvesting in baseline
MR.31	Carbon Stored in Wood Products	No. No timber harvesting in baseline
MR.32	Emissions Events in Project Area	Yes
MR.33	Emissions Events in Project Area	Yes
MR.34	Emissions from Burning from Project Activities	No. No biomass burning in project activities
MR.35	Carbon Stored in Wood Products from Project Activities	No. Timber harvesting is not a project activity
MR.36	Carbon Stored in Wood Products from Project Activities	No. Timber harvesting is not a project activity

MR.37	Carbon Stored in Wood Products from Project Activities	No. Timber harvesting is not a project activity
MR.38	Livestock Grazed in the Project Area	No. Livestock grazing is not permitted in the project area and unpermitted grazing was determined to be de minimus.
MR.39	Livestock Grazed in the Project Area	No. Livestock grazing is not permitted in the project area and unpermitted grazing was determined to be de minimus.
MR.40	Livestock Grazed in the Project Area	No. Livestock grazing is not permitted in the project area and unpermitted grazing was determined to be de minimus.
MR.41	Synthetic Fertilizer in the Project Area	No. Project activities do not include the use of synthetic fertilizers.
MR.42	Synthetic Fertilizer in the Project Area	No. Project activities do not include the use of synthetic fertilizers.
MR.43	Synthetic Fertilizer in the Project Area	No. Project activities do not include the use of synthetic fertilizers.
MR.44	Leakage Mitigation Strategies	Yes
MR.45	Commodity Production for Leakage Mitigation	No. The baseline scenario is subsistence
MR.46	Commodity Production for Leakage Mitigation	No. The baseline scenario is subsistence
MR.47	Commodity Production for Leakage Mitigation	No. The baseline scenario is subsistence
MR.48	Estimating Emissions from Activity-Shifting Leakage	Yes
MR.49	Estimating Emissions from Activity-Shifting Leakage	Yes
MR.50	Change to the Activity-Shifting Leakage Area	Yes. The Initial leakage area was reduced (229,291 ha) due to its overlap with second project area instance, however No changes have been made during the current monitoring period
MR.51	Change to the Activity-Shifting Leakage Area	Yes. The Initial leakage area was reduced (229,291 ha) due to its overlap with second project area instance, however No changes have been made during the current monitoring period
MR.52	Change to the Activity-Shifting Leakage Area	Yes. The Initial leakage area was reduced (229,291 ha) due to its overlap with second project area instance, however No changes have been made during the current monitoring period
MR.53	Change to the Activity-Shifting Leakage Area	Yes. The Initial leakage area was reduced (229,291 ha) due to its overlap with second project area

		instance, however No changes have been made during the current monitoring period
MR.54	Change to the Activity-Shifting Leakage Area	Yes. The Initial leakage area was reduced (229,291 ha) due to its overlap with second project area instance, however No changes have been made during the current monitoring period
MR.55	Change to the Activity-Shifting Leakage Area	Yes. The Initial leakage area was reduced (229,291 ha) due to its overlap with second project area instance, however No changes have been made during the current monitoring period
MR.56	Estimating $\rho_{L\text{ DEG}}$	Yes
MR.57	Estimating $\rho_{L\text{ DEG}}$	Yes
MR.58	Estimating $\rho_{L\text{ CONG}}$	No. No grassland in project area
MR.59	Estimating $\rho_{L\text{ CONG}}$	No. No grassland in project area
MR.60	Determining Emissions from Market Leakage	No. The baseline scenario is subsistence
MR.61	Determining Emissions from Market Leakage	No. The baseline scenario is subsistence
MR.62	Determining Emissions from Market Leakage	No. The baseline scenario is subsistence
MR.63	Ensuring No Leakage Within Project Proponent's Ownership	Yes
MR.65	Quantification of GERs	Yes
MR.66	Quantification of GERs	Yes
MR.67	Quantification of GERs	Yes
MR.68	Confidence Deduction	Yes
MR.69	Confidence Deduction	Yes
MR.72	Reversal Event	No. No reversal occurred
MR.73	Reversal Event as a Result of Baseline Reevaluation	No. No baseline reevaluation occurred
MR.74	Quantification of NERs for a PAA	Yes
MR.75	Quantification of NERs for a PAA	Yes
MR.76	Quantification of NERs for a PAA	Yes
MR.77	Buffer Account	Yes
MR.78	Buffer Account	Yes
MR.79	Quantification of NERs across PAAs	Yes. There is currently 3 PAA
MR.80	Quantification of NERs across PAAs	Yes. There is currently 3 PAA
MR.81	Quantification of NERs across PAAs	Yes. There is currently 3 PAA
MR.82	Vintages	Yes
MR.83	Evaluating Project Performance	Yes
MR.84	Evaluating Project Performance	Yes
MR.85	Data and Parameters Monitored	Yes

MR.86	Data and Parameters Monitored	Yes
MR.87	Data and Parameters Monitored	Yes
MR.88	Description of the Monitoring Plan	Yes
MR.89	Description of the Monitoring Plan	No. No project activity with biomass burning
MR.90	Description of the Monitoring Plan	Yes
MR.91	Description of the Monitoring Plan	Yes
MR.92	Description of the Monitoring Plan	Yes
MR.93	Description of the Monitoring Plan	Yes
MR.94	Description of the Monitoring Plan	No. No new equations were developed
MR.95	Description of the Monitoring Plan	Yes
MR.96	Description of the Monitoring Plan	No. Timber harvesting is not a project activity
MR.97	Description of the Monitoring Plan	Yes
MR.98	Description of the Monitoring Plan	Yes
MR.99	Sources of Allometry	Yes
MR.100	Sources of Allometry	Yes
MR.101	Sources of Allometry	Yes
MR.102	Sources of Allometry	Yes
MR.103	Sources of Allometry	Yes
MR.104	Validating Previously Developed Allometry	Yes
MR.105	Validating Previously Developed Allometry	Yes
MR.106	Validating Previously Developed Allometry	Yes
MR.107	Validating Previously Developed Allometry	Yes
MR.108	Validating Previously Developed Allometry	Yes
MR.109	Validating Newly Developed Allometry	No. No new allometric equations were developed.
MR.110	Validating Newly Developed Allometry	No. No new allometric equations were developed.
MR.111	Validating Newly Developed Allometry	No. No new allometric equations were developed.
MR.112	Validating Newly Developed Allometry	No. No new allometric equations were developed.
MR.113	Validating Newly Developed Allometry	No. No new allometric equations were developed.